

Rink Reports

Complete Documentation

Everything that ships with Rink Reports, in print order: role-based onboarding, the Operations & Training Manual with every module chapter, the seven-chapter training guide, and the install, first-time setup, and deployment guides.

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Part 1 — Onboarding

First-day paths for staff, supervisors, and admins.

Staff Onboarding — Here's How to Do Your Job

RinkReports · Version 5-6 · for staff

This is the short version: just the things you'll actually do. Open RinkReports on your phone or tablet, sign in, and you'll see your menu (or the bottom tabs: **Home · Reports · Menu · Account**). You only see the modules your facility turned on and gave you access to. Tap a task below for the full how-to.

You only see your own facility's data — this is automatic. Nothing to set up. If a menu item is missing, your facility either turned that module off or hasn't granted you access — ask your manager.

Submit your reports

- ☐ **Daily Reports** — Tap **Daily Reports** → pick your **Work area** → pick the **Shift** → tick each task → optional **Note** → **Submit**. Each submit is its own record; you can't edit it after, so submit a fresh one if you need to correct something. (See Daily Reports §6 — Submit a daily report.)
- ☐ **Refrigeration** — Tap **Refrigeration** → set the **Reading time** → work down each section, filling in every reading → watch the "**Normal: ...**" hints. If a value is **critically** out of range, fill in the **Corrective action** box → **Submit**. (The °F/°C toggle changes display only.) (See Refrigeration §6 — Log a refrigeration reading.)
- ☐ **Air Quality** — Tap **Air Quality** → choose the **Location** → enter each reading and watch the **range badge** (green = good). If it turns amber/red, read the banner, act, then write the required **Corrective action** note → **Submit**. (See Air Quality §6 — Record an air-quality reading.)
- ☐ **Ice Operations** — Tap **Ice Operations** → pick the tab (**Ice Make / Edging / Blade Change / Circle Check**) → fill the fields → for a **Circle Check**, mark each item **Pass/Fail** and write a note on every **Fail** → **Submit**. (See Ice Operations §6.)
- ☐ **Ice Depth** — Tap **Ice Depth** → tap a point on the rink diagram → type the depth (or use a Bluetooth caliper) → **Enter** moves to the next point → **Review & Submit** when done. (See Ice Depth §6 — Measure ice depth, and the Bluetooth caliper steps.)

File an incident or accident

- ☐ **Incidents** — Tap **Incidents** → set **When did it happen?** → pick the **Facility space(s)** → write the **Description** → set **Severity**, toggle **Ambulance called / Follow-up required**, add **witnesses** if any → **Submit** and confirm. **You can edit your own report for 24 hours**, then it locks. No photos — describe it in words. (See Incident Reporting §6.)
- ☐ **Accidents** — File from the **Accidents** menu item the same way (reporter is filled in from your login). (See the Incident chapter for the shared location list and review flow.)

Scheduling — self-service

Open **Scheduling** for your shifts and requests.

- ☐ **See your shifts** — The landing screen shows your **Next shift**, the week strip, and **Upcoming** shifts. (See Employee Scheduling §5 — Staff app.)
- ☐ **Set availability** — **Scheduling** → **Availability** → **New** → day, times, **Type** (Available / Preferred / Unavailable), optional area → **Add**. (See §6 — Submit availability.)

- ☐ **Request time off** — **Scheduling** → **Time off** → **New request** → **Starts / Ends** / optional reason → **Submit**. Cancel your own while it's pending/approved.
- ☐ **Claim an open shift** — On the landing screen, find a shift under **Open · Pick up** → **Claim shift**. It may need manager approval. (Online only.)
- ☐ **Request a swap** — **Scheduling** → **Shift swaps** → **New swap request** → pick **your shift to give up**, optionally a coworker and their shift → **Send**. (Online only.)

Working offline ("Save on this device")

RinkReports keeps working without a connection for **submitting reports**:

- When you're offline, the Submit button reads **"Save on this device"** (or "Save offline"). Tap it — your report is saved on your device and **submits automatically when you reconnect**. (See any module's "Locking, saving & offline" section, e.g. Daily Reports §8.)
- You'll see a pending count, and **Reports** → **Offline queue** lists anything still waiting to sync. You won't create duplicates.
- **What does NOT work offline**: claiming open shifts and shift swaps (those need a live connection). Submitting **availability** and **time-off** does queue offline. (△ VERIFY — claim/swap are online-only; see Employee Scheduling §8.)

If something's stuck

- **A menu item is missing** → the module is off for your facility or you don't have access. Ask your manager.
- **"No permission" / "Account not set up"** → your login isn't fully set up or lacks the submit permission. Ask your manager.
- **You made a mistake after submitting** → you can't edit most reports (incidents give you 24 hours). Submit a corrected one, or ask a manager to add a follow-up note.

That's it — submit your reports, file incidents when needed, manage your own schedule. Each chapter linked above has a full walkthrough if you want more detail.

Supervisor Onboarding — Day-to-Day Oversight

RinkReports · Version 5-6 · for supervisors (the live manager role or an elevated custom role)

This guide covers what a supervisor does day to day, roughly in the order a shift unfolds: keep the schedule moving, decide the requests that land in your queue, and review what staff submitted. Links jump into the full chapters.

⚠ **VERIFY — your rights are permission-driven, not granted by your title.** "Supervisor" is the documentation tier; the live role is **manager** (or a custom role). What you can actually do depends on the **per-module / per-action permissions** an admin has enabled for you — not on the word "manager." Two people with the same title can have different access. Every right below is a typical one; if a screen shows you "Forbidden" or hides a button, you simply haven't been granted that permission. Ask your admin. See How RinkReports is organized.

1. Start of shift — check the schedule

- ☐ **Review the week and open shifts.** Open **Scheduling Admin → Overview** for the dashboard (pending swaps/time-off, open shifts, week-at-a-glance), then **Shifts** for the grid. (⚠ **VERIFY** — the admin grid needs the scheduling **admin** permission; if you lack it you'll see a Forbidden message. See Employee Scheduling §5 — Admin grid.)
- ☐ **Watch for coverage gaps and warnings.** The grid's right rail shows open shifts needing coverage and each person's hours vs. cap. Cert blocks are hard stops; overtime/overlap are advisory warnings. (See Employee Scheduling §8 — Certification & warning gates.)

2. Through the day — decide requests as they arrive

These are the approval queues that typically route to a supervisor. Only **pending** items can be decided.

- ☐ **Time-off requests.** **Scheduling Admin → Time-Off → Approve / Deny** (optional note). The employee is notified of the decision. (See Employee Scheduling §5 — Time-off & swaps.)
- ☐ **Shift swaps.** **Scheduling Admin → Swaps** → assign a target, **Approve** (re-runs all rule checks and applies the trade/coverage), **Deny**, or **Cancel**. An applied swap can't be undone afterward.
- ☐ **Open-shift claims.** Claims that need approval appear for you to approve before the shift is assigned. (See Employee Scheduling §5 — Staff app: Open shifts.)
- ☐ **Publishing (two-person rule).** A schedule is **requested for publish** by one admin and **approved by a different one** — you cannot approve your own request. If you filed it, a colleague approves; if a colleague filed it, you can **Approve & publish** or **Reject** with a reason. (See Employee Scheduling §5 — Publishing and §8 — Draft → Published lifecycle.)

3. Review what staff submitted

Each report module has an admin **History** tab with filters and a drill-down detail panel. Supervisors review submissions and add **follow-up notes** — they do **not** edit the original (records are append-only). (⚠ **VERIFY** — History/admin tabs require that module's admin permission.)

- ☐ **Daily Reports — Daily Reports Admin → Submissions:** filter by area/employee/date, open a submission, add an admin note (history auto-deletes after 14 days). (See Daily Reports §5 — Admin tabs.)
- ☐ **Refrigeration — Refrigeration Admin → History:** review readings, out-of-range flags, and corrective-action notes; **Add follow-up note.** (See Refrigeration §5.4.)
- ☐ **Air Quality — Air Quality Admin → History:** review readings, exceedances, and corrective notes; add follow-up notes. (See Air Quality §4 — History tab.)
- ☐ **Ice Operations — Ice Operations Admin → History:** review operations and failed circle checks; add follow-up notes. (See Ice Operations §4 — Setup & configuration.)
- ☐ **Ice Depth — Ice Depth Admin → History** (and **Analytics** for trends/problem spots): review sessions; add follow-up notes. (See Ice Depth §5 — Admin tabs.)
- ☐ **Incidents — Incident Reports Admin → History:** review reports, **change status** (Submitted → In review → Resolved → Archived), and add follow-up notes. The reporter owns the first 24 hours of edits; after that only status/notes change. (See Incident Reporting §5.E and §8.)

4. Follow-ups and handoffs

- ☐ **Add follow-up notes, don't rewrite.** Follow-up notes are append-only and visible on the report; the original submission is immutable. Use them to record corrections, actions taken, or context for the next shift.
- ☐ **Escalate alerts.** Critical signals — a failed circle check, an out-of-range refrigeration/air-quality reading, an "ambulance called" incident — raise alerts (when alerts are enabled) that surface in the Communications/alerts area for acknowledgement. (See each module's "Locking, saving & offline" section.)

What a supervisor typically cannot do

Unless an admin has specifically granted it (⚠ VERIFY per facility):

- **Configure** a module's setup/settings (that's the module's `admin` permission — `facility_manager` tier).
- **Edit** a staff member's submitted report in place (everyone uses follow-up notes; incidents' 24-hour window belongs to the reporter).
- **Approve your own** publish request (always a different admin).
- Anything in the **Admin Control Center** Setup/System screens or the **Super Admin** console without the matching permission.

See the Admin onboarding guide for the configuration tasks that sit above the supervisor's day-to-day work.

Admin Onboarding — Stand Up a New Facility

RinkReports · Version 5-6 · for super_admin / org_admin / facility_manager

This is the guided first-day path for getting a brand-new rink live in RinkReports. Work top to bottom: each step builds on the one before it, and the early steps unlock the later screens. Links jump into the full module chapters where the detail lives.

Role note (⚠️ VERIFY — permission-driven, not title-driven). RinkReports has no org_admin tier in the live app — treat any org_admin step as **super_admin**. Most setup below is a **facility_manager** (admin) task, but a few steps are **super_admin-only** and are flagged ⚠️ **super_admin only**. Access is resolved per module/action, so confirm any "who can do this" claim against your configured facility. See How RinkReports is organized.

Everything you configure applies to your facility only — this is automatic. See the Admin Control Center chapter for the full back-office reference.

Phase 1 — Create the facility and its core configuration

- ☐ ⚠️ **super_admin only — Create the facility.** In **Admin Center → Facility**, click **New Facility** and fill in Name, Slug (unique), Timezone, and address/contact. A regular admin sees this screen read-only. (See Admin Control Center §5 — Facility.)
- ☐ **Seed the canonical roles.** From the **Admin Dashboard** setup checklist ("Canonical roles seeded"), open the Roles screen and **Seed default roles** to create Super Admin / Administrator / Manager / Staff. (Roles has no sidebar link — reach it from the dashboard checklist. ⚠️ VERIFY navigation. See Admin Control Center §5 — Roles.)
- ☐ **Review role permission defaults.** On the Roles screen, set each role's default View / Submit / Edit / Admin per module so new employees inherit sensible access. Add any custom per-facility roles (e.g. driver) now.
- ☐ **Turn on the modules you'll use.** In **Admin Center → Modules**, flip the on/off switch for each module (Daily Reports, Ice Depth, Ice Operations, Refrigeration, Air Quality, Incidents, Accidents, Scheduling, Communications, Facility Paperwork). This controls **staff navigation visibility only** — it doesn't grant permissions. (See Admin Control Center §5 — Modules.)
- ☐ **Set up shared lists.** Create **Departments** (for scheduling/routing) and **Facility Spaces** (the location list used by Incident, Accident, and Air Quality — use **Seed defaults** for a standard set). (See Admin Control Center §5 — Departments / Facility Spaces.)

Phase 2 — Add people and assign roles

- ☐ **Add your employees.** In **Admin Center → People**, use **Add employee** (or **Bulk add** for up to 100 at once). Set name, **Role**, and decide **Create a login & apply these permissions** (on for people who sign in; off for schedule-only staff such as minors). (See Admin Control Center §5 — People / Employees and the employee field reference in §7.)
- ☐ **Assign roles and job areas.** Choose each person's role in the same form (you can only assign roles at or below your own rank), and pick up to 4 **job areas** (one Primary) for scheduling. Set **Max weekly hours** if you want an overtime warning threshold.

- ☐ **Send invites.** For login users, the invite goes out when you create them (or via the **Invite** row action / the bulk "send invites" toggle). Seed permissions are applied automatically.
- ☐ **super_admin only — Cross-facility placement & Admin access.** Only a super_admin can place someone in a different facility and grant the **Admin module's Admin action** (Admin Center access). A regular admin cannot hand out Admin Center access. (See Admin Control Center §5 — Permissions.)

Phase 3 — Configure each module's settings

Open each enabled module's admin area (Admin Center → Module Admin) and complete its **Setup & configuration** section. Most modules offer a **Seed defaults** button to get a working starting point.

- ☐ **Daily Reports** — create **Areas** (up to **30 active**; VERIFY exact cap), build **Templates** and **Checklist Items**, and grant **Area Access** per employee. (See Daily Reports §4.)
- ☐ **Refrigeration Logs** — **Seed defaults** or build **Sections / Equipment / Fields**, set **Thresholds** (normal ranges), and configure **Settings** (out-of-range alerts, readings-per-shift cap). (See Refrigeration §4.)
- ☐ **Air Quality** — add **Reading types** and **Equipment**, choose a **Compliance** profile and **tighten thresholds** if needed (you can only tighten, never loosen — VERIFY), and set **Settings** (alerts, testing-frequency text). (See Air Quality §4.)
- ☐ **Ice Operations** — set up **Rinks, Equipment, Fuel types, Circle-check items/templates**, then choose **Visible operations** and alerts in **Settings**. (See Ice Operations §4.)
- ☐ **Ice Depth** — create **Rinks** and **Diagrams** (place measurement points), and set **Settings** (unit — inches/mm VERIFY, thresholds, colors, alerts). (See Ice Depth §4.)
- ☐ **Incident / Accident Reporting** — configure **Severity Levels** (at least one is required), **Activities**, and **Types**; the location list comes from Facility Spaces. (See Incident Reporting §4.)
- ☐ **Employee Scheduling** — set up **Job areas** (with required certifications), **Settings** (week start, approval rules, warning behavior), and **Templates / Compliance** rules. (See Employee Scheduling §4.)

Phase 4 — Permissions, retention, exports

- ☐ **Fine-tune permissions.** In **Admin Center → Permissions**, override any individual's module x action access (View / Submit / Edit / Admin). Presets (Full Access / Submitter Only / Viewer Only / No Access) write a whole row at once. Remember the **Admin-module Admin** cell is super_admin-only to grant. (See Admin Control Center §5 — Permissions.)
- ☐ **Set data retention.** In **Admin Center → Data Retention**, set keep-days per module (0 = forever, or ≥ 30) and decide on nightly auto-purge. Heed the warning for incident/accident/workers'-comp records. (See Admin Control Center §5 — Data Retention.)
- ☐ **Configure exports.** In **Admin Center → PDF/Export Settings**, set branding (logo, header/footer), layout (paper size, date format, CSV delimiter), and per-module column visibility. Run a test export. (See Admin Control Center §5 — PDF/Export Settings.)
- ☐ **Check the Setup checklist & audit log.** Return to the **Admin Dashboard** — the setup checklist should read "Ready." Spot-check the **Audit Log** to confirm your changes are recorded. (See Admin Control Center §5 — Dashboard / Audit Log.)

Quick reference — what's super_admin-only

These steps cannot be done by a facility_manager (admin) and must go to a super_admin:

- Create / edit / activate / deactivate a **facility**.
- Place an employee in a **different facility**.
- Grant the **Admin module's Admin action** (Admin Center access).
- **Hard-delete** an employee (others use Deactivate).
- Anything in the **Super Admin console** (/admin/super-admin).

(All confirmed in Admin Control Center §2 and §9. ⚠ VERIFY — org_admin has no live equivalent; treat as super_admin.)

Part 2 — Operations & Training Manual

How Rink Reports is organized, followed by one chapter per module.

RinkReports — Operations & Training Manual

RinkReports

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RinkReports is a progressive web app (PWA) for ice-rink facility operations. It replaces the clipboards and binders at the rink desk with structured, mobile-first digital flows: daily checklists, refrigeration and air-quality readings, ice operations logs, ice-depth measurements, incident and accident reports, and employee scheduling — all configured per facility from a single Admin Control Center.

You only see data for your own facility — this is automatic. There is no facility switcher for staff or facility admins; the app keeps each rink's records separate on its own. (The one exception is the platform-wide Super Admin console.)

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How RinkReports is organized

The eight modules

RinkReports is built from eight core modules. An admin turns each one on or off for the facility, and each appears in the staff menu only when enabled and the person has permission.

1. **Admin Control Center** — the back office: facility settings, employees, roles, permissions, lists, exports, retention, and the audit log.
2. **Daily Reports** — digital shift checklists, one per work area, ticked off and submitted each shift.
3. **Refrigeration Logs** — structured readings for the refrigeration plant, checked live against normal ranges.
4. **Incident Reporting** — records of something that went wrong (slip, collision, medical), reviewed and tracked by managers.
5. **Ice Operations** — the maintenance logbook: ice makes, edging, blade changes, and circle-check inspections.
6. **Air Quality** — CO / NO₂ / CO₂ readings checked live against the facility's compliance thresholds.
7. **Ice Depth** — point-by-point ice thickness measured against a tappable rink diagram.
8. **Employee Scheduling** — shift planning (admin grid) plus staff self-service (availability, time off, swaps, open shifts).

Role tiers

This manual uses a five-tier vocabulary, highest to lowest:

Doc tier	One-line description
super_admin	Platform owner. Cross-facility access; the only tier that sees the Super Admin console, creates facilities, and hard-deletes records.
org_admin	A multi-facility / organizational administrator sitting between super_admin and a single facility admin.
facility_manager	The person who runs the Admin Control Center for one facility — configures modules, employees, and settings.
supervisor	A mid-tier overseer with some elevated approval/review rights (e.g. scheduling decisions, submission review).
staff	Front-line workers who submit reports and use self-service scheduling.

⚠ **VERIFY — role mapping.** The live application does **not** use those five fixed tiers. The real model is **super_admin / admin / manager / staff**, plus optional **per-facility custom roles** (e.g. **driver**). Access is resolved **per module and per action** (View / Submit / Edit / Admin) through a permissions system — roles only seed default permissions, and an admin can override any individual person's access. The closest mapping: | Doc tier (this manual) | Live app role | Notes | ---|---|---| | super_admin | super_admin (platform flag) | Cross-facility; sees the Super Admin console. | | org_admin | (no exact equivalent) | ⚠ **VERIFY** — the app has **no separate org/multi-facility admin tier**. Treat as super_admin for now. | | facility_manager | admin (facility administrator) | Runs the Admin Center for one facility. | | supervisor | manager (or a custom role) | Some elevated rights — **but permission-driven, not title-driven**. | | staff | staff (or a custom role) | Submits reports; self-service scheduling. | Because access is permission-driven, two people with the same title can have different access if an admin has customized it. Read every "tier X can do Y" statement as "a user with the matching permission can do Y."

You only see your own facility's data

This is automatic and not configurable by staff or facility admins. There is no facility switcher, no way to view another rink's records, and nothing to set up — the separation is built in. Only the platform-wide Super Admin console works across facilities.

Module summaries

Each summary links to the full chapter. The chapters hold the screen-by-screen detail, field references, and per-module troubleshooting.

Admin Control Center — The configuration hub for one rink. Here a facility_manager sets the facility's information, turns report modules on or off, adds and manages employees, assigns roles and per-person permissions, manages shared lists (departments, facility spaces, dropdowns), and controls system settings for exports, data retention, and the audit log. Nothing a staff member sees is hardcoded — it all flows from here. Facility creation and granting Admin-module Admin access are super_admin-only.

Daily Reports — The digital shift-checklist clipboard. The facility is divided into **work areas**; each area has one or more checklist **templates** ("shifts"). Staff pick an area and shift, tick off tasks, leave an optional note, and submit. Every submission is an independent, append-only record auto-tagged with the facility's local date; reports auto-delete after 14 days. Submit access is granted area-by-area.

Refrigeration Logs — Structured readings for the refrigeration plant (compressors, pumps, condensers, brine, machine hours, alarms). Admins build sections, equipment, and reading fields, and set thresholds (normal ranges). Staff enter a round of readings, see "Normal: min – max" hints live, and must write a corrective-action note for any critically out-of-range value. A °F/°C toggle changes display only. Append-only.

Incident Reporting — Records of something that went wrong: when/where, what happened, severity, whether an ambulance was called, who was involved, and up to three witnesses. Uniquely, the reporter can **edit their own report for 24 hours**, after which it locks. No photo/file upload. Admins review in History, change status, and add follow-up notes.

Ice Operations — The maintenance logbook covering four built-in operation types: **Ice Make**, **Edging**, **Blade Change**, and **Circle Check** (a pass/fail inspection where failed items require a note and raise an alert). A facility chooses which types to show; equipment and circle-check items are admin-configured. Append-only.

Air Quality — CO / NO₂ / CO₂ (and more) readings, checked live against the facility's compliance profile with colored range badges (Within range / Corrective / Notification / Evacuation). Thresholds derive from a regulatory jurisdiction and can be **tightened but never loosened**. Over-threshold readings are exceedances and require a corrective-action note. Append-only.

Ice Depth — A two-phase (measure → review) flow that records ice thickness point-by-point against a tappable USA-Hockey rink diagram, tuned for Bluetooth calipers. Each point colors live by severity. Measures **depth, not temperature** (unit is inches or mm — no °F/°C toggle). Submitted sessions are immutable, with PDF / print / send options.

Employee Scheduling — Two sides: the **admin grid** (drag-build shifts, assign people, publish via a two-person request/approve flow) and the **staff app** (my shifts, availability, time off, claim open shifts, request swaps). Runs cert/hour-cap/overtime rule checks. **No payroll, timekeeping, or clock-in/out** — it is assignment only.

Glossary of cross-module terms

- **Area / work area** — A zone of the facility. In Daily Reports it's the checklist scope (and the "tabs" staff see); submit access is granted area-by-area. In scheduling, **job areas** are positions people are scheduled into.
- **Template / checklist** — A named, ordered set of tickable items. In Daily Reports a template is a "shift" within an area; in Ice Operations a circle-check template; in scheduling a reusable shift pattern.
- **Threshold** — An admin-defined normal range (and severity) for a numeric reading. Drives the "Normal: min - max" hint (Refrigeration), the live range badges (Air Quality), and severity colors (Ice Depth).
- **Exceedance** — A reading that crosses a threshold. Flagged on the report, counted on the confirmation screen, and (if alerts are on) raised to managers. In Air Quality an exceedance forces a corrective-action note.
- **Corrective-action note** — A required free-text note describing what was done about an out-of-range / failed reading. Required for critical refrigeration breaches, air-quality exceedances, and failed circle-check items; blocks submit until written.
- **Draft / published** — Scheduling states. Drafts are admin-only and freely editable; publishing makes shifts visible to staff and opens unassigned ones for claiming.
- **Publish-lock** — Once a shift is published it can't be freely edited or deleted; changes go through a controlled, audited path that re-checks rules and notifies affected staff.
- **Soft cancel** — Deleting a published shift marks it **cancelled** (kept for the record) rather than erasing it. Draft shifts are deleted outright.
- **Append-only** — Each submit creates a new, permanent record that staff cannot edit. To correct, you submit again (incidents' 24-hour window is the exception). Admins add follow-up notes rather than editing.
- **Follow-up note** — An append-only note an admin adds to a submitted report from the History detail panel. Cannot be edited or deleted; the original report stays intact.
- **Offline queue / "Save on this device"** — When you submit offline, the report is saved on your device and syncs automatically when you reconnect, re-running the same checks server-side. Each queued item has a unique id so a retry can't create a duplicate.
- **Facility scoping** — The automatic rule that you only ever see and act on your own facility's data. Not configurable; built in.
- **Retention / purge** — Per-module rules (Admin → Data Retention) that permanently delete records older than a set number of days, optionally each night. Destructive; gated behind a warning. (Daily Reports auto-purges after 14 days.)
- **Seed defaults** — A one-click button on many setup screens that creates a standard starter configuration (sections, reading types, roles, etc.), safe to skip and edit afterward.

Open questions / ⚠ VERIFY items

This appendix collects every ⚠ VERIFY flag from the discovery manifest and all eight module chapters. Each should be validated against a real configured facility before publishing hard claims.

Roles, tiers & permissions (Manifest; Admin Control Center; every module's "Who can use it")

1. **No org_admin tier in the live app.** The five-tier vocabulary (super_admin / org_admin / facility_manager / supervisor / staff) does not match the live model (super_admin / admin / manager / staff + custom roles). There is **no org_admin equivalent** — treat any org_admin need as super_admin. (Manifest §Role Tiers, §Gaps 1; flagged in every module's role table.)
2. **Access is permission-driven, not title-driven.** "Supervisor" / manager abilities are whatever permission rows an admin enables, not a fixed tier. Two people with the same title can differ. Validate every "tier X can do Y" claim against a configured facility. (Manifest §2, §Gaps 1; Admin Control Center §2; Refrigeration §2; Incidents §2; Ice Operations §2; Air Quality §2; Ice Depth §2; Scheduling §2.)
3. **Per-module delete rights.** "Delete = super_admin only" holds for facilities and the general pattern, but per-module delete (an admin deleting an area/template/session) should be confirmed module-by-module. (Manifest §2, §Gaps 6.)
4. **Only super_admin can grant Admin-module Admin access and edit facilities.** Confirmed in the Admin chapter as the intended guard; carry it forward as the rule for who can hand out Admin Center access. (Admin Control Center §2, §5 Permissions, §9.)
5. **Roles screen navigation.** /admin/roles is a real screen but has **no sidebar entry** — it is reached only from the Admin Dashboard's setup checklist ("Canonical roles seeded" / "Role permission defaults") or by typing the address. Confirm this is the intended navigation. (Manifest §1, §Gaps 5; Admin Control Center §3, §9.)

Daily Reports

6. **Area cap is 30, not 20.** Sales material mentions "up to 20 tabs"; the database enforces **30 active areas**. Confirm the exact number and whether "tabs" means areas or templates-within-an-area. (Manifest §4, §Gaps 2; Daily Reports §4.)
7. **No end-of-day lock.** The package described an end-of-day lock/single rolling report. The code uses an **append-only** model (one immutable record per submit, facility-date tagged, **auto-deleted after 14 days**) with **no per-day lock**. Describe it that way. (Daily Reports §8.)

Refrigeration

8. **Supervisor / org_admin rows** in the access table are mapped to the live model and should be confirmed per facility (covered by items 1–2). (Refrigeration §2.)

Incident Reporting

9. **24-hour edit window — confirmed.** The reporter can edit their own incident for 24 hours, then it locks read-only; confirmed in code (edit_window_ends_at = now() + 24h). (Incidents §8.)
10. **Incident Type not collected from staff.** Type appears as a column/filter and on the admin detail, but the current **staff** submission form does **not** present an incident-type picker. Confirm whether your facility expects staff to pick a type before training on it as a staff field. (Incidents §4, §7.)

Ice Operations

11. **Temperature unit setting removed.** Confirmed unused on live forms (no staff field collects a temperature), so the Fahrenheit/Celsius selector was removed from Ice Operations Settings; legacy History readings still display in the previously saved unit. (Ice Operations §4.)
12. **Supervisor config rights** default — confirm whether a facility grants supervisors Ice Operations config by default (covered by items 1–2). (Ice Operations §2.)

Air Quality

13. **Thresholds are tighten-only.** Facility overrides may only **lower** (tighten) a regulatory ceiling, never raise it; the app rejects looser values and clamps internally. Confirmed in code, but call it out clearly in training. (Air Quality §4, §9.)
14. **Supervisor / org_admin rows** mapped to the live model; confirm per facility (items 1-2). (Air Quality §2.)

Ice Depth

15. **Unit is inches / mm — not cm.** The training brief said "in/cm," but the app's unit options are **inches** and **mm**. (Ice Depth §4.)
16. **Offline submit when no service worker is active.** If the device can't queue offline (no service worker), the app blocks the submit and tells the user to reconnect/reload, keeping typed readings on screen. Confirm this fallback behavior. (Ice Depth §8.)
17. **Super_admin-only session delete; supervisor rights** per facility (items 1-2). (Ice Depth §2.)

Offline / PWA (Manifest §7)

18. **Offline tech is service worker + IndexedDB — not Dexie.** The package mentioned "Dexie"; no Dexie library is used. Describe it generically ("saved on your device, syncs when you reconnect") and never name Dexie. (Manifest §7, §Gaps 3.)
19. **Scheduling offline scope.** Only **availability** and **time-off** queue offline. **Swap requests, swap acceptance, open-shift claims, and the entire admin grid are online-only** by design. Confirmed in code, but the staff-side discovery once suggested swap requests might queue — treat swap/claim as online-only. (Manifest §7, §Gaps 4; Scheduling §8.)

Other

20. **Related modules not deeply mapped.** Accidents, Communications, Facility Paperwork, Departments, Spaces, Lists, Exports, and Audit Log were inventoried at a screen level only; deeper field-level detail may be needed in their own chapters. (Manifest §Gaps 7.)
21. **Dashboard & Account pages** were noted as routes but their on-screen content was not fully inventoried; capture if they need training coverage. (Manifest §Gaps 8.)

This manual links to, and does not duplicate, the module chapters in `modules/`. Where a chapter and this manual disagree, the chapter (closer to the code) wins — and the discrepancy belongs in the [△](#) VERIFY list above.

Admin Control Center

1. What this module is for

The Admin Control Center is the configuration hub for one ice rink in RinkReports. It is where the person who runs a facility sets up everything the staff app depends on: the facility's basic information, which report modules are turned on, the people who work there, the roles and permissions that decide who can do what, the shared lists that feed dropdowns across the app, and the system-wide settings for exports, data retention, and the audit trail.

Think of it as the back office. Staff members spend their time in the report flows (Daily Reports, Refrigeration, Incidents, and so on). Admins spend their time here, deciding how those flows behave. Nothing a staff member sees on their device — the modules in their menu, the areas they can submit to, the dropdown options they pick from — is hardcoded. It is all driven by what an admin configures in this Control Center.

You only see data for your own facility — this is automatic. Everything you create, edit, or view here applies to your rink and no one else's. (The one exception is the Super Admin console, described in section 5, which is reserved for the platform owner who manages every facility at once.)

2. Who can use it

RinkReports does not use a rigid ladder of five fixed job titles. Access is **permission-driven**: each person's abilities are resolved per module and per action (View / Submit / Edit / Admin), and an admin can customize any individual's access. The table below maps the documentation's five-tier vocabulary onto how the live app actually behaves. Treat the "What they can do" column as the default for that tier — an admin can widen or narrow it for any single person.

Role tier	What they can see	What they can do
super_admin	Everything, across every facility. The only tier that sees the Super Admin console and the cross-facility facility list.	Create, edit, and activate/deactivate facilities; create and edit any facility's settings; promote or demote other super admins; reset any user's password; hard-delete employees; assign a person to a different facility; everything an org_admin/facility_manager can do.
org_admin	⚠ VERIFY — the live app has no separate org/multi-facility admin tier. There is no role between super_admin and the single-facility admin. For now, treat any "org_admin" need as super_admin.	(No distinct equivalent exists in the code.)

Role tier	What they can see	What they can do
facility_manager (live role: admin)	The full Admin Control Center for their own facility. Their own facility's settings appear read-only — they cannot edit facility info.	Turn modules on/off; add/edit/deactivate/reactivate employees; assign roles at or below their own rank; set per-person permission overrides; create and edit custom roles; manage departments, spaces, lists, exports, retention, and read the audit log. Cannot create/edit facilities, hard-delete employees, or grant Admin Center access.
supervisor (live role: manager, or a custom role)	Depends entirely on the permissions an admin has granted. By default a manager has no Admin Center access unless given the admin action on the admin module.	⚠ VERIFY — supervisor abilities are not a fixed tier; they are whatever permission rows an admin enables. Two people with the title "manager" can have different access.
staff (live role: staff, or a custom role)	No access to the Admin Control Center. Attempting to open /admin shows a "Forbidden" message.	Nothing here. Staff work only in the report flows and self-service scheduling.

How the gate actually works (plain version): the Admin Control Center opens for you if you are a super admin, **or** you have the Admin action enabled on the Admin module for your facility, **or** you are an active employee whose role is Administrator or Super Admin. A deactivated account is denied everywhere, immediately.

3. How to get there

From the staff app, admins see an extra **Admin Center** link in the left sidebar (staff do not). Selecting it opens the Admin Control Center, which has its own grouped sidebar.

The Admin sidebar is organized into three groups. The Setup and System groups are the Admin Control Center itself; the Module Admin group links out to each report module's own configuration (covered in those modules' own chapters).

- **Dashboard** — /admin (setup checklist + activity overview)
- **Setup**
 - Facility
 - Modules
 - People
 - Departments
 - Facility Spaces
 - Permissions
- **Module Admin** (cross-linked, documented in each module's chapter)
 - Daily Reports Admin, Ice Depth Admin, Ice Operations Admin, Incident Reports Admin, Accident Reports Admin, Refrigeration Admin, Air Quality Admin, Scheduling Admin, Communications Admin, Facility Paperwork
- **System**
 - Lists
 - PDF/Export Settings

- Data Retention
- Audit Log
- Super Admin (only functions for super_admin; others are bounced to a Forbidden screen)

Roles is a real screen but it is **not in the sidebar**. You reach it from the Admin Dashboard's setup checklist — the "Canonical roles seeded" and "Role permission defaults" checklist items both have a button that opens it. (You can also type the address directly.) The Roles screen links back out to Permissions and People, but People and Permissions do not link to Roles. [△](#) VERIFY this is the intended navigation; the screen has no sidebar entry.

4. Setup & configuration (admins)

This section is the quick map of every admin-configurable option in the Control Center, where it lives, the valid values, and the effect. Screen-by-screen detail follows in section 5.

Setting	Where	Valid values	Effect
Module on/off	Modules	On / Off per module	Shows or hides the module in staff navigation only . Does not change anyone's permissions.
Facility info (name, slug, timezone, address, contact, active)	Facility	Text/dropdown; slug must be lowercase letters/numbers/hyphens and unique	Identifies the facility. Editable by super_admin only ; a regular admin sees it read-only.
Employee record	People	See field reference (section 7)	Creates the person, their role, job areas, and optional login.
Role assignment	People (employee form)	Any role at or below your own rank	Seeds that role's default permissions onto the person; manual overrides are kept.
Per-person permission override	Permissions, or the employee detail "Module Access" tab	View / Submit / Edit / Admin per module	Overrides the role default for that one person. Clearing falls back to the role default.
Custom roles & role defaults	Roles	Display name, key, hierarchy level, per-module action matrix	Defines a reusable permission template; changing it re-applies to that role's staff (overrides preserved).
Departments	Departments	Name, slug, color, sort order, active	Feeds the schedule department filter, shift assignment, and communication routing.
Facility Spaces	Facility Spaces	Name, slug, sort order, active	Feeds the location pickers in Incident, Accident, and Air Quality reports.

Setting	Where	Valid values	Effect
Lists (custom dropdowns)	Lists	Per-domain options (currently Timezones)	Customizes dropdown options used elsewhere (today: the facility timezone picker).
Export branding/layout/columns	PDF/Export Settings	Logo URL, header/footer text, paper size, date format, CSV delimiter, per-module column checkboxes	Controls how exported PDFs and CSVs look and which columns appear.
Data retention	Data Retention	Keep-days (0 = forever, or ≥ 30) + nightly auto-purge on/off per module	Permanently deletes records older than the threshold. Destructive.

5. Screen-by-screen walkthrough

Dashboard (/admin)

The landing screen. It shows a **Setup checklist** card (Facility info, Canonical roles seeded, Role permission defaults, First admin linked, Staff added, Invites sent) — each item has a status icon and a button that jumps you to the right screen. A badge reads "Ready" when all are done or "N left" otherwise. Below the checklist are overview cards (Active employees, and for super admins, Total facilities), a **Recent report activity** grid showing submission counts per module for the last 7 and 30 days, and an **Offline sync queue** card showing pending / synced / failed counts for offline submissions. A super admin sees a **facility switcher** dropdown in the header to pick which facility the dashboard reflects.

Facility (/admin/facility)

This screen looks different depending on who you are.

Regular admin (facility_manager): you get a **read-only** view of your facility. A "Facility settings" card states "Only super admins can edit facility settings. Contact your administrator to make changes." It lists Name, Slug, Timezone, Status, Address, City, State, Zip code, Phone, Email, Created, and Last updated, plus an "At a glance" card counting Employees / Departments / Roles. You cannot change anything here.

Super admin: you get a full "Facilities" management screen — a card per facility (name, Inactive pill if applicable, slug · timezone · created date) with **Manage employees** and **Edit** buttons, plus a **New Facility** button at the top. Editing opens the facility form; saving requires super admin.

Facility form fields: Name (required, 2+ characters), Slug (required, lowercase letters/numbers/hyphens, must be unique — typing the name auto-suggests a slug until you edit it), Timezone (dropdown, options come from the Lists screen), Address, City, State (2-letter), Zip code, Phone number, Email, and (edit only) an **Active** checkbox ("Inactive facilities are hidden from most views"). The submit button reads **Create facility / Save changes**. There is no separate deactivate button — activation is the Active checkbox in the edit form.

Modules (/admin/modules)

A single **on/off switch per module**, under a "Staff modules" card. The description spells it out: "This is a visibility switch only — it does not change per-employee permissions, and disabled modules remain protected by their own access rules." Turning a module off hides it from the staff navigation for

this facility; it does not revoke anyone's underlying permissions.

The ten toggles are: Daily Reports, Ice Depth, Ice Operations, Refrigeration, Air Quality, Incidents, Accidents, Scheduling, Communications, Facility Paperwork. (Dashboard and the Admin Center itself are never toggleable.) Flipping a switch is immediate — no Save button — and shows a confirmation toast such as "Daily Reports enabled for this facility." If a save fails, the switch rolls back and shows an error.

People / Employees (/admin/employees)

Titled "Employee / User Setup." A search box (name, email, role, code) and a status filter (**Active** / **Inactive** / **All**) sit above the employee table (columns: Name, Role, Email, Phone, Status, Actions). Two top buttons: **Bulk add** and **Add employee**.

Per-row actions: **Edit** (opens the form), **Invite** (only when the person is active, has an email, and has no login yet — sends a magic-link sign-in invite), **Preview** (view the app as that person), **Deactivate** / **Reactivate**, and **Delete** (shown only to super admins).

Add / Edit employee form (slides in from the right) — see the field reference in section 7 for the full list. Key behaviors:

- **Role** is required; the form shows a live **Role permissions** preview of what that role grants. A regular admin can only assign a role at or below their own rank.
- **Create a login & apply these permissions** (create mode, default on) sends an email invite and seeds the role's permissions. Requires an email. Uncheck it for schedule-only staff (e.g. minors) who do not log in.
- **Job areas:** pick up to 4; one can be marked **Primary**. You can create a new job area inline.
- **Employee is a minor** checkbox: when checked, the emergency contact name and phone become optional; otherwise both are required.

Deactivate opens a confirmation: "{name} will lose access immediately and stop appearing in shift assignments and routing rules. You can reactivate them later from this same list." Deactivation is a soft delete — the record and permissions are kept. **Delete** (super admin only) confirms "Delete {name}? This cannot be undone." and is a permanent hard removal.

Employee detail page (open a name) has tabs: **Profile** (read-only; edit via the list), **Certifications** (add/edit/delete certs with issuer and expiry, status badges warn when expiring or expired), **Module Access** (per-module effective level, its source, a "set override" dropdown, and a "Clear override" button), **Communication Groups** (add/remove the person from groups), and **Activity** (a read-only audit list for that person).

Bulk add (/admin/employees/bulk): a grid where you type or paste rows (First name, Last name, Email, Hire date, Role, Job areas), up to **100 at a time**, with live validation. Buttons: Add row, Paste from spreadsheet, Download template, Clear all. A **"Send login invites & apply role permissions"** switch (default on) controls whether each person gets an invite. Submitting reports each row as Added, Partial, or Failed.

Departments (/admin/departments)

Departments power the schedule's department filter, shift assignment, and communication routing. The table shows Name (with a color dot), Slug, Order (with up/down reorder arrows), Status, and Actions. **Add department** opens a form: Name (required), Slug (optional, auto-generated, lowercase/digits/hyphens), Color (default green #4DFF00), Sort order, and — in edit mode — Active. Rows can be **Edited** and **Deactivated** / **Reactivated**. There is **no delete** for departments;

deactivating hides a department from new assignments while existing shifts keep theirs.

Facility Spaces (/admin/spaces)

The shared list of physical locations that feed the location pickers in Incident, Accident, and Air Quality reports. **Seed defaults** creates a standard set (Main Rink, Lobby, Locker Room, Pro Shop, Parking Lot, and more). You can also **Add space** manually or **bulk import** from pasted CSV. Each space has Name, Slug, Sort order, and (edit only) Active. Rows offer **Edit**, **Deactivate** / **Reactivate**, and **Delete**. Delete confirms "Delete space \"{name}\"? This cannot be undone." — but a space that is referenced by existing reports cannot be deleted; the app tells you to deactivate it instead.

Permissions (/admin/permissions)

Titled "Module Access Control." It lists every active user; pick one to open their **permission matrix**. The matrix is a grid of **modules (rows) × four actions (columns): View, Submit, Edit, Admin**. The ten module rows are Daily Reports, Ice Depth, Ice Operations, Incident Reporting, Accident Reporting, Refrigeration, Air Quality, Employee Scheduling, Communications, and Admin.

- **What each action means:** View = can see the module; Submit = can create and submit entries; Edit = can modify their own or facility submissions; Admin = can change, approve, or configure that module's settings.
- **Each checkbox is a live write** — clicking it saves immediately and sets a per-person override. There is no Save button. If a save is rejected, the box rolls back.
- **Presets** (top-right buttons): **Full Access**, **Submitter Only** (View + Submit), **Viewer Only** (View), **No Access** (clears everything). A preset writes all module/action cells at once.
- **Admin guard:** only a super admin can grant the Admin module's Admin action (Admin Center access). For a regular admin, that one cell stays off even under Full Access, and they can only edit permissions within their own facility.
- A **Bulk CSV import** panel lets you paste rows with the header `user_id,facility_id,module,action,enabled`.
- A user with no facility assigned cannot have permissions set until a facility is assigned (the screen says so and links to People). A super admin user bypasses module permissions automatically; a notice explains the toggles only matter if their super-admin flag is removed.

You can also set a person's per-module override from their **employee detail → Module Access** tab.

Roles (/admin/roles)

Define reusable permission templates for your facility. **Seed default roles** (from the People area) creates the four canonical roles: Super Admin, Administrator, Manager, Staff. The Roles screen has two parts:

- **Role Manager:** rename, add, deactivate, reactivate, and reorder roles. **New role** asks for Display name, Key (lowercase letters/digits/underscores, e.g. `rink_lead`), Hierarchy level (0 = highest), and an optional Description. System roles cannot be deactivated. A **Copy permission defaults** control bulk-copies one role's module defaults onto another. Deactivating a role that still has assigned employees prompts a confirmation explaining those employees keep only their explicit overrides.
- **Role permission defaults matrix:** for each role, toggle the default **V / S / E / A** (View / Submit / Edit / Admin) per module. Changing a default re-applies to that role's current staff, but any per-person overrides set in Module Access Control are preserved.

A regular admin cannot create or modify a role that outranks their own hierarchy level.

Lists (/admin/lists)

Per-facility custom dropdown option lists. Today there is exactly one list domain: **Timezones**, which feeds the Facility settings timezone picker. **Seed defaults** loads the canonical set (safe to run repeatedly; never overwrites your edits). You can **Add**, **Edit**, **Deactivate** / **Reactivate**, and **Delete** options. Each option has a Key (for timezones, a valid IANA identifier such as *America/New_York*), a Display name, a Color, a Sort order, and (edit only) Active. An option that is in use cannot be deleted — deactivate it instead.

PDF / Export Settings (/admin/exports)

Two parts.

Run an export: pick a **Module**, a **Format** (CSV or PDF), and a **From** / **To** date range, then **Download**. The columns, delimiter, date format, and branding come from the settings below.

Export settings (one **Save export settings** button):

- **Branding** — Logo URL (recommended 300×80 px), Header text, Footer text (shown on exported PDFs).
- **Layout & Format** — Paper size (Letter / A4), Date format (MM/DD/YYYY, DD/MM/YYYY, or YYYY-MM-DD), CSV delimiter (comma / tab / semicolon), and an "Include on every export" group (Facility name, Export date, Submitted-by name).
- **Column Visibility per Module** — checkboxes per module choosing which columns appear in its export. Unchecked columns are hidden from exports but stay in the database.

Data Retention (/admin/retention)

Controls how long submitted data is kept per module, measured from each record's submission date. Summary cards show modules configured, auto-purge enabled count, and when a purge last ran. Each module row shows its current keep period and an **Edit** button.

- **Keep for (days):** a number, with preset buttons (30 days, 90 days, 180 days, 1 year, 2 years, 3 years, 5 years, Forever). Each module shows a recommended minimum (e.g. 365 for Incident/Accident/Audit Log). The value must be 0 (forever) or at least 30.
- **Enable nightly auto-purge:** when on, records older than the threshold are permanently deleted each night.
- **Manual Purge:** a "Run purge now" button reveals an inline confirmation ("Are you sure? This will permanently delete records.") before deleting immediately.

A red **Auto-purge warning** card stresses that deletion is permanent and irreversible — review your legal and regulatory retention obligations, especially for incident, accident, and workers' compensation records, before enabling it.

Audit Log (/admin/audit-log)

A **read-only**, filterable trail of create / update / delete / login / logout events. Filters: **Action**, **Entity type**, **Actor**, **From** / **To** dates, and a search by entity ID. The table columns are When, Actor, Action (color-coded badge), Entity type, and Entity ID. Selecting an entry's timestamp opens a detail panel with the full record, IP address, user agent, and collapsible **Before** / **After** snapshots. It shows up to 300 entries (newest first, last 30 days by default); narrow with filters. There is nothing to edit here.

Super Admin (/admin/super-admin)

Super_admin only — anyone else is sent to a Forbidden screen. Platform-wide management: stat cards (Total facilities, Active facilities, Total users, Super admins); a **Facilities** panel to **Activate** /

Deactivate facilities (paginated); a **Super admins & users** panel to **Reset password** (creates a recovery link), and **Promote / Revoke** super-admin status for other users (you cannot revoke your own); and an **Email invite service** health card with a **Run health check** button that verifies the deployment can send sign-in invites.

6. Step-by-step: common tasks

Turn a module on or off

1. Open **Modules** from the Admin sidebar.
2. Find the module row.
3. Click its switch. It flips immediately and shows a toast confirming the change. (Off hides it from staff navigation; permissions are unaffected.)

Add an employee and assign a role

1. Open **People**, then click **Add employee**.
2. Enter First name and Last name.
3. Pick a **Role** (you can only pick roles at or below your own rank). Review the permission preview.
4. Decide on **Create a login & apply these permissions** — leave it on (and enter an Email) for someone who logs in; uncheck it for schedule-only staff.
5. Optionally add Job areas (up to 4, mark one Primary), Employee code, Hire date, Max weekly hours, Phone.
6. If the person is a minor, check **Employee is a minor** (emergency contact becomes optional); otherwise enter the emergency contact name and phone.
7. Click **Create employee**. If a login was requested, an invite email goes out and the role's permissions are seeded.

Override one person's permissions

1. Open **Permissions** and pick the user (or open the person in **People** and go to the **Module Access** tab).
2. In the matrix, tick or untick the **View / Submit / Edit / Admin** box for the module you want to change. It saves instantly.
3. To reset to the role default, untick the overrides (or use the **No Access** preset / **Clear override**). Note: only a super admin can grant the Admin module's Admin action.

Create a custom role

1. Open the **Roles** screen (from the Admin Dashboard setup checklist).
2. In Role Manager, click **New role**.
3. Enter a Display name, a Key (lowercase letters/digits/underscores), a Hierarchy level (0 = highest; you cannot create a role above your own rank), and an optional Description. Click **Create**.
4. In the role permission defaults matrix, toggle the **V / S / E / A** boxes per module to set what the role grants. Changes apply to staff with that role (their manual overrides are kept).

Configure data retention

1. Open **Data Retention** and click **Edit** on a module.
2. Set **Keep for (days)** (use a preset or type a number; 0 = forever, or at least 30).
3. Optionally turn on **Enable nightly auto-purge**, then **Save**.

- To delete old records now, use **Run purge now** and confirm. This is permanent — heed the warning, especially for incident, accident, and workers' comp data.

Export records

- Open **PDF/Export Settings**.
- In "Run an export," pick the **Module**, **Format** (CSV or PDF), and the **From / To** dates.
- Click **Download**. The file uses the branding, layout, and column settings configured below.

7. Field reference

Employee form (People → Add/Edit employee):

Field	What it means	Valid values/units	Required?	Notes
First name	Given name	Text, up to 100 chars	Yes	
Last name	Family name	Text, up to 100 chars	Yes	
Role	The person's role	Any facility role at/below your rank	Yes	Seeds that role's permission defaults; shows a live preview.
Create a login & apply these permissions	Whether to provision a sign-in	Checkbox	No (create only, default on)	Requires an email; sends an invite. Uncheck for schedule-only staff.
Job areas	Cross-trained work areas (scheduling)	Up to 4 selections	No	One can be marked Primary; can create new areas inline.
Employee code	Internal staff code	Text	No	
Hire date	Start date	Date	No	
Max weekly hours	Per-person scheduling cap	Number 1-168	No	Blank uses the facility default; the shift grid warns when exceeded.
Email	Contact / login address	Email	No*	*Required if a login is being created.
Phone	Contact number	Phone	No	
Employee is a minor	Flags an under-age worker	Checkbox	No	When on, emergency contact becomes optional.
Emergency contact name	Who to call	Text	Required unless minor	
Emergency contact phone	Emergency number	Phone	Required unless minor	

Facility form (super_admin only):

Field	What it means	Valid values/units	Required?	Notes
Name	Facility name	Text, 2+ chars	Yes	

Field	What it means	Valid values/units	Required?	Notes
Slug	URL identifier	Lowercase letters, numbers, hyphens; unique	Yes	Auto-suggested from the name.
Timezone	Local time zone	IANA zone from the Lists picker	No	Defaults if blank/invalid.
Address / City / State / Zip code	Mailing address	Text (State is 2-letter)	No	
Phone / Email	Facility contact	Phone / email	No	Email must be valid if provided.
Active	Whether the facility is live	Checkbox (edit mode)	No	Inactive facilities are hidden from most views.

Retention row (Data Retention):

Field	What it means	Valid values/units	Required?	Notes
Keep for (days)	Retention period	0 (forever) or ≥ 30 days	Yes	Per-module recommended minimums shown (e.g. 365 for incident /accident/audit).
Enable nightly auto-purge	Automatic deletion	Checkbox	No	Permanently deletes records older than the threshold each night.

8. Locking, saving & offline

Admin configuration is online-only. The Admin Control Center is a desk task and is not part of the offline submission queue. You must be connected to use it; there is no "save on this device" here.

Saving behaves in two patterns. Some screens save **instantly on click** — module switches, the permission matrix checkboxes and presets, and the role default matrix all write immediately with no Save button (a failed write rolls the control back and shows an error). Forms (Facility, Employee, Department, Space, List option, Retention, Export settings) collect your input and save when you press the submit button (**Create...** / **Save changes** / **Save export settings**).

What "locked" means here. Admins do not edit staff submissions in place. The Audit Log is permanently read-only — it is an immutable record. Facility records and employees use soft deactivation (kept for history) rather than deletion wherever possible; true deletes are reserved for super admins and, for facilities and referenced spaces/list options, are blocked or steered to "deactivate instead." Data Retention purges are the one deliberately destructive action, gated behind a warning and an inline confirmation.

9. Troubleshooting & FAQ

Why can't I edit my facility's name or address? Editing facility settings is reserved for super admins. As a facility admin you get a read-only view. Ask your super admin to make the change.

Why don't I see the Super Admin screen working? It only functions for super admins. If you open it without that flag you are sent to a Forbidden screen. This is by design.

Why can't I grant someone Admin Center access? Only a super admin can enable the Admin module's Admin action. For a regular admin, that one cell stays off even when you apply Full Access.

Why can't I assign the Administrator role to someone? You can only assign roles at or below your own rank. A manager can't create or promote into an admin-tier role; a super admin is unrestricted.

Why can't I find the Roles screen in the sidebar? It has no sidebar entry. Open it from the Admin Dashboard's setup checklist ("Canonical roles seeded" or "Role permission defaults"). [△ VERIFY](#) — this is the only built-in path besides the direct address.

Why can't I delete this employee / space / list option? Hard-deleting employees is super-admin only — others should use Deactivate. A facility space or list option that is referenced by existing records can't be deleted at all; deactivate it instead so history stays intact.

I turned a module off but the person can still reach it by URL — why? The Modules switch only controls **navigation visibility**. The module is still protected by its own access rules; to actually remove someone's access, change their permissions in the Permissions screen.

I deactivated someone but they say they can still see things. Deactivation takes effect immediately for new sign-ins and blocks access, but an already-open session may need a refresh. A deactivated account cannot sign in, submit, or view data until reactivated.

Why is the permission matrix empty / greyed out for a user? A user with no facility assigned has nowhere to store permissions yet — assign them a facility from People first. A super-admin user bypasses module permissions entirely, so the toggles are informational unless their super-admin flag is removed.

Why did my export download the wrong columns? Column visibility is per module in PDF/Export Settings. Unchecked columns are intentionally hidden from exports (the data still exists in the database).

Source (footnote)

Key files verified for this chapter (paths for maintainers only; not part of the user-facing content):

- Access gating: `src/lib/auth/require-admin.ts`, `src/lib/auth/get-is-admin.ts`, `src/proxy.ts`, `src/lib/supabase/session.ts`
- Permission model: `src/lib/permissions/actions.ts` (modules × View/Submit/Edit/Admin, presets), `src/lib/permissions/levels.ts`, `src/lib/permissions/index.ts`
- Navigation & shell: `src/components/admin/nav-config.ts`, `src/components/admin/sidebar-nav.tsx`, `src/app/admin/layout.tsx`, `src/app/admin/page.tsx`
- Modules: `src/app/admin/modules/page.tsx`, `_components/module-toggles.tsx`, `actions.ts`, `src/lib/modules/module-keys.ts`
- Facility: `src/app/admin/facility/page.tsx`, `_components/{facility-form,read-only-view,facilities-table,new-facility-button,edit-facility-section}.tsx`, `actions.ts`, `types.ts`
- People/Employees: `src/app/admin/employees/{page.tsx,actions.ts,types.ts}`, `_components/{employee-form,employees-client,role-permission-preview,seed-roles-button}.tsx`, `[id]/_components/employee-detail.tsx`, `[id]/actions.ts`, `bulk/_components/bulk-add-client.tsx`, `bulk/{actions.ts,_lib/validation.ts}`

- Permissions: `src/app/admin/permissions/{page.tsx,user-permission-actions.ts,validators.ts}, [userId]/page.tsx, _components/permission-matrix.tsx`
- Roles: `src/app/admin/roles/{page.tsx,actions.ts}, _components/{role-manager,roles-matrix}.tsx`
- Departments / Spaces: `src/app/admin/departments/**, src/app/admin/spaces/**`
- Lists: `src/app/admin/lists/**, _lib/facility-dropdowns.ts`
- Exports: `src/app/admin/exports/{page.tsx,actions.ts,types.ts}, _components/{export-settings-form,run-export-panel}.tsx`
- Retention: `src/app/admin/retention/{page.tsx,actions.ts,types.ts}, _components/retention-row.tsx`
- Audit Log: `src/app/admin/audit-log/**`
- Super Admin: `src/app/admin/super-admin/{page.tsx,actions.ts,types.ts}, _components/{facilities-panel,super-admin-users-panel,invite-service-health-card}.tsx`

Daily Reports

1. What this module is for

The Daily Reports module is RinkReports' digital replacement for the paper "shift checklist" clipboard. Your facility is divided into **work areas** — front-of-house and operational zones such as Front Desk, Pro Shop, Custodial, Skate Sharpening, Concessions, Event Set Up, Learn to Skate, Public Skate, Locker Rooms, and Building Services. (Those ten are the example areas Tenuity Ice Pavilion uses; your facility's list is whatever your administrator has set up.)

For each area, an administrator builds one or more **checklists** (called "shifts" or "templates"). When you work a shift, you open Daily Reports, pick your work area and shift, tick off each task as you complete it, optionally leave a note, and submit. Managers can then review every area's reports for the day in the Admin Center.

Each work area is logged **independently** — submitting the Concessions checklist has nothing to do with whether the Locker Rooms checklist has been done. Every submission is a separate, permanent record (see Section 8 for exactly how saving and "locking" work — there is no single end-of-day report that fills up across areas).

You only see data for your own facility — this is automatic. There is no way to view or submit reports for another rink.

2. Who can use it

Daily Reports access is **permission-driven**, not strictly tier-driven: an administrator grants each person the ability to **view**, **submit**, **edit**, or **administer** the module, and — uniquely for this module — grants submit access **area by area** (the "Area Access" matrix). Two people with the same job title can therefore have different access if an admin has customized it.

The five-tier vocabulary used across this documentation maps onto the live app's roles as below. Because the real model is permission-based, treat the "Can do" column as the typical default, not a hard rule.

Role tier	Can see	Can do
super_admin	Everything, platform-wide	Full configuration plus delete; always has area access.
org_admin	△ VERIFY — no separate org-admin role exists in the live app; treat as super_admin.	△ VERIFY — same as super_admin in practice.
facility_manager (the live admin role)	All areas, all submissions, and the full admin console for their own facility	Create/edit areas, templates, checklist items; grant area access; review, edit, annotate, and delete submissions. Always has submit access regardless of the Area Access matrix.

Role tier	Can see	Can do
supervisor (the live manager role, or a custom role)	What their permissions allow	⚠ VERIFY — has no special daily-reports powers by default; sees and submits only where granted, exactly like staff unless an admin has enabled the module's admin action for them.
staff	The areas they've been granted submit access to, plus recent History for the facility	Submit daily reports for their granted areas; view History (read-only). No access to the admin console.

Anyone whose account is **deactivated** is denied everywhere. A signed-in user with **no area access granted** sees a "No areas assigned" message and can submit nothing until an admin grants access.

3. How to get there

Staff: In the left sidebar (or the bottom **Reports / Menu** tab on a phone), tap **Daily Reports**. This opens your reporting console at the Daily Reports page. From there, **View history** (top-right) opens the read-only History page.

The Daily Reports menu item only appears if your administrator has the module turned on for the facility and you have view/submit permission.

Administrators: Open the **Admin Center** (the "Admin Center" link appears in the staff sidebar for admins), then choose **Daily Reports Admin** under the "Module Admin" group.

4. Setup & configuration (admins)

The Daily Reports admin area is organized into five tabs: **Areas**, **Templates**, **Checklist Items**, **Area Access**, and **Submissions**. The first four are setup; Submissions is review (covered in Sections 5 and 6). A reminder banner notes that reports auto-delete after 14 days.

Daily Reports configuration is a facility_manager-level (admin) task. It is also cross-referenced from the **Admin Control Center**, where the **Modules** page carries the master ON/OFF switch that shows or hides Daily Reports in staff navigation.

Areas (the "tabs" staff see)

An **area** is one of the work zones staff pick from. On the **Areas** tab you can:

- **Add area** — give it a **Name** (required), an optional **Slug** (a URL-safe short name; auto-generated from the name if left blank — lowercase letters, digits, and hyphens), a **Color** swatch, and a **Sort order**.
- **Edit** an area, including toggling it **Active**.
- **Reorder** with the up/down arrows (this controls the order staff see).
- **Deactivate / Reactivate** — deactivating hides the area from staff without deleting its history.
- **Delete** — only succeeds if the area has no submissions; otherwise you're told to deactivate instead.
- **Bulk upload areas** — import many at once from a file.

The area cap. A facility may have at most **30 active areas** at one time. This limit is enforced by the database, not just the screen. The Areas tab shows a live "X / 30 active" badge; once you hit 30, the

Add area button reads "Cap reached" and reactivating another area is blocked until you deactivate one. (Note: some sales material mentions "up to 20 tabs" — the actual enforced cap is **30 active areas**.)

Templates (the "shifts" / checklists)

A **template** is a named checklist that belongs to a single area — for example "Opening checklist" or "Hourly inspection." On the **Templates** tab, first pick an area from the **Area** dropdown, then:

- **Add template** — **Name** (required), optional **Description**, and **Sort order**.
- **Edit**, **Deactivate / Reactivate**, and **Delete** (delete is blocked if any submission references it — deactivate instead).

An area can have several templates; the staff member chooses which one applies to the shift they're logging.

Checklist Items

A **checklist item** is a single tickable task within a template. On the **Checklist Items** tab, pick the **Area** and then the **Template**, then:

- **Add item** — **Label** (required; the text staff see, e.g. "Inspect ice surface for cracks") and an optional **Description** (a sub-line of guidance).
- **Edit**, reorder (up/down), **Deactivate / Reactivate**, and **Delete**.
- **Bulk upload** items into a template from a file.

A template with no items is allowed — staff can still submit it (it just records the area/shift and any note).

Area Access

The **Area Access** tab is a grid of **staff (rows) × areas (columns)** with a checkbox in each cell. Checking a box lets that staff member submit daily reports **in that area**. Notes:

- Changes save immediately as you toggle each box (no separate Save button).
- **Admins and super admins always have access**, regardless of the boxes.
- You can **Download CSV** (a template of the current grid) and **Import CSV** to set access in bulk; the import reports counts of granted / revoked / skipped rows and any issues.

5. Screen-by-screen walkthrough

Staff console (the Daily Reports page)

When you open Daily Reports you see a single scrolling page:

- **Header** — a breadcrumb (Reports > Daily Reports), the page title, and a **View history** button.
- **Meta-chip strip** — your name, your facility, today's date, and a **live clock** that updates every second.
- **Shift setup card** — two dropdowns:
 - **Work area** — lists only the areas you're allowed to submit to, each shown with its color swatch.

- **Shift** — lists the templates for the area you picked, with each one's item count beside it (e.g. "Opening checklist · 8 items"). If an area has exactly one shift, it is **auto-selected** for you; otherwise you'll see "Choose shift type..." until you pick.
- **Checklist card** — once a shift is chosen, its items appear as a list of large checkboxes (each with its optional description line). At the top is a "**X / Y complete**" counter and a **progress bar** that fills (and tints to the area's color) as you tick items. A template with no items shows "No checklist items on this template. You can still submit."
- **Note card** — an optional free-text box ("Anything to flag for managers?").
- **Sticky submit bar** — pinned to the bottom of the screen. It shows your progress ("X/Y complete") and a **Submit** button. The button stays disabled until a shift is selected. When you're offline it adds "· offline — will sync when reconnected."

After you submit, you land on a "**Submitted!**" confirmation screen with a green check, the area and shift name, the timestamp, an "X of Y items checked" summary, and **Submit another** plus a sign-out option.

History page

View history opens **Daily Report History** — a read-only list of recent daily reports for your facility (newest first). Each row shows the area (with its color dot), the shift/template name, the timestamp, who submitted it, and an "X/Y complete" badge. A note at the bottom reminds you reports auto-delete after 14 days. You cannot edit anything from here.

Admin tabs (Submissions review)

In **Daily Reports Admin → Submissions**, managers get a filterable table of submissions:

- **Filters** — by area, by employee, and by date range (defaults to the last 14 days).
- **Table** — Submitted time, Area, Template, Employee, Items (checked/total), Notes count, and an **Open** link.
- **Detail panel** (after clicking Open) — shows the full checklist with each item's checked/unchecked state, the list of notes, and admin actions: **toggle any item**, **add an admin note**, **edit or delete existing notes**, and **delete the submission**. There is a **Back to list** link to return to the filtered table.

6. Step-by-step: common tasks

Submit a daily report (staff)

1. Open **Daily Reports** from the sidebar (or Reports tab).
2. In **Work area**, pick the zone you're logging.
3. In **Shift**, pick the checklist (it may already be chosen if there's only one).
4. Tick each task as you complete it; watch the progress bar.
5. Optionally type a **Note** for managers.
6. Tap **Submit** in the sticky bar.
7. Confirm on the **Submitted!** screen, or tap **Submit another** to log a different area.

Add a new work area (admin)

1. Go to **Daily Reports Admin → Areas**.
2. Click **Add area** (if the "X / 30 active" badge shows you're at the cap, deactivate one first).

3. Enter a **Name**, optionally adjust the **Slug**, pick a **Color**, set a **Sort order**.
4. Click **Create area**. It now appears as an option in the staff Work area dropdown for anyone granted access.

Build a checklist template (admin)

1. Go to **Daily Reports Admin → Templates** and choose the **Area** from the dropdown.
2. Click **Add template**, give it a **Name** (and optional description), then **Create template**.
3. Switch to the **Checklist Items** tab, choose the same **Area** and your new **Template**.
4. Click **Add item** for each task (Label required), or use **Bulk upload** to paste many at once.
5. Reorder items with the up/down arrows so they read top-to-bottom the way staff work.

Grant area access to an employee (admin)

1. Go to **Daily Reports Admin → Area Access**.
2. Find the employee's row and the area's column.
3. **Check the box** — it saves immediately. (To grant many at once, use **Download CSV**, edit the `can_submit` column, and **Import CSV**.)

The employee will now see that area in their Work area dropdown the next time they open Daily Reports.

Review or annotate a submission (admin)

1. Go to **Daily Reports Admin → Submissions** and use the filters to narrow the list.
2. Click **Open** on a row.
3. Add an **admin note**, adjust a checkbox if needed, or **Delete submission**. Notes you add are visible to admins and to staff on the report.

7. Field reference

Staff console

Field	Type	Required	Notes
Work area	Dropdown	Yes	Only areas you have submit access to; shows a color swatch.
Shift	Dropdown	Yes	Templates within the chosen area; shows each one's item count; auto-selected if only one exists.
Checklist items	Checkboxes	No	One per active item on the template; drive the progress bar. A template may legitimately have zero items.
Note	Text area	No	Free text for managers; saved with the submission.

Area form (admin)

Field	Type	Required	Notes
Name	Text	Yes	What staff see in the Work area dropdown.
Slug	Text	No	Auto-generated from the name if blank; lowercase letters, digits, hyphens only.
Color	Color picker	No	Accents the area in dropdowns, the checklist, and History.
Sort order	Number	No	Lower numbers appear first.
Active	Checkbox	No (edit only)	Inactive areas are hidden from staff but keep their history; counts against the 30-active cap only while active.

Template form (admin)

Field	Type	Required	Notes
Name	Text	Yes	E.g. "Opening checklist."
Description	Text	No	Optional sub-line shown in the admin table.
Sort order	Number	No	Order of shifts within the area.
Active	Checkbox	No (edit only)	Inactive templates are hidden from staff.

Checklist item form (admin)

Field	Type	Required	Notes
Label	Text	Yes	The task text staff tick off.
Description	Text	No	Optional guidance line under the label.
Sort order	Number	No (edit only)	New items are auto-placed at the end.
Active	Checkbox	No (edit only)	Inactive items don't appear on new submissions.

8. Locking, saving & offline

There is no end-of-day "report" that locks. Unlike a single paper sheet that fills up over a shift, RinkReports treats each submit as its own complete record. Specifically:

- **Saved vs submitted.** While you're filling in the checklist, nothing is stored on the server — your ticks live in the page until you tap **Submit**. Pressing Submit creates the report.

- **Append-only.** Every Submit creates a **new, permanent record**. Staff cannot go back and edit a submitted report; if something was wrong, you simply submit again — the new submission is a separate row, and both are kept for the audit trail. Each report is automatically tagged with your facility's local calendar date so managers can group a day's reports, but a new day doesn't "close" the previous one — there is no lock step.
- **What admins can change.** Managers don't reopen a staff report for the staffer to edit; from the Submissions detail panel they can toggle items, add/edit/delete admin notes, or delete the whole submission.
- **Auto-delete.** Daily reports are kept for **14 days** and then automatically purged. History and the admin Submissions list only show what's still within that window.

⚠ **VERIFY** — "the report locks/submits at end of day." The training package described an end-of-day lock. The code does **not** implement a per-day lock or a single rolling report; it uses an append-only model (one immutable record per Submit, tagged with the facility-local date, auto-deleted after 14 days). Describe it that way.

Offline ("Save on this device"). Daily Reports works without a connection:

- If you tap Submit while offline, the report is **saved on your device** and queued. The confirmation reads "**Saved on this device**" and explains it will submit automatically once you're back online — and that the same checks run then. You can keep working.
- The sticky bar shows an "offline — will sync when reconnected" hint so you know you're in offline mode.
- When your connection returns, the queued report submits on its own. Each queued report carries a unique identifier so a retry can't create a duplicate, and the same area, shift, and permission checks that run online are re-applied when it syncs. If access was revoked while you were offline, the queued report can be rejected at sync time.

9. Troubleshooting & FAQ

"No areas assigned" / the Work area dropdown is empty. You haven't been granted submit access to any area. Ask an admin to check the **Area Access** grid in Daily Reports Admin.

"Account not ready" or "Your account isn't fully set up yet." Your employee record isn't active or fully provisioned. Contact your administrator.

I can't find Daily Reports in my menu. Either the module is turned off for your facility (an admin controls this on the **Modules** page) or you lack view permission.

The Shift dropdown says "No shifts available." The selected area has no active templates yet. An admin needs to add at least one template to that area.

I ticked the wrong box / submitted by mistake. You can't edit a submitted report. Submit a corrected one — both are kept — and, if needed, ask a manager, who can adjust or delete the submission from the Submissions tab.

Admin: "Cap reached" when adding an area. Your facility already has 30 active areas (the database limit). Deactivate an area you no longer use, then add the new one.

Admin: "Cannot delete area/template with existing submissions." Deletion is blocked once reports reference the item. Use **Deactivate** instead — it hides the area or template from staff while preserving history.

My report didn't show up. If you submitted offline, it's queued and will appear once it syncs. Also remember History only shows the last **14 days**; older reports are auto-deleted.

Can I see another facility's reports? No. You only see data for your own facility — this is automatic.

Source

```
src/app/reports/daily/ (page.tsx, actions.ts, _components/daily-report-console.tsx,
history/page.tsx, [areaSlug]/[templateId]/done/page.tsx, _lib/submit.ts, _lib/compute.ts,
_lib/offline.ts); src/app/admin/daily-reports/ (page.tsx, actions.ts, area-access-actions.ts,
types.ts, _components/areas-tab.tsx, templates-tab.tsx, items-tab.tsx, area-access-tab.tsx,
submission-detail.tsx, area-form.tsx, item-form.tsx);
supabase/migrations/000000000000007_daily_reports_schema.sql (30-active-area cap trigger, 14-day
retention). Cross-referenced with docs/training/00-MANIFEST.md.
```

Refrigeration Logs

RinkReports training guide — Refrigeration Logs module

1. What this module is for

The **Refrigeration Logs** module is where staff record the readings for your rink's refrigeration plant — compressors, pumps, condensers, brine/supply-return temperatures, machine hours, alarms, and anything else your facility has set up to track. Each time someone walks the plant and takes a "round" of readings, they open this form, enter the numbers, and submit.

The module does three things for you:

- **Captures readings** for each piece of equipment in a structured, repeatable way, so every round looks the same and nothing gets skipped.
- **Checks each reading against a normal range** as it's entered, and shows a "Normal: min – max" hint right under the field so staff know immediately whether a value looks right.
- **Flags out-of-range readings** and — for the most serious ("critical") ones — requires the person to write down what corrective action they took before they can submit. If your facility has turned on out-of-range alerts, managers also get notified.

Every submitted report is a permanent record. You don't edit a past report — you submit a new round. Admins can review every submission and add follow-up notes, but the original reading is never changed.

You only see data for your own facility — this is automatic. There's no facility switcher, and you can't see another rink's refrigeration logs.

2. Who can use it

Access is **permission-driven**, not strictly tied to a job title. The table below shows the typical default for each tier, but an admin can grant or remove any of these on a per-person basis. The four actions that matter are **View**, **Submit** (file a reading), **Edit**, and **Admin** (configure the module).

Tier	Submit a reading	Review history / add follow-up notes	Configure (Setup & Settings)
super_admin	Yes	Yes	Yes
org_admin △ VERIFY	No exact equivalent in the live app — treat as super_admin.	—	—
facility_manager (the admin role)	Yes	Yes	Yes
supervisor (the manager role, or a custom role) △ VERIFY	Typically yes (needs the submit permission)	Only if granted the admin action	Only if granted the admin action
staff	Yes (needs the submit permission)	No access to the admin History/Setup screens	No access

Notes that hold true in code:

- To open and submit the staff form, you need the `submit` permission for the refrigeration module **and** an active employee account assigned to your facility. Without it, you see a "No permission" message.
- The whole **Refrigeration Admin** area (Setup / History / Settings) requires **admin-level access** (the global super-admin flag, or the `admin` permission for your facility). Non-admins are shown a "Forbidden" message rather than the screens.
- A **deactivated** account is denied everywhere — it can't sign in, submit, or view.

⚠ **VERIFY** — the rows for supervisor and the `org_admin` tier are mapped to the live `super_admin` / `admin` / `manager` / `staff` model from the manifest. Because access is per-person and per-action, confirm against your configured facility before treating any of these as a hard rule.

3. How to get there

Staff (submitting a reading):

- In the left sidebar (or the mobile **Menu** tab), choose **Refrigeration**. The address is `/reports/refrigeration`.
- If the module is turned off for your facility, or you lack the submit permission, the menu item won't be there (or you'll get a "Not available" message).

Admins (configuring or reviewing):

- Open the **Admin Center** (the "Admin Center" link appears in the staff sidebar for admins), then choose **Refrigeration Admin** under **Module Admin**. The address is `/admin/refrigeration`.
- The admin screen opens on the **Setup** tab by default, with **History** and **Settings** tabs alongside it.

4. Setup & configuration (admins)

All refrigeration configuration lives on the **Refrigeration Admin** page (`/admin/refrigeration`), split across three tabs: **Setup**, **History**, and **Settings**. Changes here change what staff see on the reading form. For the broader admin picture — modules on/off, permissions, exports, retention — see the **Admin Control Center** chapter.

Getting started fast: Seed defaults

If your facility has **no sections yet**, the Setup tab shows a "**No sections yet**" card with a **Seed defaults** button. Clicking it creates six standard sections — **Compressors, Pumps, Condensers, Supply / Return, Machine Hours, Alarms** — plus a default settings row. You can then rename, add to, or delete any of them. You can also skip seeding and build everything by hand with **Add section**.

Sections

A **section** is a card on the staff form (for example, "Compressors" or "Brine"). On the Setup tab, sections are listed down the left; each entry shows its name and a count of "N equipment, N fields." Pick one to manage it.

For each section you can:

- **Add section** — name (required) and an optional slug (auto-generated from the name).
- **Rename** — change name, slug, and sort order (controls the order cards appear in).
- **Deactivate / Activate** — a deactivated section is hidden from the staff form but kept for history.

- **Delete** — removes the section. This is blocked if equipment, fields, or reports still reference it; the app suggests deactivating instead.

Equipment

Within a section, **Equipment** is the list of individual units — e.g. "Compressor 1," "Compressor 2." The count of compressors (or any equipment) is entirely admin-configurable: add as many as your plant has. Each piece of equipment gets its own labeled group of fields on the staff form.

For each section you can **Add equipment** (name, optional slug) and, per item, **Edit** (name, slug, sort order), **Deactivate / Activate**, or **Delete**.

Fields

Fields are the actual reading inputs. A field can be **section-level** (not tied to a specific unit) or attached to a **specific piece of equipment**. For each field you set:

- **Label** — what staff see (e.g. "Suction pressure").
- **Key** — an internal identifier (auto-generated from the label if left blank).
- **Type** — one of:
 - **numeric** — a number entry; the only type that supports thresholds and the "Normal:" range hint.
 - **text** — free text.
 - **boolean** — a yes/no checkbox.
 - **select** — a dropdown; you supply the options, one per line as `key|Label`.
 - **computed** — a read-only value calculated automatically on submit from other numeric fields in the same section. Staff see "Calculated automatically on submit" and never type into it.
- **Unit** — e.g. psi, °F. (Temperature units drive the °F/°C toggle behavior — see §5.)
- **Sort order** — reorder fields with the ↑ / ↓ buttons on each row.

Each field can be **Deactivated / Activated** or **Deleted**, and fields are also marked required where configured (a required field shows a red mark to staff and blocks submit if empty).

Thresholds (normal ranges)

A **threshold** defines the normal range for a numeric field and how serious it is to fall outside it. Open a numeric field's **Thresholds** panel (the "Thresholds" button on the field row) to add one. Each threshold has:

- **Min** and/or **Max** — the normal range. You can set just one side. This is exactly what staff see as the "Normal: min - max" hint.
- **Severity** — **warn**, **high**, or **critical**.
- **Scope** — for a section-level field, you can scope a threshold to **All equipment** or to one specific unit. An equipment-specific threshold wins over a section-wide one for that unit.

Severity matters at submit time:

- A reading outside **any** active threshold is flagged **out-of-range** and counted on the confirmation screen and in History.
- A reading outside a **critical** threshold additionally **forces the staff member to write a corrective-action note** before they can submit (see §5).

Thresholds can be **Edited**, **Deactivated / Activated**, or **Deleted**.

Settings (the Settings tab)

There is one settings row per facility, controlling alerting and the per-shift cap:

- **Enable out-of-range alerts** — a checkbox. When on, submitting a report with any out-of-range reading raises an alert to managers (a banner reminds staff: "Out-of-range readings will trigger an alert to managers.").
- **Default alert severity** — **warn / high / critical** — used when a triggered threshold doesn't carry its own severity.
- **Readings per shift** — a number from **1 to 99**, or **blank for unlimited**. This is the expected number of reading rounds per shift. When set, it (a) caps the Round number staff can enter and (b) shows staff the "of N" indicator and a "This facility logs N readings per shift" hint. The server rejects a round number above this cap.

Click **Save settings** to apply.

5. Screen-by-screen walkthrough

5.1 Staff reading form (/reports/refrigeration)

Header. At the top: a "Reports > Refrigeration" breadcrumb, the page title **Refrigeration**, and two buttons — **Back** (returns to the previous page) and **Dashboard** (jumps to your dashboard).

Meta-chip strip. A row of small chips just below the header, each with an icon: **your name**, **facility name**, **today's date**, the **live time** (it ticks each second), and the current **outdoor temperature** for your facility (e.g. "72°F · location"). If weather isn't available it reads "Temp unavailable."

Offline banner. If you're offline, a small note appears: "You're offline. Your report will be saved on this device and submitted automatically when you reconnect." (See §8.)

Alerts banner. If your facility has out-of-range alerts turned on, a note reminds you that out-of-range readings will alert managers.

Log Information card. This card holds the round's basics and the unit toggle:

- **°F / °C toggle** — a switch in the card header. It changes only how temperatures are **displayed**; values are always stored canonically in **°F**. Flipping it converts any temperature values you've already typed, once, to the new display unit. Non-temperature fields (like *psi*) are unaffected.
- **Facility** and **Employee** — shown read-only.
- **Reading time** — a date-and-time picker, pre-filled with now. Adjust it if you're logging a round you took earlier.
- **Shift (optional)** — free text, e.g. "AM / PM / Overnight."
- **Round # (optional)** — the round number for this shift. If your facility set a readings-per-shift cap, the label reads **"Round # (optional) — of N"** and a hint below says how many readings the facility logs per shift. Entering a round above the cap is rejected on submit.

Section cards. One card per active section, in the order admins set. Within a card:

- **Section-level fields** appear first in a responsive grid.
- Each **piece of equipment** gets its own sub-heading (e.g. "Compressor 1") with its fields grouped beneath it.

Field types as they appear:

- **numeric** — a number box (decimal keypad on mobile) with the unit shown beside it and appended to the label, e.g. "Suction pressure (psi)." Temperature fields show "(°F)" or "(°C)" matching the toggle.
- **text** — a single-line text box.
- **boolean** — a checkbox with the field label.
- **select** — a dropdown of the admin-defined options.
- **computed** — a greyed, read-only box reading "Calculated automatically on submit."

Normal-range hints. Under each numeric field that has a threshold, a muted line reads "**Normal: min - max unit**" (or " $\geq$ min" / " $\leq$ max" if only one bound is set). When you flip the °F/°C toggle, temperature ranges convert to match.

Corrective-action box. If a numeric value breaches a **critical** threshold, a highlighted **Corrective action** box appears under that field with a "(required)" mark. You must describe the action taken; submit is blocked until you do. (Warn/high out-of-range readings are still flagged, but don't force a note.)

Notes. A general **Notes (optional)** card near the bottom for any overall comments about the round.

Submit. A full-width button at the bottom:

- Online: "**Submit refrigeration report.**"
- Offline: "**Save on this device.**"
- While saving: "**Submitting...**"

If you try to submit with a missing required field, an unparseable number, or a missing critical corrective-action note, the form highlights the problem and jumps you to the first one.

Your recent submissions. Below the form, if you've submitted in the last 30 days, a read-only list shows each report's time, a "N values" badge, and an "N out-of-range" badge where applicable.

5.2 Confirmation screen (/reports/refrigeration/done)

After a successful online submit you land on a "**Submitted!**" screen with a green checkmark, the submission timestamp, a "N values recorded" chip, and — if any reading was out of range — an "N out-of-range" chip. Two buttons: **Submit another** and **Back to home**.

(When you submit **offline**, you instead see an inline "**Saved on this device**" confirmation with a **Back to dashboard** button — the report syncs later.)

5.3 Admin — Setup tab

Covered in detail in §4. Left column: the section list (with equipment/field counts and an "off" badge for inactive sections) plus an **Add section** card. Right column: the selected section's **Equipment**, **Section-level fields**, per-equipment field groups, and inline **Thresholds** panels — each with Add / Edit / Deactivate / Delete controls.

5.4 Admin — History tab

A filterable, read-only table of submitted reports. Filters include **employee**, a **date range** (defaults to the last 14 days), an **out-of-range** filter (yes/no), and a **notes search**. Columns: **Submitted**, **Submitter**, **Values**, **OOR** (out-of-range count badge), **Notes** (excerpt), and a **View** link. **Load more** appears when there are additional rows. There's also an **Export** button in the page header.

Clicking **View** opens the **report detail**: who submitted it and when, the reading-taken time / shift / round, the submitter's notes, then **Recorded values** grouped by equipment (each value shown with

its unit and an **Out of range / OK** status), and a **Follow-up notes** section. Admins can **Add follow-up note** here — notes are append-only and the original report is immutable (you cannot edit a submitted reading).

5.5 Admin — Settings tab

The single settings form described in §4: out-of-range alerts on/off, default alert severity, and readings-per-shift.

6. Step-by-step: common tasks

Log a refrigeration reading (staff)

1. Open **Refrigeration** from the menu.
2. In **Log Information**, confirm or adjust the **Reading time**; optionally type the **Shift** and **Round #**.
3. (Optional) Flip the **°F / °C** toggle to your preferred temperature display.
4. Work down each section card, filling in the fields for every piece of equipment. Watch the **"Normal: ..."** hints.
5. If a reading turns **critically** out of range, fill in the **Corrective action** box that appears.
6. Add any overall comments in **Notes**.
7. Tap **Submit refrigeration report** (or **Save on this device** if offline).
8. On the **Submitted!** screen, choose **Submit another** or **Back to home**.

Add a compressor (or any equipment) / add a section (admin)

1. Go to **Admin Center > Refrigeration Admin > Setup**.
2. To add a unit: pick the section (e.g. "Compressors"), then in **Equipment** use **Add equipment** — type the name (e.g. "Compressor 3") and **Add equipment**.
3. To add a whole new section: use **Add section** in the left column.
4. Add the **fields** each new section/equipment needs (see next task).

Add a reading field

1. On **Setup**, select the section.
2. In the **Section-level fields** block (or a specific equipment's "Fields for ..." block), use **Add field**.
3. Enter the **Label**, choose the **Type** (numeric / text / boolean / select / computed), set the **Unit** if any, and — for **select** — list options as key|Label, one per line.
4. **Add field**. Reorder with **↑ / ↓** as needed.

Set a normal range / threshold

1. On **Setup**, find the **numeric** field and click **Thresholds**.
2. In **Add threshold**, set **Min** and/or **Max**, pick the **Severity** (warn / high / **critical**), and (for a section-level field) choose the **Scope** — All equipment or one unit.
3. **Add threshold**. Staff will now see "Normal: min – max" under that field, and critical breaches will demand a corrective-action note.

Set readings per shift

1. Go to **Refrigeration Admin > Settings**.
2. In **Readings per shift**, enter a number from **1-99** (or leave blank for unlimited).

- 3. **Save settings.** Staff now see "Round # — of N" and can't submit a higher round number.

Turn on out-of-range alerts (admin)

- 1. Go to **Settings**, check **Enable out-of-range alerts**, and pick a **Default alert severity**.
- 2. **Save settings.** Submissions with out-of-range readings now raise a manager alert that requires acknowledgement.

Review a report and add a follow-up note (admin)

- 1. Go to **History**, filter as needed, and click **View** on a report.
- 2. Read the recorded values and any corrective-action notes.
- 3. In **Add follow-up note**, type your note and **Add note**. The original report stays unchanged.

7. Field reference

Round / log fields (staff form, Log Information card):

Field	Type	Required	Notes
Reading time	Date & time	No (defaults to now)	When the round was taken.
Shift	Text	No	Free text, e.g. AM / PM / Overnight.
Round #	Whole number	No	Capped at the facility's readings-per-shift value; shows "of N" when set.
Notes	Long text	No	General comments about the round.
Corrective action	Long text	Yes, when a value breaches a critical threshold	Appears inline under the offending numeric field.

Reading fields are entirely admin-defined per facility, so there is no fixed list. Each follows its configured type:

Field type	What staff enter	Normal/expected range shown?
numeric	A number, in the field's unit	Yes — "Normal: min – max unit," from the field's threshold(s)
text	Free text	No
boolean	A yes/no checkbox	No
select	One option from a dropdown	No
computed	(nothing — read-only, calculated on submit)	Shows a range only if a threshold is set on the computed value

Default seeded sections (if an admin clicks Seed defaults): Compressors, Pumps, Condensers, Supply / Return, Machine Hours, Alarms. Specific equipment and fields within them are configured per facility.

The exact reading labels, units, and normal ranges at your rink depend on how your administrator configured Setup and Thresholds. Use the on-screen "Normal: ..." hints as the source of truth for what's expected.

8. Locking, saving & offline

Append-only. A submitted refrigeration report is a permanent record. There is **no edit** of a past report — you submit a **new** round instead. Admins don't change submitted values either; they **add follow-up notes**, which are themselves append-only. The detail screen states plainly: "The original report is immutable."

Saving online. Tapping **Submit refrigeration report** files the report, records every value (with out-of-range flags), saves any corrective-action notes against the exact readings they address, optionally raises a manager alert, and takes you to the **Submitted!** screen.

Offline ("Save on this device"). RinkReports is a progressive web app, so the form works without a connection:

- When you're offline the Submit button reads **"Save on this device,"** and an on-screen note explains the report is queued locally.
- After saving offline you see a **"Saved on this device"** confirmation. The report is held on your device and **submits automatically when you reconnect**, re-running the same checks (required fields, critical notes, and the readings-per-shift cap) on the server.
- A pending/failed indicator and the offline-queue page let you see anything still waiting to sync.
- Because each queued report carries a unique id, a retry can't create a duplicate.

Don't close the tab with unsaved readings still on screen — the app warns you before a refresh or close would discard them (until the report has been queued or submitted).

9. Troubleshooting & FAQ

"No permission" when I open Refrigeration.

You don't have the **submit** permission for this module. Ask an admin to enable it for you in the Permissions area.

"Account not set up."

Your sign-in isn't linked to an active employee record at a facility. Contact your administrator.

"Not configured yet" / "Refrigeration reporting isn't configured yet."

No active sections or fields exist for your facility. An admin needs to seed defaults or build sections, equipment, and fields on the Setup tab.

The Refrigeration menu item is missing.

Either the module is switched off for your facility (admins control this on the Modules page) or you lack the view/submit permission.

It won't let me submit — a corrective-action box is red.

A numeric reading is **critically** out of range. Fill in the **Corrective action** box for that field, then submit.

My round number was rejected.

It exceeded the facility's **readings-per-shift** cap. Use a number within "of N," or ask an admin to raise the cap on the Settings tab.

I flipped to °C and my entered numbers changed.

That's expected — the toggle converts displayed temperatures. Values are stored in °F regardless; only the display changes.

I made a mistake on a submitted report.

You can't edit it. Submit a corrected new round. For the record, an admin can add a follow-up note explaining the correction.

Do out-of-range readings always alert managers?

Only if an admin has enabled **out-of-range alerts** on the Settings tab. When enabled, an alert is raised (requiring acknowledgement) summarizing the out-of-range readings.

Can I see another rink's logs?

No. You only see data for your own facility — this is automatic.

Source

Staff flow: `src/app/reports/refrigeration/page.tsx`,
`src/app/reports/refrigeration/_components/submission-form.tsx`,
`src/app/reports/refrigeration/done/page.tsx`, `src/app/reports/refrigeration/actions.ts`,
`src/app/reports/refrigeration/_lib/compute.ts`, `src/app/reports/refrigeration/_lib/submit.ts`.
Admin flow: `src/app/admin/refrigeration/page.tsx`, `src/app/admin/refrigeration/types.ts`,
`src/app/admin/refrigeration/_components/setup-tab.tsx`, `.../settings-tab.tsx`, `.../history-tab.tsx`,
`.../report-detail.tsx`, `.../seed-defaults-card.tsx`. Role/permission context:
`docs/training/00-MANIFEST.md`.

Incident Reporting

Product: RinkReports — the Next.js progressive web app (PWA) for ice-rink operations. **Brand colors:** #4DFF00 (primary green) · #002244 (navy). **Data scoping:** You only see data for your own facility — this is automatic. You never switch facilities or see another rink's incidents; RinkReports keeps each facility's records separate on its own.

1. What this module is for

Incident Reporting is how staff record something that went wrong on the ice or in the building — a slip, a collision, an altercation, an equipment problem, a guest who needed medical help. A report captures **when and where** it happened, **what happened**, **how serious it was**, whether an **ambulance was called**, who was **involved**, and any **witnesses**. Once filed, the report goes to managers/admins to review, track follow-ups, and mark resolved.

Two things make this module different from most other RinkReports forms:

- **You can correct your own report for 24 hours** after submitting. After that window closes, it locks and becomes read-only (see §8). Most other report types are one-and-done with no edit window.
- There is **no photo or file upload** anywhere in this module. You describe what happened in words; you do not attach images, video, or documents.

2. Who can use it

Access is **permission-driven**, not strictly tier-driven: an administrator can turn each person's access on or off per action (view / submit / edit / admin). The table below shows the typical default mapping. Where a label feels rigid, treat it as the default and confirm against your configured facility.

Role tier (doc vocabulary)	Typical access to Incident Reporting
super_admin	Full: file reports, edit own within 24h, and full admin (History, Types, Severities, Activities, status changes, follow-up notes).
org_admin	⚠ VERIFY — the live app has no separate org_admin tier . Treat as super_admin.
facility_manager (live role: admin)	Full admin for their own facility: review every report in History, change status, add follow-up notes, and configure Types / Severities / Activities. Can also file reports.
supervisor (live role: manager or a custom role)	⚠ VERIFY — typically can file reports and edit their own within 24h. Admin review/config only if granted the admin action for this module.
staff (live role: staff or a custom role)	File incident reports and edit their own within the 24-hour window. No access to the admin console (History/Types/Severities/Activities, status, follow-up notes).

Notes confirmed in code:

- To even open the staff incident page, you need the **submit** permission for Incident Reporting. Without it you see a "No access" message ("You don't have permission to submit incident reports.").
- The admin console requires admin-level access; staff who try to reach it are shown a Forbidden message, not the admin screens.
- A **deactivated** account cannot file, view, or edit anything.

3. How to get there

Staff (to file or view your own reports):

- Open the staff app and choose **Incidents** from the sidebar (desktop) or the **Menu** tab (mobile bottom bar: Home · Reports · Menu · Account).
- This opens **Incident Reports** (/reports/incidents): the new-report form, followed by **Your recent reports** (your own submissions from the last 30 days).
- The Incidents menu item only appears if your facility has the module turned on (admins control this on the Modules page) and you have permission.

Admins (to review and configure):

- Open the **Admin Center**, and under **Module Admin** choose **Incident Reports Admin** (/admin/incident-reports).
- The admin page opens on the **History** tab, with tabs across the top for **History** · **Incident Types** · **Severity Levels** · **Activities**.

4. Setup & configuration (admins)

Configuration lives in the **Incident Reports Admin** screen (/admin/incident-reports). Original submitted reports are **immutable** — admins never edit a staffer's report in place; they change its **status** and add **follow-up notes** (see §5).

The header carries two shortcuts: **Manage locations** (jumps to the shared Facility Spaces list) and an **Export** button for this module.

Severity Levels tab

Severity levels rank how serious an incident was (e.g. Critical, High, Medium, Low). This is the **required** picker on the staff form — a facility must have at least one active severity or staff see "Not configured yet."

- **Add severity** opens a side panel with: **Key** (lowercase letters, digits, underscores — e.g. high, very_high; must be unique in your facility), **Display name** (what staff see), **Color**, and **Sort order**.
- Each row can be **Edited**, **Deactivated** / **Reactivated**, or **Deleted**. Deactivating hides it from the staff picker while keeping it on past reports.
- **Delete is blocked** if any reports already use that severity — you'll be told "Cannot delete; in use by N reports. Deactivate instead."
- If no severities exist yet, a **Seed defaults** card offers the four standard levels: **Critical**, **High**, **Medium**, **Low**. Seeding is safe to run more than once (it skips duplicates).

Activities tab

Activities describe what was going on when the incident happened ("Activity at the time" on the staff form) — e.g. Public Skating, Hockey, Figure Skating. This picker is **optional** for staff, and staff can

always choose "**Other...**" and type a free-text activity instead.

- **Add / Edit / Deactivate / Reactivate / Delete** work the same way as severities (Key + Display name + Color + Sort order). Delete is blocked when an activity is in use; deactivate instead.
- **Seed default activities** offers Public Skating, Hockey, Figure Skating, Learn to Skate, and Maintenance.
- A **bulk import** card accepts a simple CSV (display_name, key, color, sort_order, one per line). Duplicate keys are skipped, not overwritten.

Incident Types tab

Incident Types categorize reports (e.g. Slip / Fall, Equipment, Altercation).

- **Add incident type** captures **Name**, **Slug** (auto-derived from the name if left blank), **Color**, and **Sort order**, with the same Edit / Deactivate / Reactivate / Delete controls. Delete is blocked when a type is in use.
-  **VERIFY** — **Type is shown and filtered on in the admin History view, but the current staff submission form does not present an incident-type picker.** Type appears as a column/filter and on the report detail; it is not collected from staff in the live form. Confirm whether your facility expects staff to pick a type before training on it as a staff field.

Facility Spaces (the "where" list) — lives elsewhere

The **Facility space** picker on the staff form is **not** configured inside this module. It is fed by the shared **Facility Spaces** list in the Admin Center (**Setup → Facility Spaces**, /admin/spaces), which is the same list used by Accident and Air Quality. Use the **Manage locations** button on the Incident Reports Admin header to jump straight there. Add, rename, reorder, or deactivate spaces there and they appear (or disappear) in the staff picker automatically. Staff can always add an "**Other**" free-text space when the right location isn't listed.

Cross-reference — Admin Control Center. Module visibility (whether Incidents shows in staff navigation at all) is set on the **Modules** page; who can file or administer is set under **Permissions / Roles**; the location list is **Facility Spaces**. See the Admin Control Center chapter for those shared screens.

5. Screen-by-screen walkthrough

A. The staff incident form (/reports/incidents)

The page opens with an "Incident Reports" header, then the form, then **Your recent reports** below. The form is grouped into cards:

Card 1 — When & where

- **When did it happen?** (required) — a date-and-time picker, pre-filled with the current date/time. Adjust it to when the incident actually occurred.
- **Facility space** (required) — a dropdown you can **multi-select**; pick one or more spaces. If there are many spaces a **search box** appears. At the bottom is an **Other** option — tick it and a free-text box appears to **describe the space**. You must pick at least one space or fill in an "Other" space.

Card 2 — What happened

- **Description** (required) — a free-text box, up to **500 characters** (a live counter shows how many you've used). Describe what happened in detail.

- **Activity at the time** (optional) — a dropdown of admin-defined activities, plus "**Other...**" which reveals a free-text box.
- **Severity** (required) — a dropdown of admin-defined severity levels.
- **Immediate actions taken** (optional) — a free-text box for what was done right after.
- **Was an ambulance called?** — a Yes/No toggle (defaults to No). Turning it on raises a higher-priority alert to the facility (see §8).
- **Number of people involved** (optional) — a numeric field; accepts a whole number (0 or more) only.
- **Follow-up required** — a Yes/No toggle (defaults to No).

Card 3 — Witnesses (optional)

- Add up to **3 witnesses**. Use **Add a witness / Add another witness**; each row has **Name**, **Phone**, **Email**, and a **Brief statement**, and a **Remove** button. A counter shows n/3.
- Rule per witness: if you start a witness you must give a **name** and **at least one contact** (phone or email). Empty witness rows are simply ignored.

Reporter identity is automatic. The form does **not** ask for your name or phone — RinkReports fills the reporter from your login so it can't be spoofed.

Submit & the "Are you sure?" dialog. The submit button reads **Submit incident report**. Tapping it first checks the required fields, then opens a confirmation dialog: "**Submit this incident report?**" with the reminder that you can edit it for 24 hours, after which it becomes read-only. Choose **Confirm & submit** to file it, or **Go back** to keep editing.

B. The "Reported" confirmation screen (.../done)

After submitting you land on a confirmation screen with a green check and "Reported — Thank you, [your name]. Your report has been submitted." It shows a summary (Status, Severity, when it happened, when submitted, and any "Other space"), a note that "You can edit this report for 24 hours," and three buttons: **Edit report**, **Submit another**, **Back to home**.

C. Your recent reports list

Below the form, **Your recent reports (Last 30 days)** lists your own submissions. Each entry shows the time submitted, a colored **severity** pill, a **status** badge (Submitted / In review / Resolved / Archived), any "Other" location, and a short excerpt of the description. Click **Incident** on a row to open that report.

D. The single-report view / 24-hour edit (/reports/incidents/[id])

Opening one of your reports does one of two things:

- **Within 24 hours, and it's yours:** the **Edit Incident Report** form opens, pre-filled with everything you entered, with an "Editable until [date/time]" note. Make changes and **Save changes**; a confirmation dialog ("Save changes to this report?") reminds you the change is recorded in the report's history. After saving you see a "Changes saved" banner and can keep editing until the window closes.
- **After 24 hours, or if it isn't yours:** a **read-only** view of the report. If it's yours, it says "The 24-hour edit window for this report has closed, so it's now read-only." A **Back to incident reports** button returns you to the list.

E. Admin: Incident Reports Admin (/admin/incident-reports)

History tab — a filterable table of every incident at your facility (most recent first). Columns: Submitted, Reporter, Type, Severity, Location, Status, and a **View** link. Filters let you narrow by **status, type, severity, employee, location, and a date range (from/to)**. Click **View** to open the detail panel.

Report detail panel — opens from History → View. It shows:

- The **Original report** (read-only, never editable here): Type, Severity, Activity (or the "(other)" text), Occurred-at, Reporter name & phone, **Ambulance called** (highlighted when Yes), People involved, Follow-up required, Facility spaces (including any "other"), Description, and Immediate actions.
- **Witnesses** — each name with phone/email and statement.
- **Status** — a **Change status** dropdown (Submitted → In review → Resolved → Archived) and a timeline of when each status was reached (Submitted / Reviewed / Resolved / Archived).
- **Follow-up notes** — an append-only thread. Type into **Add follow-up note** and **Add note**. Notes show author and time and **cannot be edited or deleted** once added.
- **Change log** — an append-only audit trail of edits (create/update) with who and when. Visible to admins only.

Incident Types · Severity Levels · Activities tabs — the configuration lists described in §4.

6. Step-by-step: common tasks

File an incident report (staff)

1. Sidebar/Menu → **Incidents**.
2. Set **When did it happen?**
3. Open **Facility space** and tick one or more spaces (or tick **Other** and describe it).
4. Write the **Description** (required, up to 500 characters).
5. Optionally pick **Activity** (or "Other..."), set **Severity** (required), add **Immediate actions**, toggle **Ambulance called** and **Follow-up required**, and enter **people involved**.
6. Optionally add up to **3 witnesses** (each needs a name + phone or email).
7. Tap **Submit incident report**, then **Confirm & submit** in the dialog.
8. You land on the **Reported** screen.

Edit an incident within 24 hours (reporter)

1. From the **Reported** screen tap **Edit report**, or open the report from **Your recent reports**.
2. If you're inside the window, the edit form opens pre-filled. Change anything (spaces, severity, witnesses, etc.).
3. Tap **Save changes** → **Save changes** in the confirmation dialog.
4. You'll see "Changes saved" and can keep editing until the window closes. After 24 hours the report locks to read-only.

Change an incident's status (admin)

1. Admin Center → **Incident Reports Admin** → **History**.
2. **View** the report.
3. In the **Status** section, pick a new value (Submitted / In review / Resolved / Archived). It saves immediately and stamps the matching timestamp.

Add a follow-up note (admin)

1. Open the report detail (History → View).
2. Scroll to **Follow-up notes**, type your note in **Add follow-up note**, and click **Add note**.
3. The note is permanent — it can't be edited or deleted.

Add a new severity (admin)

1. Admin Center → **Incident Reports Admin** → **Severity Levels**.
2. Click **Add severity**.
3. Enter a **Key** (e.g. high), a **Display name** (e.g. High), a **Color**, and a **Sort order**.
4. **Create severity**. It appears in the staff Severity dropdown right away. (To get four standard levels at once, use **Seed defaults** when the list is empty.)

Add an activity (admin) — same flow on the **Activities** tab (**Add activity**), or use **Seed default activities** / the **CSV bulk import**.

7. Field reference

Field	Where	Required?	Notes
When did it happen?	Staff form	Yes	Date & time picker; defaults to now.
Facility space	Staff form	Yes	Multi-select from Facility Spaces; or tick Other + free text. At least one space or an "Other."
Other space	Staff form	Conditional	Free text; required if Other is ticked.
Description	Staff form	Yes	Free text, max 500 characters (live counter).
Activity at the time	Staff form	No	Admin-defined list, plus " Other... " free text.
Severity	Staff form	Yes	Admin-defined list; facility must have at least one.
Immediate actions taken	Staff form	No	Free text.
Was an ambulance called?	Staff form	No (defaults No)	Yes raises a critical, acknowledgement-requiring alert to the facility.
Number of people involved	Staff form	No	Whole number, 0 or more.
Follow-up required	Staff form	No (defaults No)	Yes/No toggle.
Witnesses (up to 3)	Staff form	No	Each: Name + (Phone or Email) required; Statement optional.
Reporter name / phone	(Automatic)	—	Filled from your login; not asked on the form, not editable.
Status	Admin detail	—	Submitted / In review / Resolved / Archived; admin-set.

Field	Where	Required?	Notes
Follow-up notes	Admin detail	—	Append-only; can't be edited or deleted.
Incident Type	Admin History/detail	—	Admin-configurable category; Δ VERIFY whether staff pick it (not in the live staff form).
Severity / Activity — Key	Admin config	Yes	Lowercase letters, digits, underscores; unique per facility.
Display name	Admin config	Yes	What staff see in the dropdown.
Color / Sort order	Admin config	No	Color swatch and ordering in the list.

8. Locking, saving & offline

Submitted status. A new report is saved with status **Submitted**. Admins later move it through **In review** → **Resolved** → **Archived** as they handle it.

The 24-hour edit window (confirmed). When a report is created, RinkReports stamps an edit-window end time of **submission + 24 hours**. While the current time is at or before that mark, **the reporter (and only the reporter) can edit their own report** — change details, spaces, severity, witnesses, etc. Every change is written to the report's change log. **Once 24 hours pass, the report locks:** the reporter sees a read-only view ("The 24-hour edit window for this report has closed"), and any attempt to save is rejected ("The edit window for this report has closed."). After that, only admins act on the report — and even they **never edit the original**; they change **status** and add **follow-up notes** instead. The original report and its follow-up notes/change log are immutable.

Ambulance alerts. Filing a report with **Ambulance called = Yes** raises a high-priority, acknowledgement-requiring alert so it surfaces above routine notifications. This is best-effort and never blocks your submission.

Offline. RinkReports is a PWA, so you can file an incident with no connection. When you're offline:

- The submit button changes to **Save offline** and a banner notes "This report will be saved on your device and submitted automatically when you reconnect."
- After confirming, you see **"Saved on this device"** with a **Submit another report** button. The report is queued locally and submitted automatically once you're back online — re-running the same checks as an online submission, with no duplicates.
- **Editing an existing report is online-only.** The offline queue handles new incident reports; the 24-hour edit/update path runs against the live record and isn't queued.

9. Troubleshooting & FAQ

"You don't have permission to submit incident reports." Your account lacks the **submit** permission for this module. Ask an admin to enable it (Permissions / Roles).

"Account not ready" / "Your account isn't fully set up yet." Your login isn't linked to an active employee record. Contact your administrator.

"Not configured yet" — no severities. A facility needs at least one active **Severity Level** before staff can file. An admin should add one (or **Seed defaults**) on the Severity Levels tab.

I can't find the right location in the Facility space list. Tick **Other** in the dropdown and type the space. Admins can add it permanently in **Facility Spaces** (Admin Center → Setup → Facility Spaces).

The activity I need isn't listed. Choose **"Other..."** and type it in. Admins can add activities on the Activities tab.

I need to fix my report but the edit form won't open. The **24-hour window has closed** — the report is now read-only. Ask a manager/admin to add a **follow-up note** with the correction; the original can't be changed.

Can I attach a photo? No. This module has no photo or file upload — describe the incident in the Description and Immediate-actions fields.

Why can't I edit someone else's report? You can only edit your **own** report, and only within its 24-hour window. Others' reports are read-only to you.

My note/status change disappeared from the dropdown. Follow-up notes are append-only and permanent (they can't be edited or deleted). A status change saves immediately and stamps its timestamp; if it didn't apply you'll get an on-screen error.

I submitted while offline — did it go through? It's saved on your device and will submit automatically when you reconnect. You'll find it in **Your recent reports** once it syncs.

Source (footnote)

Staff flow: `src/app/reports/incidents/page.tsx`,
`src/app/reports/incidents/_components/submission-form.tsx`,
`src/app/reports/incidents/[id]/page.tsx`, `src/app/reports/incidents/done/page.tsx`,
`src/app/reports/incidents/actions.ts`, `src/app/reports/incidents/_lib/compute.ts`,
`src/app/reports/incidents/_lib/submit.ts`, `src/app/reports/incidents/types.ts`.

Admin flow: `src/app/admin/incident-reports/page.tsx`, `src/app/admin/incident-reports/actions.ts`,
`src/app/admin/incident-reports/types.ts`, and `_components/` (`history-tab.tsx`, `history-filters.tsx`,
`report-detail.tsx`, `types-tab.tsx`, `severities-tab.tsx`, `severity-form.tsx`, `activities-tab.tsx`,
`activity-form.tsx`, `type-form.tsx`, `seed-defaults-card.tsx`, `status-badge.tsx`).

Schema / 24-hour window:

`supabase/migrations/000000000000103_incident_reports_redesign_columns.sql` (`edit_window_ends_at ...`
`default (now() + interval '24 hours')`), `...104_incident_report_children.sql`,
`...102_incident_activities.sql`, `...105_facility_spaces_incident_admin_write.sql`.

Ice Operations

1. What this module is for

Ice Operations is the digital logbook for the day-to-day maintenance your crew performs on the ice and on the machines that maintain it. Instead of paper sheets pinned by the resurfacers, your team logs each job on a phone or tablet — and every entry is automatically stamped with who did it and when.

The module covers four kinds of jobs ("operation types"):

- **Ice Make** — a resurfacing run: which rink, which machine, water used, machine hours, and how much snow was taken.
- **Edging** — time spent running the edger around the boards.
- **Blade Change** — swapping the blade on a machine, recording the old blade's hours and the new blade's ID.
- **Circle Check** — a structured pre-use safety/condition inspection of a machine, with a pass/fail checklist. Failed items raise an alert so a manager knows right away.

Two important things to understand up front:

- **Your facility chooses which of these four operation types it uses.** An admin can hide any of them, so your tabs may show all four or just one or two. (The four types themselves are built in — they can't be renamed or added to — but visibility is yours to control.)
- **Equipment is your own list.** The machines you pick from (resurfacers, edgers, blade sets, etc.) and their "types" are set up by your admin for your facility — nothing is hard-coded.

You only see data for your own facility — this is automatic.

2. Who can use it

Access is permission-driven, not a fixed rank. The table shows the typical default for each tier; an admin can widen or narrow any one person's access per action (view / submit / configure). Treat the tiers as defaults, not hard rules.

Tier	Typical access to Ice Operations
super_admin	Full access everywhere, plus all admin configuration.
org_admin	△ VERIFY — the live app has no separate org-admin tier; treat as super_admin.
facility_manager (the admin role)	Submit reports and run all admin setup, history, and settings for their facility.
supervisor (the manager role, or a custom role)	Submit reports if granted the submit permission. Admin setup only if also granted the module's admin permission. △ VERIFY whether your facility grants supervisors config rights by default.
staff (or a custom role such as driver)	Submit reports if granted the submit permission. No access to the admin setup/history/settings screens.

To log an operation you must have an active employee account at the facility and the `submit` permission for Ice Operations. If you lack it, the form shows a "No permission" notice instead of the tabs. If your account isn't fully set up, you'll see an "Account not set up" notice — contact your

administrator.

3. How to get there

- **Staff side (logging jobs):** open **Ice Operations** from the left sidebar (or the **Menu** tab on mobile). The link only appears if the module is turned on for your facility and you have access. The URL is `/reports/ice-operations`; it lands you on the first enabled operation tab automatically.
- **Admin side (configuring):** admins open the **Admin Center**, then **Ice Operations Admin** under Module Admin (`/admin/ice-operations`). This is the only way in — form configuration lives entirely in the admin area, so there is no configuration shortcut on the staff screens.

4. Setup & configuration (admins)

Everything in this section lives in **Admin Center → Ice Operations Admin**, which has three tabs: **Setup**, **History**, and **Settings**. The page header reminds you that submitted reports are immutable — admins review and annotate, they don't rewrite. For broader admin topics (employees, permissions, exports, retention, the Modules on/off switch), see the **Admin Control Center** chapter.

If your facility has nothing configured yet, the Setup tab offers a **Seed defaults** button. It creates a starter settings row plus five starter circle-check items (four for ice resurfacers, one for the edger). You can edit or add to everything afterward.

Setup tab — what you configure

The Setup tab is a stack of cards. Each card lists existing items with **Edit / Deactivate / Delete** buttons and an "Add" form at the bottom. "Deactivate" hides an item from staff without erasing its history; "Delete" removes it (with a confirmation prompt).

- **Rinks** — the surfaces ice operations are performed on (name and an optional slug, with a sort order). Rinks are required for Ice Make.
- **Fuel types** — power sources for resurfacers (e.g. Electric, Gas, Propane). Each fuel type can anchor one circle-check template.
- **Equipment** — your machines, grouped by **equipment type**. The built-in types are **Ice Resurfacer, Edger, Blade Set, Hand Edger, and Other**. For each machine you can set a name, type, slug, model, serial number, an **hours count** (admin-maintained — it does **not** auto-update from submissions), an optional fuel type, and sort order. The equipment type is what decides which operation tab a machine shows up under (resurfacers → Ice Make / Circle Check, edgers → Edging, blade sets → Blade Change).
- **Circle-check items** — the individual inspection items staff see during a circle check. Each item has a label, optional description, and an "applies to" scope (All equipment, or one equipment type); staff answer every item as **Pass / Fail**. Items can be reordered with up/down arrows, toggled active, deleted, and **bulk-imported** (up to 50 per facility).
- **Circle-check templates** — an optional, more structured alternative to the flat item list. You create **one template per fuel type** (up to **4 templates** per facility), give it a name/description, and add its own list of checklist **fields** (also bulk-importable). When an operator picks a resurfacer whose fuel type matches a template, that template's fields replace the generic item list for that check.

Settings tab

- **Visible operations** — checkboxes for Ice Make, Circle Check, Edging, Blade Change. Check the ones your facility uses; **leaving all unchecked shows every operation** (the default). This is what controls which tabs staff see.

- (Removed) **Temperature unit** — the staff forms no longer collect a temperature, so the °F/°C selector was removed from Settings. Legacy temperature readings on older submissions still display in the facility's previously saved unit (default °F).
- **Enable ice operations alerts** — when on, a failed circle check raises a facility alert.
- **Default alert severity** — warn / high / critical, used when a circle-check failure alert is raised without its own severity.

5. Screen-by-screen walkthrough

The Ice Operations shell

Every staff screen sits inside a shared shell:

- A **page header** with the "Ice Operations" title and breadcrumb (Reports > Ice Operations).
- Header buttons: **Show Activity Feed** (toggles a "Recent activity" panel of the last few submissions — each row shows the operation type, the rink/machine, a "failed" badge if a circle check had failures, and the time), **Back**, and **Dashboard**.
- A **tab row** with one tab per **enabled** operation type. The tabs follow a fixed order: Ice Make, Circle Check, Edging, Blade Change.

If a needed prerequisite is missing, you'll see a friendly notice inside the shell instead of a form — e.g. "No rinks configured" (Ice Make needs at least one rink), "No machines configured" (no equipment of the right type), or "No checklist items" (Circle Check with no items and no template). These point you to ask an administrator.

Auto-timestamp & operator: you never type who you are or stamp the time by hand. Each form captures the current date/time when it loads (shown as "When it happened"), and the submission is recorded against **your** logged-in employee account. The confirmation screen shows "Submitted by" with your name.

Ice Make form

Fields: **Rink** (required), **Machine** (required — only resurfacers appear), **Water Used (gallons)**, **Machine Hours**, **Snow Taken (%)** (0-100), **Time On**, **Time Off**, and **Notes**. Submit button reads **Submit resurface**.

Edging form

Fields: **Machine** (required — only edgers appear), **Hours Run**, and **Notes**. Submit button reads **Submit edging report**.

Blade Change form

Fields: **Machine** (required — only blade sets appear), **Old Blade Hours**, **New Blade ID** (free text — serial number or ID), and **Notes**. The person who performed the change is recorded automatically as you (the logged-in user); there is no "who did it" dropdown. Submit button reads **Submit blade change**.

Circle Check form

1. Pick a **Machine** (required). Until you do, the checklist area says "Select a machine to view the checklist."
2. The app loads the right checklist:

- If the machine has a fuel type whose **template** exists, that template's fields are shown ("Using template: ...").
 - If the machine has **no** fuel type assigned but templates exist, a **Fuel type** dropdown appears so you can pick one for this check.
 - Otherwise, the generic **circle-check items** that apply to that machine's type are shown.
3. For each item, tap **Pass** (green) or **Fail** (red). When you mark an item **Fail**, a required "**What's wrong?**" note box appears beneath it.
 4. Add optional **General Notes**.
 5. **Submit circle check**. The button stays disabled until every item has been answered (a counter shows how many remain) and every failed item has a note — items are never recorded as passed by default.

A failed circle check is flagged on the confirmation screen, recorded with a failed-item count, and — if alerts are enabled — raises a manager alert listing the failed items.

Confirmation ("done") screen

After any submit you land on a **Submitted!** screen with a green checkmark, the operation label, and (for a circle check with failures) a "_N_ failed items" badge. Below it: when it happened, when it was submitted, the rink, the equipment, and who submitted it. Buttons: **Submit another** and **Back to home**.

6. Step-by-step: common tasks

Log an Ice Make (resurface)

1. Open **Ice Operations** → **Ice Make** tab.
2. Choose the **Rink** and the **Machine**.
3. Fill in water used, machine hours, snow %, and time on/off as appropriate (these are optional).
4. Add notes if needed → **Submit resurface**.

Run a Circle Check

1. Open **Ice Operations** → **Circle Check** tab.
2. Select the **Machine** (and a **Fuel type** if prompted).
3. Tap **Pass** or **Fail** on every checklist item (all must be answered); write a note for every **Fail**.
4. Add general notes if needed → **Submit circle check**. Failures notify your managers.

Log a Blade Change

1. Open the **Blade Change** tab → select the **Machine**.
2. Enter the **Old Blade Hours** and the **New Blade ID** → **Submit blade change**.

Enable or disable an operation type (admin)

1. Admin Center → **Ice Operations Admin** → **Settings**.
2. Under **Visible operations**, check the types you want staff to see (uncheck to hide). Leaving all unchecked shows every type.
3. **Save settings**. The staff tabs update accordingly.

Add a resurfer (or other machine) to equipment (admin)

1. Admin Center → **Ice Operations Admin** → **Setup**.

2. In the **Equipment** card's "Add equipment" form, enter a **Name**, choose the **Type** (e.g. Ice Resurfacer), optionally set model/serial/hours and a **Fuel type**, then **Add equipment**.
3. The machine now appears in the matching operation tab for staff.

Build a circle-check template (admin)

1. Setup tab → add a **Fuel type** (if you haven't).
2. In **Circle check templates**, add a template for that fuel type, then add its **fields** (typed in, or bulk-uploaded).
3. Assign that fuel type to the relevant resurfacers in the **Equipment** card so operators load the template automatically.

7. Field reference

Operation type	Field	Required	Notes
Ice Make	Rink	Yes	Active rinks only
	Machine	Yes	Ice Resurfacer equipment only
	Water Used (gallons)	No	Decimal
	Machine Hours	No	Decimal
	Snow Taken (%)	No	0–100
	Time On / Time Off	No	Time of day
	Notes	No	Free text
Edging	Machine	Yes	Edger equipment only
	Hours Run	No	Decimal
	Notes	No	Free text
Blade Change	Machine	Yes	Blade Set equipment only
	Old Blade Hours	No	Decimal
	New Blade ID	No	Free text (serial/ID)
	Performed by	(auto)	Recorded as the logged-in user
	Notes	No	Free text
Circle Check	Machine	Yes	Ice Resurfacer equipment only
	Fuel type	Only if machine has none	Loads the matching template
	Checklist items	Per template/config	Pass/Fail each
	Failure note ("What's wrong?")	Yes for each failed item	Blocks submit if missing
	General Notes	No	Free text

All four types also capture an automatic **occurred-at timestamp** and the **operator** (you).

8. Locking, saving & offline

- **Append-only / no editing.** Every submission is a permanent record. The form itself says "You can't edit this after submitting." There is no edit button — to correct something, submit a new entry. Admins can review entries and add **follow-up notes** from the History tab, but they cannot alter the original.
- **Offline.** All four forms work offline. If you submit while your device is offline, the report is **saved on this device** (you'll see a "Saved on this device" confirmation) and **queued**. When you reconnect, it syncs automatically and the exact same checks run server-side then — including the rink/machine and failed-note validation. Each queued item is tracked so a retry can't create a duplicate. You can keep working in the meantime; pending items also appear in your offline queue.
- **Validation is the same online and offline.** Required machine, a valid occurred-at time, and a note on every failed circle-check item are enforced either way.

9. Troubleshooting & FAQ

A tab I expected is missing. Your facility has that operation type unchecked in Settings → Visible operations. An admin can re-enable it.

"No machines configured" / "No rinks configured." No equipment of the needed type (or no rink, for Ice Make) has been set up yet. Ask an admin to add it in Setup.

"No checklist items" on Circle Check. No circle-check items and no template fields are configured. An admin adds items (or a template) in Setup.

The Submit button on a circle check is greyed out. You marked an item **Fail** but didn't write a note. Add a note in the "What's wrong?" box under each failed item.

The Circle Check shows no checklist after I pick a machine. The machine's fuel type may have no template yet, or no generic items apply to its type. Pick a fuel type if prompted, or ask an admin to configure items/templates.

I'm offline — will my report be lost? No. It's saved on your device and syncs when you're back online.

I made a mistake after submitting. You can't edit a submitted entry. Submit a corrected new entry; for context, an admin can add a follow-up note to the original in History.

"You don't have permission to submit." Your account doesn't have the Ice Operations `submit` permission. An admin grants it from Permissions.

Who got my circle-check failure alert? Failed circle checks raise a facility alert (if alerts are enabled in Settings) at the configured default severity, listing the failed items — your managers see it in the Communications/alerts area.

Source

- Staff: `src/app/reports/ice-operations/page.tsx`,
`src/app/reports/ice-operations/[operationType]/page.tsx`, `src/app/reports/ice-operations/[operationType]/_components/{ice-make-form,edging-form,blade-change-form,ice-ops-shell,offline-queued-card,use-offline-submit,shared}.tsx|ts`,
`src/app/reports/ice-operations/[operationType]/done/page.tsx`,
`src/app/reports/ice-operations/actions.ts`,
`src/app/reports/ice-operations/_lib/{submit,compute,offline}.ts`,
`src/app/reports/ice-operations/types.ts`.
- Admin: `src/app/admin/ice-operations/page.tsx`, `src/app/admin/ice-operations/_components/{setup-tab,settings-tab,history-tab,history-filters,submission-detail,seed-defaults-card,circle-check-import}.tsx|ts`, `src/app/admin/ice-operations/{actions,types}.ts`.
- Manifest: `docs/training/00-MANIFEST.md`.

Air Quality

Part of the **RinkReports** training package. RinkReports is a progressive web app (PWA) for ice-rink operations. Brand colors: **#4DFF00** (primary green) and **#002244** (navy). **You only see data for your own facility — this is automatic.** You never switch facilities or see another rink's air-quality readings; the app keeps each facility's records separate on its own.

1. What this module is for

The Air Quality module records the air-quality readings staff take around the rink — typically carbon monoxide (CO), nitrogen dioxide (NO₂), and carbon dioxide (CO₂), measured in parts per million (ppm) — and checks each reading against your facility's compliance thresholds the moment you type it.

What makes Air Quality different from a plain log:

- It is **threshold-aware and live**. As you enter a number, the form shows a colored badge — **Within range**, **Corrective action**, **Notification**, or **Evacuation** — based on the regulatory profile your administrator selected.
- It tracks a **regulatory jurisdiction**. The thresholds, the sampling cadence ("how often must we test this week?"), and the escalation instructions all come from a compliance profile, so the form reflects the rules your facility is actually held to.
- A reading that crosses a threshold is flagged as an **exceedance**, recorded on the report, and (if your facility has alerts turned on) raises an alert for managers.
- It is **append-only**. Once you submit, the report is a permanent record — you can't edit it, you submit a new reading instead. Admins add follow-up notes rather than changing the original.

Temperature is one of several reading types defined in the database (alongside CO, NO₂, etc.), each carrying its own unit and decimal precision. Because the readings list is metadata-driven, there is **no single global °F/°C toggle** here — each reading type simply shows the unit your administrator configured for it.

2. Who can use it

Access is **permission-driven**: what you can do depends on the per-module permissions an admin has set for you, not strictly on your job title. Two people with the same title can have different access if an admin customized it. The table below shows the typical default mapping.

Role tier	Typical Air Quality access
super_admin	Full access to everything, across the platform. ⚠ VERIFY — there is no separate "org_admin"; treat as super_admin.
org_admin	No access as a distinct tier — the live app has no org_admin role. ⚠ VERIFY.
facility_manager (the admin role)	Full module admin: Setup, Compliance, History, Settings, follow-up notes, exports. Requires the air_quality admin permission (or the super-admin flag).

Role tier	Typical Air Quality access
supervisor (the manager role, or a custom role)	Submit and view readings; configure only if granted the admin permission. ⚠ VERIFY — exact rights depend on the permission matrix, not a fixed tier.
staff (or a custom role)	Submit air-quality readings (needs the air_quality submit permission). No access to the admin Setup/Compliance/History/Settings screens.

Notes that hold true in the code regardless of title:

- To open the staff reading form at all, you need the **submit** permission for **air_quality**. If you don't have it, the page shows "No permission."
- If your account isn't linked to an active employee record, the page shows "Account not set up."
- If the facility hasn't configured any locations yet, the page shows "Not configured yet — talk to your administrator."
- The whole admin area (`/admin/air-quality`) requires facility-manager-level admin access; staff who browse to it are shown a "Forbidden" message.

3. How to get there

Staff (submitting a reading):

- In the left sidebar (or the mobile **Menu** tab), choose **Air Quality**. This opens the reading form at `/reports/air-quality`.
- The menu item only appears if your facility has the Air Quality module switched on and you have permission to use it.

Admins (configuring the module):

- Open the **Admin Center** (the extra link in the staff sidebar, visible to admins), then under **Module Admin** choose **Air Quality Admin** (`/admin/air-quality`).
- The admin screen has four tabs across the top: **Setup**, **Compliance**, **History**, **Settings**.

4. Setup & configuration (admins)

All configuration lives in the **Air Quality Admin** screen and is a facility-manager task. See the **Admin Control Center** chapter for cross-cutting setup (Facility Spaces, Modules on/off, Permissions, Exports).

First-time setup: Seed defaults

If nothing has been configured yet, the Setup tab shows a **Seed defaults** card. Clicking **Seed defaults** creates the standard ice-rink reading types — **CO**, **NO₂**, and **CO₂** — so you have a working form in one click. You can edit or add to them afterward.

Setup tab — Equipment & Reading types

The Setup tab has two halves:

- **Locations** come from your shared **Facility Spaces** list (the same locations used by Incident and Accident reports). You don't create them here — you add, rename, or bulk-import them under **Facility Spaces** (`/admin/spaces`), and they appear here automatically. On this tab you click a location only to scope equipment to it.

- **Equipment** — monitors/sensors. You can add equipment that is **facility-wide** (not tied to a location) or **scoped to a specific location**. For each you can set a name, an optional slug, an optional model and serial number, and a sort order. Each item can be **Edited**, **Deactivated/Activated**, or **Deleted**. Equipment scoped to a location only appears as a choice on the reading form when that location is selected.
- **Reading types** — what staff actually measure (e.g. CO, NO₂, CO₂, temperature). For each reading type you set:
 - **Label** (e.g. "CO") and an optional **Key** (auto-generated from the label if blank).
 - **Unit** (e.g. ppm).
 - **Decimals** (0-6) — how many decimal places the input accepts.
 - **Required on reports** — whether staff must fill this reading in before they can submit.
 - **Sort order** — reading types can also be reordered with the ↑ / ↓ buttons.
 - Reading types can be **bulk-uploaded** and can be Edited / Deactivated / Deleted.
 - **Threshold tiers are not set here** — they come from the facility's compliance profile on the Compliance tab.

Compliance tab — profile, metrics, thresholds, escalation, roles

This is where the form's "live feedback" gets its rules.

- **Compliance profile (jurisdiction)**. Pick the regulatory profile for the facility from a dropdown. The reading form, the threshold tiers, the sampling cadence, and the escalation steps all derive from it. A badge shows whether it is **Binding regulation** or **Guidance**, and whether it uses the **Single-sample** or **1-hour TWA** measurement method.
- **Metrics tracked**. Check which of the profile's metrics (CO, NO₂, ...) are active for your facility.
- **Threshold overrides — tighten only**. Each metric shows the profile's regulatory ceiling for each tier (**Corrective**, **Notification**, **Evacuation**). You may enter a stricter (lower) value to hold your facility to a tighter standard, or leave a field blank to use the regulatory ceiling. **You can tighten a threshold but you can never loosen it**. The screen states this plainly ("A value must be at or below the floor"), and the app **enforces it**: if you enter an override above the regulatory floor, the save is rejected with a message like "...override is looser than the regulatory floor...; overrides may only tighten." Internally the engine also clamps any too-loose value back to the regulatory ceiling, so a looser number can never take effect even by accident.
- **Escalation steps**. For each tier you can type the contacts/actions operators should take (e.g. who to call at the Notification level). This text is shown in the reading form when a reading reaches that tier. Leave blank to use the built-in default guidance.
- **Role access**. Optional comma-separated role keys for "May submit readings" and "May view logs." This is **advisory only** — actual access is still governed by each user's module permissions, not by what you type here.
- **Save compliance profile** writes all of the above at once.

Below the profile panel, a **Compliance rules** section lists the facility's stored rules grouped by jurisdiction (these back the engine and any "sustained exceedance" evacuation criteria).

Settings tab

One row of facility-wide settings:

- **Testing frequency** — free text guidance shown to staff on the reading form (e.g. "CO every 2 hours during sessions").

- **Default jurisdiction** — pre-fills the jurisdiction field when adding a compliance rule (offers existing jurisdictions as suggestions).
- **Enable air quality alerts** — master on/off switch for the alerts raised when a reading exceeds a threshold.
- **Default alert severity** — warn, high, or critical; used when a triggered threshold has no severity of its own.

History tab (review)

A filterable, read-only list of submitted reports with a drill-down detail panel. Admins **cannot edit** a submitted report; they can **add follow-up notes**, which are themselves append-only ("Notes are append-only and cannot be edited or deleted. The original report stays intact"). Filters include date range, employee, location, equipment, reading type, and whether the report had an exceedance.

5. Screen-by-screen walkthrough

Staff reading form (/reports/air-quality)

From top to bottom, what you see:

1. **Page header** — breadcrumb (Reports → Air Quality), the eyebrow "Air quality reading," the title, and a reminder that after you submit, the report can't be edited.
2. **Location** (required) — a dropdown of your facility's spaces. Pick where the readings were taken. You can't submit until you choose one. Changing the location clears any equipment choice that doesn't belong to the new location.
3. **Compliance context panel** (shown when a compliance profile is configured) — displays the profile's name, a **Binding** or **Guidance** badge, and a **1-hr TWA** or **Single sample** method badge, plus any guidance note. It also includes:
 - A **frequency tracker**: a badge reading **On schedule** or **Behind by N**, and a line like "2 of 3 this week · weekend 0/1 (weekend sample needed)" — telling you how many readings are still required this period.
 - A **Reading type** selector — **Routine**, **Post-resurfacing**, or **Post-edging**. (This labels why the reading was taken and drives the frequency tracking; it is distinct from the "Reading Kind" / corrective concept below.)
4. **Equipment** (optional) — appears when monitors are configured for the selected location (or facility-wide). Pick the monitor you used.
5. **Readings** — one input per active reading type, in the admin-defined order. Each shows its **label**, its **unit** beside the box, and a **required** marker (*) where applicable. The numeric keypad opens on mobile, and the input respects the configured decimal precision.
 - For any reading tied to a tracked metric, a small grey hint shows the tier ceilings (e.g. "Corrective > 35 · Notification > 100 ppm"), and once you type a number a **live range badge** appears: **Within range** (green), **Corrective action** (amber), **Notification** (red), or **Evacuation** (red). This is the threshold-aware feedback — it updates as you type.
6. **1-hour TWA calculator** (only when the profile uses the TWA method) — a collapsible helper to enter the required samples (e.g. 13 readings, one every 5 minutes); it averages them and fills the reading for you when you click **Use average**.
7. **Compliance banner + corrective-action note** — if any reading reaches Corrective level or higher, a colored banner appears with the escalation instructions for that level, **and a required**

"Corrective action taken" note box appears. You cannot submit an over-threshold reading until you describe the corrective steps you took. (This is the "corrective reading kind" — an exceedance forces a corrective note.)

8. **Notes (optional)** — anything to flag for the manager.
9. **Monitoring log sections** (collapsible, optional) — supplementary record-keeping: **Equipment & tester info** (date of test, tester certification, CO/NO₂ monitor type/model/calibration date — with a warning if a calibration is over a year old, ventilation last-inspection date); **Section 1: General information & equipment status** (arena operating status, ice resurfacers and other fuel-burning equipment with fuel types, ventilation status, last-maintenance notes); **Section 2: Air quality measurements** (routine and post-edging measurement rows with location/time/CO/NO₂/temp/note); **Section 4: Additional recommendations & notes** (electric-equipment consideration, staff-trained and public-signage checkboxes, unusual observations).
10. **Submit** — the button reads **"Submit readings for [location]"** when online, or **"Save offline"** when offline. It stays disabled until a location is chosen, every required reading is filled, and any required corrective note is written; helper text below explains what's still missing.

Confirmation screen (/reports/air-quality/done)

A green checkmark and **"Submitted!"**, the location name and timestamp, a chip showing how many **readings were recorded**, and — if anything crossed a threshold — a chip showing the **exceedance count and severity**. Buttons: **Submit another** and **Back to home**.

Admin tabs (/admin/air-quality)

Setup, Compliance, History, Settings — covered in §4. The History detail panel shows the submitter's notes, each reading with its at-submit verdict, and an append-only follow-up-note thread; an **Export** action is available from the page header.

6. Step-by-step: common tasks

Record an air-quality reading (staff)

1. Open **Air Quality** from the sidebar/menu.
2. Choose the **Location**.
3. If a compliance panel appears, glance at the **frequency tracker** to see whether more readings are due this period, and set the **Reading type** (Routine / Post-resurfacing / Post-edging).
4. Optionally pick the **Equipment** (monitor) you used.
5. Enter each reading. Watch the **live range badge** — green means within range.
6. If a badge turns amber or red, read the **escalation banner**, take the action it describes, then fill in the required **Corrective action taken** note.
7. Add optional **Notes** (and any monitoring-log details your facility wants).
8. Tap **Submit readings**. You land on the **Submitted!** screen showing the reading and exceedance counts.

Tighten a threshold (admin)

1. Go to **Air Quality Admin → Compliance**.
2. Make sure the correct **jurisdiction profile** is selected.
3. Under **Threshold overrides (tighten only)**, find the metric and tier, and type a value **at or below** the shown regulatory ceiling. (Leaving it blank keeps the regulatory ceiling.)

- Click **Save compliance profile**. If you accidentally enter a value above the floor, the save is rejected — you can only tighten.

Add a reading type (admin)

- Go to **Air Quality Admin → Setup**.
- In the **Reading types** card, use the **Add reading type** form: enter a **Label**, optional **Key**, **Unit**, **Decimals**, and tick **Required on reports** if staff must always fill it.
- Click **Add reading type**. Reorder with ↑ / ↓ if needed. (Bulk upload is available for many at once.)

Set up compliance escalation (admin)

- Go to **Air Quality Admin → Compliance**.
- Select the **jurisdiction profile**.
- Under **Escalation steps**, type the contacts and actions for each tier — **Corrective**, **Notification**, **Evacuation**. (Blank fields fall back to built-in default guidance.)
- Click **Save compliance profile**. This text now appears in the staff reading form's banner whenever a reading reaches that tier.

7. Field reference

Staff reading form

Field	Type	Required	Notes
Location	Dropdown (facility spaces)	Yes	Must be chosen before you can submit.
Reading type	Dropdown: Routine / Post-resurfacing / Post-edging	Shown with a compliance profile	Labels why the reading was taken; drives frequency tracking.
Equipment	Dropdown	No	Only monitors valid for the chosen location (or facility-wide).
Each reading (e.g. CO, NO ₂ , CO ₂ , Temp)	Numeric, decimal	Per its "Required" flag	Unit and decimal precision come from the reading type. Shows tier-ceiling hint + live range badge.
Live range badge	Computed	—	Within range / Corrective action / Notification / Evacuation, per the effective (override-tightened) thresholds.
1-hour TWA calculator	Sample entry helper	No	Only for profiles using the TWA method; averages samples into the reading.
Corrective action taken	Text	Yes, when any reading is over threshold	You cannot submit an exceedance without it.
Notes	Text	No	For the manager.

Field	Type	Required	Notes
Date of test / tester certification / monitor info / sections 1, 2, 4	Mixed (collapsible)	No	Supplementary monitoring-log detail.

Thresholds shown (per metric, from the compliance profile):

Tier	Meaning on the form
Within range	Reading is at or below the corrective ceiling — green badge.
Corrective	Reading exceeds the corrective ceiling — amber badge; corrective note required.
Notification	Reading exceeds the notification ceiling — red badge; notify per escalation steps.
Evacuation	Reading exceeds the evacuation ceiling — red badge; follow evacuation protocol.

Admins set each tier's ceiling per metric on the Compliance tab. The **effective** ceiling is the regulatory floor, optionally **tightened** (lowered) by a facility override — never raised.

8. Locking, saving & offline

- **Append-only.** A submitted air-quality report is permanent. You cannot edit it — submit a new reading instead. Admins don't edit submissions either; they add **follow-up notes**, which are themselves append-only.
- **Exceedances.** On submit, the server re-evaluates every reading against the facility's effective thresholds (the same engine the form previews with). Readings over threshold are flagged as **exceedances**, the report records the highest severity, and — if alerts are enabled in Settings — an alert is raised for managers. A required corrective-action note must accompany any exceedance. A separate "sustained exceedance" rule can raise a critical, evacuation-level alert when readings stay high over time.
- **Offline.** Air Quality works offline. If you're offline when you submit, the button reads **Save offline**, the readings are queued on your device, and you see a "Saved on this device" confirmation. When you reconnect, the queued reading submits automatically and the **same exceedance checks run then** — so an over-threshold reading is flagged on sync, not silently accepted. Each queued item has a unique id, so a retry can't create a duplicate. (See the **Offline** chapter for the queue and pending/failed indicators.)

9. Troubleshooting & FAQ

The Air Quality menu item isn't showing. Either your facility has the module switched off (an admin controls this on the Modules page) or you don't have the `air_quality` view/submit permission. Ask your administrator.

I opened the form and it says "No permission" / "Account not set up" / "Not configured yet."
 "No permission" = you lack the submit permission. "Account not set up" = your login isn't linked to an active employee record. "Not configured yet" = the facility has no locations yet (an admin adds them in Facility Spaces).

The Submit button is greyed out. You must choose a **Location**, fill in **every required reading**, and — if any reading is over threshold — write the **Corrective action taken** note. The helper text under the button tells you which one is missing.

A reading turned red and now there's a note box I can't skip. That reading exceeded a threshold. Take the action in the escalation banner, then describe what you did in the **Corrective action taken** box. This is required before an over-threshold reading can be saved.

I can't raise a threshold above the regulatory number (admin). That's by design — **thresholds can be tightened but never loosened**. Enter a value at or below the regulatory ceiling, or leave it blank to use that ceiling.

Why is there no °F/°C switch? Temperature is just one of several database-defined reading types, each with its own configured unit. There's no single global temperature toggle; the unit shown next to each reading is whatever the admin set for that reading type.

Can I fix a reading I just submitted? No — reports are append-only. Submit a corrected new reading; an admin can add a follow-up note to the original explaining the correction.

Will my reading still get checked if I submit it offline? Yes. It's queued on your device and re-evaluated against the thresholds when it syncs, so exceedances are still flagged.

Source

Staff: `src/app/reports/air-quality/page.tsx`,
`src/app/reports/air-quality/_components/submission-form.tsx`,
`src/app/reports/air-quality/actions.ts`, `src/app/reports/air-quality/done/page.tsx`,
`src/app/reports/air-quality/types.ts`, `src/app/reports/air-quality/_lib/{compliance.ts,submit.ts,compute.ts,load-compliance.ts,sustained.ts}`. Admin: `src/app/admin/air-quality/page.tsx`,
`src/app/admin/air-quality/types.ts`, `src/app/admin/air-quality/actions.ts`, `src/app/admin/air-quality/_components/{setup-tab.tsx,compliance-tab.tsx,compliance-profile-panel.tsx,settings-tab.tsx,history-tab.tsx,report-detail.tsx,seed-defaults-card.tsx}`. Permission/auth:
`src/lib/permissions/check.ts`, `src/lib/auth`. Threshold tighten-only enforcement: `validateOverrides` and `effectiveMetricTiers` in `_lib/compliance.ts`, called from `saveComplianceProfileConfig` in `src/app/admin/air-quality/actions.ts`.

Ice Depth

1. What this module is for

Ice Depth is how RinkReports records **how thick the ice is** across a sheet — point by point, against a rink diagram, so you can see at a glance where the ice is at target, too thin, or too thick.

It is built for the way the job is actually done: you walk the rink with a depth gauge (manual or a Bluetooth caliper), tap each measurement point on a digital rink diagram, and enter the reading. Each point lights up in color the instant you enter a value — green when the depth is in range, red when it's below the minimum, amber when it's above target — so problem spots are obvious before you even finish.

This module measures **depth, not temperature**. There is no °F/°C toggle anywhere in it. Readings are recorded in the facility's chosen unit (inches or millimeters).

A few things this module deliberately does **not** do:

- **No photo documentation.** There is no camera, attachment, or image-upload step. The "documentation" is the annotated diagram and the per-point readings.
- It is a **two-phase flow** — first you measure, then you review and submit — not a single long form.

Once you submit, the session becomes a permanent, read-only record with a printable diagram, a PDF, and the option to send it to your facility's distribution list.

You only see data for your own facility — this is automatic.

2. Who can use it

Access is permission-driven, not strictly tied to a job title. The table below shows the typical default for each tier; an admin can grant or remove any of these per person.

Tier	Submit a depth session	Review history / analytics	Configure rinks, diagrams, points, settings
super_admin	Yes	Yes	Yes (plus the only tier that can permanently delete a submitted session)
org_admin	⚠ VERIFY — no separate org-admin tier exists in the app; treat as super_admin	⚠ VERIFY	⚠ VERIFY
facility_manager (admin)	Yes	Yes (Admin → Ice Depth)	Yes (Admin → Ice Depth)
supervisor (manager or custom role)	Yes, if granted the Ice Depth submit permission	Only if granted Ice Depth admin access	Only if granted Ice Depth admin access
staff	Yes, if granted the Ice Depth submit permission	No access	No access

Notes:

- To **submit** a depth session you need the Ice Depth submit permission. Without it, the page shows a "No permission" message instead of the form.

- The whole **Admin → Ice Depth** area (Rinks, Diagrams, History, Analytics, Settings) requires facility-admin access for the Ice Depth module.
- **Deleting** a submitted session is restricted to super_admin — the Delete button only appears for them, and the server re-checks. Everyone else, sessions are immutable.
- A deactivated or not-yet-set-up account is denied everywhere, with an "Account not set up" message.

3. How to get there

Staff (to submit a reading):

- In the staff app, open **Ice Depth** from the left sidebar (or the **Menu** tab on mobile). The menu item only appears if your facility has the Ice Depth module turned on and you have access.
- Opening Ice Depth takes you straight to a rink diagram — the app automatically lands you on your facility's **default rink and its default diagram**. If your facility has more than one rink or diagram, a **Rink** and **Diagram** picker appears at the top so you can switch.

Admins (to configure or review):

- Open the **Admin Center** (the "Admin Center" link in the staff sidebar), then choose **Ice Depth Admin** under Module Admin. This opens the tabbed admin screen at `/admin/ice-depth`.

4. Setup & configuration (admins)

All Ice Depth setup lives under **Admin → Ice Depth**, which has five tabs: **Rinks, Diagrams, History, Analytics, Settings**. Everything here applies only to your own facility. See the **Admin Control Center** chapter for how admin access, modules, and permissions are managed.

If Ice Depth has never been set up, the Diagrams and Settings tabs show a **"Seed defaults"** card to create the initial settings/rink/diagram for you to start from.

Rinks tab

A **rink** is a sheet of ice. Create one with **New rink** (name, optional slug, sort order). For each rink you can:

- **Rename**, change **sort order**.
- **Make default** — the rink staff land on first (only an active rink can be the default).
- **Deactivate / Activate** — hide a rink from staff without deleting it.
- **Delete** — only allowed once the rink has no diagrams; otherwise you're told to move or delete its diagrams first.

The **first rink you create automatically becomes the facility default**.

Diagrams tab (the diagram + measurement points)

A **diagram** (called a "layout" internally) belongs to a rink and defines the rink picture plus where the measurement points sit. There is a cap of **8 active diagrams** per facility (shown as "X / 8 active").

- **New diagram** — choose its rink (required), a name, optional slug/description, and an **aspect ratio** (width ÷ height; default 0.425, roughly a vertical NHL rink). You must create a rink first.
- Selecting a diagram opens the **point-placement editor**:
 - A **USA Hockey rink** drawing fills the panel. An optional **Logo URL** can be set per diagram (square PNG/SVG with transparent background) and renders at center ice.

- A mode toolbar with three modes:
 - **Place** — click anywhere on the rink to drop a new numbered point.
 - **Select** — click a point to open its editor (set a **Label**, fine-tune **X/Y** position as 0-1 fractions, move up/down in order, **Deactivate/Activate**, or **Delete**).
 - **Drag** — drag a point to reposition it; release to save.
- **Renumber 1..N** compacts the active points' numbers into order.
- A cap of **60 active points** per diagram is enforced.
- Per diagram you can also **rename**, set **description / logo / aspect ratio / sort order**, **Make default** (the diagram staff land on for that rink), **Deactivate/Activate**, or **Delete** (deleting cascades the points; if sessions reference it, deactivate instead). Deleting a point keeps the snapshots on any existing measurements.

Settings tab

One row per facility controls how depths are classified and displayed:

- **Measurement unit** — inches or mm. (⚠ VERIFY — the training brief said "in/cm," but the app's unit options are **inches** and **mm**, not centimeters.)
- **Low threshold** — at or below this value, a reading is flagged **low (below min)**.
- **High threshold** — above this value, a reading is flagged **high (above target)**.
- **Low / OK / High colors** — the color each severity shows in admin views (History detail, Analytics).
- **Enable alerts on submitted sessions, Alert on** (low / high / any), and **Default alert severity** — when on, a submitted session that contains the chosen severities can trigger a notification to your facility's configured recipients.

Existing sessions **snapshot** the unit and thresholds at the moment they were submitted. Changing settings later does **not** reclassify old history.

5. Screen-by-screen walkthrough

Layout (rink + diagram) picker

Opening Ice Depth redirects you onto the default rink's default diagram. If your facility has more than one rink or more than one diagram, a top bar shows a **Rink** dropdown and a **Diagram** dropdown. Switching either jumps you to that diagram's measure screen. A **sync chip** (online/offline indicator) sits in the header, and a back arrow returns to Ice Depth.

Measure phase (the interactive rink)

This is the first of two phases. The screen shows:

- A point-progress line — "**Point X of Y**" while you're entering, or "**Y points — tap to enter**" before you start — and a "**X recorded**" counter, plus a green progress bar.
- A collapsible "**Using a Bluetooth caliper?**" helper (see §6).
- The **USA Hockey rink diagram** with a numbered **chip at every measurement point**.

To record a point:

1. **Tap a chip.** A small **popover** opens anchored to that point, showing the point number, its label, a number field (with the facility's unit beside it), and **Skip / Save & Next** buttons.
2. **Type the depth** (or let a Bluetooth caliper type it). The field only accepts digits and a single decimal point, up to 3 decimal places.

3. As you type, the popover border, the point chip, and a small label update **live by severity** — **Optimal** (green), **Below min** (red), or **Above target** (amber).
4. **Press Enter** (or Tab) to save and jump to the next point. On the last point the button reads **"Save & Review"** and takes you to the review screen. **Esc** closes the popover without advancing. **Skip** moves to the next point without recording this one.

You can revisit any point at any time by tapping its chip again — entering a new value overwrites the old one. When at least one point has a value, the big **"Review & Submit"** button at the bottom becomes active and jumps you to the review phase. (Nothing is saved to the server during the measure phase — it's all on your screen until you submit.)

Review & Submit phase

"Step 3 of 3." This is your last chance to check the readings before they become a permanent record:

- A **summary card** with a small annotated rink thumbnail, the **average depth**, and **severity pills**: "N optimal," "N thick," "N below min," and "N skipped" as applicable.
- A **per-point list** — each point's number (colored by severity), its label, its status (Optimal / Below min / Above target / Not recorded), and the recorded value.
- A **Notes (optional)** box.
- A green **Submit** button. Its label reflects completeness — **"Submit Reading"** when every point is recorded, or **"Submit (X of Y recorded)"** otherwise. (When offline it reads **"Save Offline."**)
- A **"← Back to measure"** link to return and adjust before finalizing.

Done / confirmation screen

After submitting you land on the confirmation screen for that session:

- A green checkmark and a **Submitted** badge, the diagram name, and the submitted timestamp.
- **Stat pills**: Optimal, Below min, Above target, and Total measurements.
- The **annotated rink diagram** with every measured point colored by severity, plus any submitter **Notes**.
- Action buttons:
 - **Download PDF** — downloads a PDF of the session report.
 - **Print Diagram** — opens the browser print dialog; print styling hides the app chrome so only the rink diagram (with name and timestamp) prints, full-page on US Letter portrait.
 - **Send Report** — emails/sends the report to your facility's Ice Depth recipients (see §6). Ice Depth does **not** auto-send on submit; sending stays under your control.
 - **Submit Another** — returns to Ice Depth to start a new session.
 - **Back to Dashboard**.
 - **Back to Form** (top) returns to the measure screen for that diagram.

Admin tabs (Admin → Ice Depth)

- **Rinks** — manage sheets of ice (see §4).
- **Diagrams** — draw diagrams and place/label measurement points (see §4).
- **History** — a filterable list of submitted sessions (filter by diagram, employee, date range, and whether the session had low/high readings, with "Load more" paging). Click a session to open a **detail panel**: the annotated diagram at its snapshot severities, submitter info, unit and thresholds as snapshotted, low/high badges, submitter notes, and an **append-only follow-up notes** thread you can add to. Super admins also see a **Delete session** button (with a confirmation dialog).

- **Analytics** — per-diagram trends over a date range: summary stats (sessions, readings, average depth, % below min, % above target), a "**Problem spots**" heat-map of the rink colored by each point's most common condition, a **per-point breakdown** table (worst-first), and a **daily activity** strip.
- **Settings** — unit, thresholds, colors, alerting (see §4).

6. Step-by-step: common tasks

Measure ice depth across the rink

1. Open **Ice Depth**. Pick the **Rink** and **Diagram** at the top if prompted.
2. **Tap the first point** on the rink diagram.
3. **Type the depth** and press **Enter** to save and advance to the next point. Use **Skip** to leave a point unrecorded.
4. Watch each chip color in: green = optimal, red = below min, amber = above target.
5. When done (or after recording at least one point), tap **Review & Submit**.
6. On the review screen, check the per-point list and average, add **Notes** if needed, then tap **Submit Reading**.

Use a Bluetooth caliper

RinkReports doesn't pair the caliper for you — calipers in this class (e.g. iGaging Absolute Origin) connect at the OS level as a **Bluetooth keyboard**.

1. **Pair once** in your phone/tablet's Bluetooth settings: hold the caliper's **DATA** button until it appears (it shows up as a keyboard), then connect.
2. Set the caliper to the **same unit** as the form (shown in the in-app helper).
3. In the form, **tap a point** to open its popover, then press the caliper's **DATA** button — the reading fills in and the form jumps to the next point automatically. Keep pressing DATA to walk the whole rink hands-free.

Tip: the in-app "**Using a Bluetooth caliper?**" panel above the rink has these steps. The caliper's Enter (and Tab) terminator both save and advance, just like pressing Enter yourself.

Create a rink layout (admin)

1. **Admin → Ice Depth → Rinks → New rink**. Name it and save. (The first rink becomes the default.)
2. Go to the **Diagrams** tab → **New diagram**, pick the rink, name it, set the aspect ratio, and create.
3. Open the new diagram. In **Place** mode, **click the rink** to drop each measurement point.
4. Switch to **Select** mode to **label** points and fine-tune positions; use **Drag** mode to nudge them. Use **Make default** so staff land on this diagram for that rink.
5. Confirm in **Settings** that the unit and low/high thresholds are right for this facility.

Download, print, or send a report

On a session's **Done** screen (or reach a past session through Admin → History):

- **Download PDF** to save/share a PDF copy.
- **Print Diagram** to print the annotated rink (the app chrome is hidden automatically).
- **Send Report** to push it to the facility's configured Ice Depth recipients. If none are set up, you'll see "No recipients are configured for ice depth — set up a send list in Admin → Communications." The button confirms how many recipients it reached.

7. Field reference

Field	Where	What it is
Rink	Layout picker	Which sheet of ice you're measuring (only if more than one exists).
Diagram	Layout picker	Which point layout to use for this rink (only if more than one exists).
Measurement point	Measure phase	A numbered, optionally labeled spot on the rink. Tap to enter its depth.
Depth value	Point popover	The reading, in the facility's unit (inches or mm). Digits + one decimal point, up to 3 decimals; must be zero or positive.
Notes (optional)	Review phase	Free text for anything worth flagging (conditions, equipment, resurface schedule).
Severity (auto)	Everywhere	Computed from the reading vs. thresholds: Optimal (in range), Below min ($\leq$ low threshold), Above target ($>$ high threshold). Not editable.
Average depth (auto)	Review / Done	Average of all recorded readings in the session.
Stat pills (auto)	Review / Done	Counts of optimal / below-min / above-target / skipped / total points.
Admin — Settings		
Measurement unit	Settings	inches or mm.
Low / High threshold	Settings	Boundaries that classify each reading's severity.
Low / OK / High color	Settings	Colors used in admin History and Analytics.
Enable alerts / Alert on / Default alert severity	Settings	Whether and when a submitted session triggers a notification.
Admin — Rink	Name, slug, sort order, default, active	Defines a sheet of ice.
Admin — Diagram	Name, slug, description, rink, logo URL, aspect ratio, sort order, default, active	Defines a rink picture and its point set.
Admin — Point	Label, X (0-1), Y (0-1), order, active	A single measurement location on a diagram.

8. Locking, saving & offline

Adjust freely before you finalize. During the **measure** phase, nothing is sent to the server — every reading lives on your screen. You can re-tap any point to change its value, skip points, and bounce between **measure** and **review** ("← Back to measure") as many times as you like. The only thing that commits a session is the **Submit** button on the review screen.

A submitted session is immutable. Once submitted it becomes a permanent, read-only record. There is no "edit this session." If something was wrong, you record a **new** session. Admins don't edit submitted sessions in place — on the History detail panel they can add **append-only follow-up notes**, and that's it. The only way a session is removed is a **super-admin hard delete**, which is gated behind a confirmation dialog and cannot be undone.

Snapshots protect history. Each session stores the unit, thresholds, and each point's number/label/position at submit time. Editing a diagram or changing Settings afterward never changes how past sessions look or are classified — deleted points even keep their snapshots on old measurements.

Offline. Ice Depth works offline:

- If you're offline when you submit, the button reads **"Save Offline"** (or "Save on this device"), and the session is **queued on your device**. You'll see a "Saved on this device" confirmation; the session submits automatically when you reconnect, and the same severity checks and rules run then.
- A queued session is replayed exactly once (it can't create a duplicate) and lands the same record an online submit would.
- ⚠ VERIFY: if your device can't queue offline at all (no service worker active), the app blocks the submit and tells you plainly to reconnect or reload — your typed readings stay on screen rather than being lost.
- Note: offline covers **submitting** a session. Browsing previously submitted sessions or the Done page requires a connection.

9. Troubleshooting & FAQ

"No permission" when I open Ice Depth. Your account doesn't have the Ice Depth submit permission. Ask an admin to grant it (Admin → Permissions).

"Account not set up." Your login isn't linked to an active employee record yet. Contact your administrator.

"Not configured" / "No diagrams." Your facility hasn't created a rink and diagram yet, or no diagram is assigned to your rinks. An admin sets these up under Admin → Ice Depth → Rinks and Diagrams.

A point chip won't turn green even with a value in it. Green means "Optimal." If the chip is red or amber, the reading is below the low threshold or above the high threshold for your facility — that's expected behavior flagging a thin or thick spot, not an error.

My Bluetooth caliper types into the wrong place / nothing happens. Make sure you've **tapped a point first** so its popover (the input) is focused, then press the caliper's DATA button. The caliper must be paired as a keyboard in your device's Bluetooth settings, and set to the same unit as the form.

I entered the wrong depth. Before submitting, just tap that point again and type the correct value, or use "← Back to measure" from the review screen. After submitting, the session is locked — record a new one.

Can I add a photo? No. Ice Depth has no photo or attachment feature by design; the record is the annotated diagram and the readings.

Why is there no °F/°C switch like Refrigeration has? Because this module measures **depth**, not temperature. The unit (inches or mm) is set once by an admin in Settings.

"Send Report" says no recipients. No distribution list is configured for Ice Depth. An admin sets one up under Admin → Communications.

Do changes to thresholds change my old reports? No. Each session snapshots its unit and thresholds at submit time; later settings changes never reclassify history.

Why does the average / counts look off vs. what I measured? Skipped points aren't counted in the average or severity totals — only recorded readings are. The summary shows how many points were skipped.

Source

- Staff flow: `src/app/reports/ice-depth/page.tsx`, `src/app/reports/ice-depth/[layoutSlug]/page.tsx`, `src/app/reports/ice-depth/_components/{submission-form.tsx,diagram-nav.tsx,sync-chip.tsx}`, `src/app/reports/ice-depth/actions.ts`, `src/app/reports/ice-depth/_lib/{compute.ts,submit.ts,offline.ts}`, `src/app/reports/ice-depth/types.ts`
- Done / PDF / send / print: `src/app/reports/ice-depth/[layoutSlug]/done/page.tsx`, `src/app/reports/ice-depth/[layoutSlug]/done/_components/{send-report-button.tsx,print-diagram-button.tsx}`, `src/app/reports/ice-depth/[layoutSlug]/done/pdf/route.ts`
- Admin: `src/app/admin/ice-depth/page.tsx`, `src/app/admin/ice-depth/_components/{rinks-tab.tsx,layouts-tab.tsx,layout-editor.tsx,history-tab.tsx,session-detail.tsx,analytics-tab.tsx,settings-tab.tsx,seed-defaults-card.tsx}`, `src/app/admin/ice-depth/{actions.ts,types.ts}`, `src/app/admin/ice-depth/_lib/analytics.ts`
- Shared rink rendering: `src/components/ice-depth/usa-rink.tsx`, `src/components/ice-depth/rink-geometry.ts`

Employee Scheduling

Brand: RinkReports. Primary green **#4DFF00**, navy **#002244**. **You only see data for your own facility — this is automatic.** You never switch facilities or see another rink's schedule; RinkReports keeps each facility's records separate for you.

1. What this module is for

Employee Scheduling is how a facility **plans who works which shift** and how staff **manage their own availability and requests**. It has two sides:

- **The admin scheduling grid** — managers build a week of shifts by dragging on a calendar, assign people to them, and then publish the schedule so staff can see it.
- **The staff scheduling app** — each employee sees their upcoming shifts, sets weekly availability, requests time off, picks up open shifts, and asks to swap shifts with a coworker.

Important — there is no payroll, timekeeping, or clock-in/out here. Scheduling is **assignment only**: it says who is scheduled to work when. It does **not** record hours actually worked, clock anyone in or out, produce timesheets, or calculate pay. The one pay-adjacent number, "**max weekly hours**," is purely an **overtime/scheduling warning threshold** — when an assignment would push someone over it, the grid shows a warning. It changes nothing about pay.

The module also runs **scheduling rule checks** as you assign people: required-certification gaps, double-booking, approved time-off overlap, minor hour caps, overtime thresholds, and required breaks. Most are advisory warnings; a missing required certification is a hard block (see §8).

2. Who can use it

Access is **permission-driven**, not a fixed rank ladder. The table below is the typical default mapping; because an admin can customize any person's permissions per module, two people with the same title can differ. ⚠ **VERIFY** any place a tier label feels too rigid for a specific facility.

Tier (doc vocabulary)	Real app role	Scheduling access (default)
super_admin	super_admin (platform-wide)	Full admin grid + all approvals across the system; sees everything an org/facility admin sees.
org_admin	(no exact equivalent — ⚠ VERIFY ; treat as super_admin)	No separate tier exists in RinkReports.
facility_manager	admin (facility administrator)	Full admin grid: build/assign/publish shifts, approve publish requests, decide time-off and swaps, configure settings and job areas. Needs the admin permission for this facility.
supervisor	manager (or a custom role)	Mid-tier. Whatever the admin grid permission grants them; commonly review/decision rights. ⚠ VERIFY per facility — gated by permissions, not title.

Tier (doc vocabulary)	Real app role	Scheduling access (default)
staff	staff (or a custom role such as driver)	Staff scheduling app only (my shifts, availability, time off, claim open shifts, request/accept swaps). Needs the scheduling view permission; submitting requests needs submit. No access to the admin grid.

- The **admin grid** (/admin/scheduling) requires the **admin** permission (or **super_admin**). Non-admins who try are shown a **Forbidden** message.
- The **staff app** (/reports/scheduling) requires an active employee account with the scheduling view permission. Without **view**, the page shows "You don't have access to scheduling yet." Submitting availability or time off additionally needs **submit**; if you only have **view**, you can see your schedule but not file requests.
- A **deactivated** account is denied everywhere.

3. How to get there

Staff side — /reports/scheduling ("Scheduling" in the staff menu / mobile bottom tabs):

- The landing screen shows a **Next shift** hero, a 7-day week strip, your **Upcoming** shifts, any **Open · Pick up** shifts you can claim, **Your claims · Awaiting approval**, and **Quick links** to My schedule, Time off, Availability, Shift swaps, and Notifications.

Admin side — /admin/scheduling (Admin Center → Module Admin → **Scheduling Admin**):

- A sticky sub-navigation runs across the top: **Overview · Shifts · Templates · Publish history · Publish requests · Time-Off · Swaps · Compliance · Job areas · Settings · Notifications**.
- **Overview** is the dashboard (KPI cards, pending swaps/time-off, open shifts, week-at-a-glance, module tiles).
- **Shifts** is the drag-and-drop grid where you actually build the schedule.

4. Setup & configuration (admins)

Before staff see a useful schedule, an admin sets up a few things. These all live under **Scheduling Admin**.

Job areas (positions) — /admin/scheduling/job-areas

"Job areas" are the **positions** people can be scheduled into (e.g. Skate Rental, Front Desk, Resurfacer).

- **Add** a job area by typing a name and pressing **Add area** (or **Bulk upload**).
- **Reorder** with the up/down arrows, **rename** inline, and **deactivate** (toggle) or **delete**.
 - Deactivating hides an area from new assignments without affecting people already assigned.
 - Deleting is only allowed when **no one is assigned**; otherwise deactivate instead.
- **Required certifications**: under each area you can add cert names (e.g. CPR). RinkReports will then **hard-block** assigning anyone who lacks a current matching certification to a shift in that area (a manager can override — see §8).

Assigning people to job areas (cross-reference)

Adding an employee, choosing their role, and seeding permissions is covered in the **Admin Control Center** chapter. The **scheduling-specific** parts of the employee form are:

- **Job areas** — multi-select, **up to 4** per employee (cross-training). One can be marked the **Primary job area**. These determine which positions the person can be scheduled/qualified for, and which areas they can pick when setting availability.
- **Max weekly hours** — an optional **per-employee cap**. Leave blank for "No individual cap." This is the overtime-warning threshold only (not payroll).

Scheduling settings — /admin/scheduling/settings

One settings form controls scheduling behavior facility-wide. There is also a **Seed defaults** button that fills in sensible starting values and a few default compliance rules.

Key settings (see §7 for the full list):

- **Week start day, Default shift minutes.**
- **Minor max weekly hours, Overtime weekly hours, Minimum break minutes / after hours** — feed the advisory warnings.
- **Swaps require manager approval, Swap request expiry (hours)** (default 72), **Open shifts: first-come, first-served.**
- **Notify employees when schedule is published, Notify on overtime warnings.**
- **Allow staff to submit weekly availability** (on/off).
- **Require employees to be assigned to a shift's job area.**
- **Block scheduling-grid saves that raise warnings** — when **on**, advisory warnings (hours cap, overlap, etc.) block the save; when **off**, they are advisory and a manager can confirm and save anyway. (Missing required certs always block regardless.)

Templates & compliance

- **Templates** (/admin/scheduling/templates) are reusable shift patterns you can apply to a week. You can also save a block straight from the grid (see §5).
- **Compliance** (/admin/scheduling/compliance) lists the rule set (minor hours, overtime, required break) that drives the warnings; rules can be enabled/disabled, edited, and reordered.

5. Screen-by-screen walkthrough

Admin grid — /admin/scheduling/shifts

The grid header reads: "Drag in a day column to create a shift; drag a block to move it, or its edges to resize. Click a shift to assign, duplicate, or delete."

- **Views:** Day / Week / Month toggle, with Previous/Next and **Today**. Month view is read-only — click a day to drop into the editable week.
- **Drag-create:** drag in a day column to paint a new shift; an **assign popover** opens.
- **Move / resize:** drag a block to move it; drag its top/bottom edge to change start/end. Changes save immediately (optimistically).
- **Assign popover** (create or edit): set **Start time / End time**, choose an **Employee** (or Open / unassigned), and a **Job area**. As you choose, RinkReports checks for conflicts live ("Checking for conflicts...") and shows any cert block or advisory warnings. You can also **Save as template** from here.

- **Click a shift** → a detail panel on the right lets you **assign**, **duplicate**, **edit**, or **delete** it.
- **Position (job-area) filter**: a filter chip row narrows the board to one job area so the whole view — grid, hours, crew tally — stays consistent.
- **Right rail**: Open shifts needing coverage, Swap requests, Time-off, and a Crew roster with each person's scheduled hours vs. their cap.
- **Export** writes the visible week to CSV. A KPI strip shows scheduled hours, shift count, open shifts, and pending swaps.
- **Request publish · (week)** button — files a publish request for the visible week (see below).

Publishing

- **Publish requests** (/admin/scheduling/publish/requests): the queue of pending requests. Each shows the range, who filed it, and notes. A different admin clicks **Approve & publish** (confirm dialog: "Approve and publish all draft shifts in this window?") or **Reject** with a required reason. **You cannot approve or reject your own request** — the buttons are hidden and the system blocks it, showing "You filed this request — a different admin must approve or reject it."
- **Publish history** (/admin/scheduling/publish): an append-only log of every publish event (when, range, shift count, who published).

Time-off & swaps (admin)

- **Time-Off** (/admin/scheduling/time-off): pending employee requests; **Approve** / **Deny** (with an optional note) — only pending requests can be decided. The employee is notified of the decision.
- **Swaps** (/admin/scheduling/swaps): pending swap requests; a manager can **assign a target**, **Approve** (re-runs all rule checks and applies the trade or a one-way coverage hand-off), **Deny**, or **Cancel**. An applied swap can't be denied or cancelled afterward.

Staff app — /reports/scheduling

- **My schedule** — your upcoming published shifts.
- **Availability** (/reports/scheduling/availability) — your weekly availability rows. Add an entry: **Day of week**, **Start/End time**, **Type** (Available / Preferred / Unavailable), an optional **Area/department you want to work** (only areas you're assigned to), optional **Effective from/to** dates, and notes. Edit or remove entries. If your facility has turned availability submission off, you'll see a message instead.
- **Time off** (/reports/scheduling/time-off) — **New request** with **Starts**, **Ends**, and an optional reason. Track status (pending/approved/denied) and cancel your own (see §8).
- **Open shifts** — on the landing screen, published shifts with no one assigned appear under **Open · Pick up** with a **Claim shift** button. Some are marked **Approval req.** A claim you make appears under **Your claims · Awaiting approval** until a manager approves it.
- **Shift swaps** (/reports/scheduling/swaps) — **New swap request**: pick **your shift to give up**, optionally a **coworker**, and optionally **their shift to take** (a trade) or leave it as just covering yours. Lists show **Outgoing** and **Incoming** requests with status; you can **Accept** an incoming pending swap directed at you, or **Cancel** your own pending/accepted one.
- **Notifications** (/reports/scheduling/notifications) — schedule alerts (publish, decisions, swap notices); mark one or all read.

6. Step-by-step: common tasks

Place and edit a shift (admin)

1. Go to **Scheduling Admin → Shifts**. Navigate to the right week.
2. **Drag** in a day column to create a shift (or click **Add shift**).
3. In the popover, set **Start/End time**, pick an **Employee** (or leave Open), and a **Job area**. Resolve any warnings, then **Save**.
4. To edit later, **drag** to move/resize, or **click** the shift and use the detail panel (assign / duplicate / edit / delete).

Publish a schedule

1. Build the week's draft shifts on the grid.
2. Click **Request publish · (week)**, add optional notes for the approver, and **File request**.
3. A **different** admin opens **Publish requests**, clicks **Approve & publish**, and confirms. Drafts in that window become **published**; any unassigned published shifts open for claims.

Request a publish & approve it (two-person flow)

1. Admin A files the publish request (above).
2. Admin B opens **Publish requests → Approve & publish** (or **Reject** with a reason). Admin A cannot approve their own — that's by design.

Submit availability (staff)

1. **Scheduling → Availability → New availability** (or per-day).
2. Choose day, times, **Type**, optional **area** and effective dates, then **Add availability**. (Queues offline if you're disconnected — see §8.)

Request time off (staff)

1. **Scheduling → Time off → New request**.
2. Set **Starts** and **Ends**, add an optional reason, **Submit request**. Track its status; cancel while pending/approved if plans change.

Claim an open shift (staff)

1. On the **Scheduling** landing screen, find a shift under **Open · Pick up**.
2. Tap **Claim shift**. If it needs approval, it moves to **Your claims · Awaiting approval** until a manager approves. (Online only.)

Request a swap (staff)

1. **Scheduling → Shift swaps → New swap request**.
2. Pick **your shift to give up**, optionally a **coworker** and **their shift**, add a note, **Send swap request**.
3. The coworker (if chosen) is notified and can **Accept**; a manager then approves it if your facility requires approval. (Online only.)

7. Field reference

Admin shift (assign popover / detail)

Field	What it is
Start time / End time	The shift's hours. End must be after start.

Field	What it is
Employee	Who is assigned, or Open / unassigned .
Job area	The position the shift is for (drives qualification & cert checks).
Break minutes	Unpaid break length, subtracted from scheduled hours in tallies.
Role label	Optional free-text label shown on the shift.
Notes	Optional shift notes.
Status	Draft, Published, or Cancelled (set by the publish / cancel flows, not typed directly).
Override reason	Optional note recorded when a manager overrides a certification block.

Scheduling settings

Setting	Effect
Week start day	First day of the scheduling week.
Default shift minutes	Default length for a newly created shift.
Minor max weekly hours	Cap used to warn when a minor is over-scheduled.
Overtime weekly hours	Threshold for the overtime warning.
Minimum break minutes / after hours	When a long shift should include a break (advisory).
Swaps require manager approval	If on, an accepted swap still needs a manager to approve/apply it.
Swap request expiry (hours)	Undecided swaps lapse to "expired" after this (default 72, capped at the shift start).
Open shifts: first-come, first-served	How open-shift claims are resolved.
Notify on publish / on overtime	Whether staff get notified on those events.
Allow staff to submit weekly availability	Turns the staff availability form on/off.
Require employees to be assigned to a shift's job area	Enforces job-area qualification.
Block scheduling-grid saves that raise warnings	On = advisory warnings block; off = advisory only (manager confirms).

Staff availability

Field	What it is
Day of week	Sunday–Saturday.
Start / End time	The window for that day.
Type	Available / Preferred / Unavailable.
Area / department you want to work	Optional; must be a job area you're assigned to.
Effective from / to	Optional date range the entry applies to.
Notes	Optional.

Staff time-off / swap

Field	What it is
Starts / Ends (time off)	The requested off period. End must be after start.
Reason (time off)	Optional.
Your shift to give up (swap)	The shift you want covered/traded.
Coworker (swap)	Optional target coworker.
Their shift to take (swap)	Optional — makes it a trade rather than one-way coverage.
Note (swap)	Optional message.

8. Locking, saving & offline

Draft → Published lifecycle

- New shifts are **drafts** — admins build and rearrange them freely; staff don't see drafts.
- A schedule is released through a **publish request** that a **second admin approves** (requester ≠ approver). Approving turns drafts in that window **published** and visible to staff; any **unassigned** published shift becomes an **open shift** staff can claim.

Publish-lock (what changes after publishing)

- Once a shift is **published it is locked** from free editing or deletion. Admins can still change a published shift, but only through a **controlled, audited path** that re-checks the scheduling rules and notifies affected staff (the editor shows "This shift is published — saving republishes it and notifies affected staff").
- **Deleting a published shift is a soft cancel** — it is marked **cancelled**, not erased, so there's a record. (Draft shifts are deleted outright.)

Approvals as locks

- **Time-off** and **swaps** move **pending** → **approved/denied** (swaps also pass through **accepted** and end at **approved & applied**). You can **cancel your own pending (or approved) time-off**, and **cancel your own pending/accepted swap**, but you **cannot reverse a manager's decision**, and an **applied swap** can't be undone here.

Certification & warning gates (when assigning)

- A **missing/expired required certification** for the shift's job area is a **hard block**. Only a facility manager can **override**, and every override is **logged** (with an optional reason).
- Other signals — overtime, minor hours, double-booking, approved-time-off overlap, missing break, job-area mismatch — are **advisory**. They warn and require a **Confirm & save**, unless the facility turned on "block saves that raise warnings," in which case they block too.

What queues offline vs. online-only

RinkReports is a PWA: some staff actions are **saved on your device and sync when you reconnect**. Scheduling's offline scope is deliberately narrow:

- **Queues offline (staff): Submit availability** and **Request time off**. When offline, the form saves locally and shows "Saved offline — will sync when you're back online," replaying once reconnected.

- **Online-only (by design):** the **entire admin grid** (drag-create / edit / move / delete / publish), **claiming an open shift**, **swap requests**, **accepting a swap**, and **cancelling a swap**. These depend on live shift state and on cert/hour-cap/publish-lock checks that can't be safely replayed later, so they require a connection. (Confirmed in code: swap request/accept/cancel and open-shift claim all run as direct online actions with no offline queue.)

9. Troubleshooting & FAQ

"Where do I see hours worked or run payroll?"

You don't — that isn't in RinkReports. Scheduling is **assignment only**: it plans who works when. There's no clock-in/out, timesheet, or pay calculation. "Max weekly hours" is only a warning threshold for over-scheduling.

"I can't approve my own publish request."

Correct — a publish must be approved by a **different** admin than the one who filed it. Ask a colleague with admin access to approve it.

"The grid won't let me assign this person."

If it's a **certification block**, the person lacks a required cert for that job area — only a facility manager can override (it's logged). If it's an **advisory warning** (overtime, overlap, etc.), click **Confirm & save** — unless your facility blocks warning saves, in which case fix the conflict first.

"I claimed/swapped a shift while offline and nothing happened."

Claiming and swaps are **online-only**. Reconnect and try again. (Availability and time-off do queue offline.)

"Staff say they can't see the schedule."

Check that the schedule for that week has been **published**, the staff member has the scheduling view permission, and their account is **active** and assigned to your facility. To let them file requests too, they also need **submit**.

"A published shift disappeared from history after delete."

It didn't — deleting a **published** shift is a **soft cancel** (status cancelled), so the record remains. Only **drafts** are removed outright.

"Why can't this employee be scheduled in this position?"

They may not be assigned to that **job area**, or the area may be **deactivated**. Add the area to their employee record (up to 4) or reactivate the area under **Job areas**.

Source (footnote)

```
src/app/reports/scheduling/ (page.tsx, actions.ts, swaps/, time-off/, availability/, notifications/,
my-schedule/, _components/{availability-form, time-off-form, swap-form, swap-action-button,
claim-open-shift-button}.tsx); src/app/admin/scheduling/ (page.tsx, shifts/page.tsx +
_components/{week-board, assign-popover, publish-button}.tsx, publish/page.tsx,
publish/requests/_components/requests-client.tsx, job-areas/_components/job-areas-client.tsx,
settings/_components/settings-form.tsx, _components/scheduling-nav.tsx, _lib/{grid-actions,
publish-request-actions, enforcement, governance-actions}.ts);
src/app/admin/employees/_components/employee-form.tsx (job areas, max weekly hours);
```

`src/app/api/offline-sync/route.ts` (scheduling replay = availability + time-off only);
`docs/training/00-MANIFEST.md`.

Part 3 — Training Guide

Staff report modules, scheduling for staff and admins, and the admin console.

1. Welcome & Getting Started

1.1 What is Rink Reports?

Rink Reports is your facility's operations platform. It replaces paper logbooks, clipboards, and whiteboards with one app that runs on any phone, tablet, or computer — and keeps working when the rink's Wi-Fi doesn't. Staff use it to submit operational reports (daily checklists, ice depth readings, refrigeration rounds, air-quality logs, incident and accident reports, and more), manage their work schedule, and receive facility communications. Administrators use the Admin Center to configure every module, review submissions, manage employees and permissions, and publish schedules.

Everything in the app is scoped to your **facility**: you only ever see your own rink's data, and every timestamp you see is shown in your facility's local time zone.

1.2 Signing in

- There is **no self-signup**. Accounts are created by an administrator in the admin console. If you don't have a login, ask your manager.
- Go to the app's address and sign in with your email and password at the **/login** page. Password resets go through an emailed reset link (the **update-password** flow).
- If you're signed in but see "**Account not ready — Your account is being set up**", your login exists but your employee record hasn't been activated yet. Contact your administrator.
- If a page says "**No permission**" or you land on a **Forbidden** page, your account works but you haven't been granted access to that module. That's a permissions question for your admin, not a bug.

1.3 Finding your way around

Desktop (large screens). A left sidebar lists every module your facility has enabled, in this order: Dashboard, Daily Reports, Ice Depth, Ice Operations, Dasher Boards, Refrigeration, Air Quality, Incidents, Accidents, Scheduling, Communications, Facility Paperwork. Admins also see an **Admin Center** entry below a divider. The Communications and Scheduling items show a red badge with your unread count.

Mobile (phones). A bottom tab bar gives you four tabs: **Home** (the dashboard), **Reports** (jumps to the first report module your facility has enabled), **Menu** (opens the full module list), and **Account**.

The header (always visible). Shows your name, your facility, a live date and clock, and the current outdoor temperature for your facility's location (this is the weather outside — not an ice or building temperature). On the right: the **sync status badge** (see chapter 4 on offline work), a **light/dark theme toggle**, and your user menu with **My Account** and **Sign out**.

Modules your facility hasn't enabled simply don't appear — no tile, no sidebar entry. If a module you expect is missing, your facility admin has it turned off or you don't have permission to view it.

1.4 The Dashboard

The dashboard is your home screen: a grid of large, color-coded tiles, one per module, under a greeting ("Hi, {your name}").

Status bubbles. Tiles carry a small red or green "monitoring light" so you can see at a glance whether a module needs attention. What red means per module:

Module	Red bubble means
Refrigeration	The latest report had a reading out of range
Air Quality	The latest report recorded an exceedance
Ice Operations	The latest submission had a failed circle-check item
Ice Depth	The latest session tripped the facility's low/high alert setting
Incidents	Reports are waiting for admin review (with a count)
Accidents	An accident was submitted in the last 2 days
Communications	You have unread messages or unacknowledged alerts
Scheduling	You have unread notifications or there are open shifts
Dasher Boards	Open safety-critical (severity A) issues or failed checks

These lights come from thresholds your administrators configured — the app never uses hardcoded limits.

Personalizing your dashboard. Every tile has an "x" in its corner: hiding a tile removes it from your dashboard only. It doesn't change your permissions, and the module stays reachable from the sidebar. Hidden tiles collect at the bottom of the page as "+" pills — tap one to restore it.

My areas today. If your facility uses daily-report area assignments, a widget at the top shows how many of your assigned areas are complete today and which one is next up.

1.5 Your account settings

Open **My Account** from the header menu (or the Account tab on mobile). You can edit:

- **Contact** — email and phone number (phone is required).
- **Mailing address** — street, city, state, postal code, country.
- **Emergency contact** — name and phone.
- **Notifications** — a single master switch: "Receive text message notifications." Turning it off stops all SMS to you.

The **Save Changes** button only lights up once you've changed something.

Changing your email is a two-step process: the app sends a confirmation link to your new address, and your old email keeps working until you click it. An amber notice reminds you while the change is pending.

A few things people look for here that live elsewhere: **password changes** go through the emailed reset flow, the **light/dark theme** toggle is in the header, your **calendar sync link** is in Scheduling → My schedule, and **dashboard tile hiding** is done on the dashboard itself.

1.6 Roles and permissions in one minute

- Your access is granted **per module and per action** (view, submit, edit, admin) by your facility's administrators. Roles (staff, manager, admin, plus custom roles your facility defines) mainly seed sensible defaults — the actual source of truth is the permission grants.
- Some modules also scope access **per area** (for example, which daily-report work areas you can submit for).
- **Admins** see the Admin Center. Some admin modules additionally require a module-scoped admin grant (for example, scheduling administration) — so an admin can be given the console without every module in it.
- **Super admins** manage facility-level settings that ordinary admins can't touch.

If you can see a module but a button you expect is missing or disabled, that's almost always a permission tier — ask your admin rather than retrying.

2. Staff Report Modules

This chapter walks through every report module a staff member uses day to day. Each section covers what the module is for, how to reach it, the submission workflow, and the details worth knowing before your first shift with the app.

2.1 Daily Reports

What it's for. The shift checklist tool. You pick a work area and a shift type, tick off the checklist items your admins configured, add an optional note, and submit. It also hosts custom "report forms" and a read-only history.

Getting there. The Daily Reports tile or sidebar item. Header buttons: **Assignments** (supervisors only), **Report forms** (when your facility uses custom forms), and **View history**.

Submitting a daily report:

1. In the **Shift setup** card, choose your **Work area** (you only see areas you're allowed to submit for) and then the **Shift** template — each option shows how many checklist items it has.
2. The **Checklist** card appears with a progress bar ("3 / 12 complete"). Tap each row to check it off; rows tint in the area's color when done.
3. Add a **Note (optional)** — "Anything to flag for managers?"
4. The sticky bar at the bottom shows your live progress and the **Submit** button.

After submitting you get a confirmation screen with the area, template, timestamp, and how many items you checked, plus **Submit another** and **Sign out** buttons — handy at a shared kiosk.

"My areas today" (assignment routing). If your facility routes areas to people, the module opens with your assigned areas as cards — each marked **Done** or **Not started** — plus a collapsible **Open areas** section anyone can complete. An area counts as done once any of its templates has a submission for the day. Assignment updates ("Assigned to Zamboni Room for Monday...") appear in a bell banner with a **Mark read** button.

Assignments board (supervisors). Staff with the edit or admin tier see every active area for the day, who's assigned (including "(from schedule)" and "(default owner)" sources), and **Done/Incomplete** badges. Buttons: **Assign / Reassign** (a checkbox picker of employees), **Open up** (clears assignees so anyone can complete it), and **Re-sync from schedule**. Near the end of the day, incomplete assigned areas sort to the top under a red "Day closes soon" warning. Assignment changes are online-only — they're never queued offline.

Report forms. Admin-built custom forms with titled instances ("e.g. Friday Night Game"). Tap **Add Report**, give it a title, then fill in the form's fields (text, numbers, selects, checkboxes, dates, times) and either **Save draft** or **Submit report**. Drafts are yours alone; submitted reports lock. Each instance keeps a frozen snapshot of the form as it looked when created, so later admin edits don't change reports already in flight. When a day is locked, everything goes read-only with an amber lock notice.

History and corrections. **View history** lists recent submissions (read-only, scoped to your facility and areas), with chips for completion counts and **Corrected/Correction** markers. On your own submissions you'll see a **File correction** link: you re-tick the checklist, enter a required reason ("What was wrong?"), and file it. The original stays on record — corrections supersede, they never erase. Daily report history auto-deletes after 14 days.

If you see "No areas assigned," you haven't been granted any submit areas for this module — talk to your supervisor.

2.2 Incident Reports

What it's for. Reporting a non-injury operational incident — something that happened at the facility that managers need to review. You can edit your report for 24 hours; after that it's read-only.

Getting there. The Incidents tile or sidebar item.

The form — three cards:

1. **When & where** — when it happened (defaults to now) and the **Facility space(s)**: a searchable multi-select of your facility's spaces, plus an **Other** option with a free-text description.
2. **What happened** — the **Description** (required, 500 characters with a live counter), **Activity at the time**, **Severity** (required), optional **Incident type**, **Immediate actions taken**, an ambulance-called Yes/No toggle, **Number of people involved**, and a **Follow-up required** toggle.
3. **Witnesses (optional)** — up to 3. Each witness needs a name and at least one contact (phone or email).

Before submitting, a dialog confirms: "You can edit this report for 24 hours after submitting; after that it becomes read-only."

After submitting you land on a confirmation screen showing status, severity, and timestamps, with **Edit report**, **Submit another**, and **Back to home** buttons. Open the report again within 24 hours to make changes — every edit is recorded in the report's audit trail. After the window closes (or for anyone who isn't the reporter), the report renders read-only.

The module home lists **your recent reports** from the last 30 days with severity pills and status badges (Submitted / In review / Resolved / Archived) so you can see when a manager has picked yours up.

Offline: new reports queue on your device; editing is online-only.

2.3 Accident Reports

What it's for. Reporting a personal injury: who was hurt, how badly, which body parts, where and when, witnesses, and whether it's a workers' comp claim.

Getting there. The Accidents tile or sidebar item.

The form — five numbered sections:

1. **Person involved** — name, contact (phone or email), and age (all required).
2. **What happened** — **Severity** as colored pill buttons, **Primary injury type**, **Medical attention** (some options warn "Selecting this option will alert managers"), **Body parts affected** on an interactive body diagram, and the required description.
3. **Where & when** — date/time (defaults to now), location (facility spaces), and the activity at the time.
4. **Witnesses** — up to 5, each with a name and optional contact and statement.
5. **Workers' comp** — only needed when the injured person is an employee filing a claim. Ticking the checkbox reveals your facility's workers' comp instructions and a required "I have read and understand" acknowledgement; you can't submit a comp claim without it.

The **body diagram** deserves a minute of practice: tap regions to mark front, back, or both. Midline regions cycle front → back → both; paired regions (arms, legs) are selectable per side. The selections travel with the report.

A sticky bar shows "**Auto-saved · review before submitting**" and the **Submit report** button. The app also warns you before closing the tab with an unfinished report. Like incidents, accident reports are **editable for 24 hours** — after submitting you land directly on the report page with a green "Submitted" banner and, while the window is open, an "Editable for N more hours" notice above the edit form.

2.4 Ice Depth

What it's for. Walking the rink with a depth gauge and recording ice thickness at each configured point on a rink diagram, then reviewing and submitting the session.

Getting there. The Ice Depth tile — it opens straight into your default rink's diagram. If your facility has multiple rinks or diagrams, dropdowns at the top let you switch.

Phase 1 — Measure.

- The screen shows a USA Hockey rink diagram with numbered point chips at each measurement location (plus door markers and your facility logo). A progress bar tracks "Point 3 of 12 · 2 recorded".
- **Tap a point** to open its entry popover: type the depth, watch the live severity word (**Optimal / Below min / Above target**) and color, then **Save & Next** (or **Skip**). Pressing **Enter** or **Tab** also saves and advances.
- Chips recolor as you go: green = optimal, red = below minimum, amber = above target.

Using a Bluetooth caliper (there's a built-in helper for this): pair the caliper once in your device's Bluetooth settings — it shows up as a keyboard. Then tap a point and press the caliper's **DATA** button; the reading types itself in and advances to the next point, hands-free. Set the caliper to the same unit as the form.

Phase 2 — Review & submit. Tap **Review & Submit** (enabled once at least one point is recorded). You'll see the average depth in large type, pills counting optimal / thick / below-min / skipped points, the full per-point list, and a notes field. The submit button tells you the truth about coverage — "Submit (10 of 12 recorded)" — so partial sessions are explicit.

The done screen shows a SUBMITTED badge, stat pills, and the read-only diagram with your values, plus **Download PDF**, **Print Diagram** (prints just the diagram, no app chrome), **Send Report** (emails the configured recipients), **Submit Another**, and **Back to Dashboard**.

Thresholds (the low/high limits and the unit) come from your facility's ice-depth settings — the defaults are 1 and 1.5 inches.

2.5 Ice Operations

What it's for. The ice maintenance log: five quick-entry forms in one module. Nothing here can be edited after submitting, so double-check before you tap.

Getting there. The Ice Operations tile. Tabs across the top: **Ice Make · Circle Check · Edging · Blade Change · Propane Tank Change** (your facility can disable individual tabs). A **Show Activity Feed** button reveals the last 8 submissions facility-wide.

The five forms:

- **Ice Make Activity** — log a resurfacing run: rink, machine, **water used** (with a Gal / L / % Tank unit toggle — % Tank only works when the machine has a tank capacity on file, and switching units converts your number), machine hours, snow taken (%), time on/off, notes.
- **Digital Circle Check** — pick the machine and answer every checklist item with **Pass** or **Fail**. Items start unanswered — nothing is pre-checked, so a recorded "pass" always means someone actually looked. A **Fail** requires a "What's wrong?" note. Submit stays disabled until every item is answered ("Tap Pass or Fail on every item to submit — 4 remaining"). If the machine has no fuel type on file you'll pick one first so the right checklist template loads.
- **Edging** — machine, hours run, notes.
- **Blade Change** — machine, old blade hours, new blade ID (serial), notes.
- **Propane Tank Change** — machine and machine hours; the date is recorded automatically when you submit.

Machine dropdowns show each machine's running hours ("Zamboni 552 — 1,240 hrs") and are filtered to the right equipment type per form. The done screen shows the operation, timestamps, rink/equipment, and — for circle checks — a red "N failed items" chip.

Offline: all five forms queue, and the log keeps the time you did the work, not the time it later syncs.

2.6 Refrigeration

What it's for. The plant-room reading round: every configured reading, per section and per piece of equipment, with out-of-range values flagged as you type.

Getting there. The Refrigeration tile. If your facility has enabled alerts, an amber banner reminds you: "Out-of-range readings will trigger an alert to managers."

The form:

- **Log Information** card — your name and facility (read-only), the reading time (defaults to now), and optional **Shift** and **Round #** fields (if your facility logs, say, 3 rounds per shift, the form says so).
- **°F / °C toggle** in the card header. Flipping it converts every temperature you've already typed, and re-labels every field and range hint. Values are always stored in °F regardless of what you display.
- **One card per section**, with fields for the section itself and a labeled subgroup for each piece of equipment. Numeric fields show a **normal-range hint** underneath ("Normal: 12 – 18 psi") based on your facility's thresholds.
- **Critical readings require a note.** If a value breaches a critical threshold, a red **Corrective action** box appears inline and the report won't submit until you describe what you did about it.
- **Notes (optional)**, then the full-width **Submit refrigeration report** button.

The done screen shows the timestamp plus chips for "N values recorded" and "N out-of-range". The module home lists your last 30 days of submissions with out-of-range badges.

2.7 Air Quality

What it's for. Logging CO and NO2 (and any other configured readings) per location, checked live against your facility's jurisdiction-aware compliance thresholds — with the full regulatory monitoring log attached.

Getting there. The Air Quality tile. Note in the header: after you submit, the report can't be edited.

The workflow, top to bottom:

1. Pick the **Location** (required before anything else).
2. If your facility has a **jurisdiction profile**, a compliance card shows its name, a **Binding** or **Guidance** badge, the measurement method (**1-hr TWA** or **Single sample**), a **frequency tracker** ("On schedule" / "Behind by 2" with your weekly and weekend sample counts), and a **Reading type** dropdown.
3. Pick the monitoring **Equipment** if monitors are configured.
4. Enter each **reading**. Fields bound to a compliance threshold show a tier hint ("Corrective > 25 · Notification > 50 · Evacuation > 100 ppm") and a live badge the moment you type: **Within range**, **Corrective action**, **Notification**, or **Evacuation**.
5. For TWA jurisdictions, a collapsible **1-hour TWA calculator** takes your readings at 5-minute intervals, averages them, and fills the main field via **Use average**.
6. Any over-threshold reading triggers a **compliance banner** with your facility's escalation instructions (for example, Evacuation: "Evacuate the facility now and contact the fire department immediately") and a **required Corrective action taken** note — the submit button stays disabled until you fill it in.
7. The **Monitoring log** — collapsible sections mirroring the regulatory form: equipment and tester info (monitor models, calibration dates — with a warning if calibration is over a year old), general information and equipment status, routine and post-edging measurement tables, and additional recommendations (staff trained, public signage checkboxes).

The submit button labels itself "**Submit readings for {Location}**" once you've picked one. The done screen shows readings recorded and, when applicable, a severity-colored exceedance chip.

Offline: submissions queue, and the same exceedance checks run when they sync.

2.8 Dasher Boards

What it's for. A single-screen field tool for the rink perimeter. Tap any board panel, glass panel, or door on the diagram to report a problem; open issues stay pinned to the asset until they're fixed. A formal **inspection walk** is an optional overlay that ends with a sign-off.

Getting there. The Dasher Boards tile — straight to your default rink's condition map, with a rink switcher in the header.

Reading the diagram. Legend: **red** = open severity-A issue · **yellow** = open B/C issue · **coral** = flagged fail with no issue yet · **lime** = door. A **Glass** switch toggles the glass layer, and the diagram zooms. The toolbar shows "Last walked {date} by {name}" and — for submitters — a **Start inspection walk** button.

Reporting an issue (the everyday flow — no walk needed). Tap an asset to open its sheet. The **Report issue** form is right there: pick **Board** or **Glass** if the position has both, choose severity — **A — Safety critical**, **B — Needs repair** (the default), or **C — Cosmetic** — pick a category, describe the problem, and **Submit issue**. Severity A adds two required extras: an **Action taken** note and a **Supervisor** to notify, stated up front.

The asset sheet also shows: open issues on that asset (with **Acknowledge** for severity A — supervisor tier — and **Mark fixed**), a walk-only Pass/Fail **condition check**, the read-only **replacement spec** (dimensions, thickness, material), and a collapsible **issue history**.

Checklist — due today. When inspection items are due, a card lists them grouped by cadence (daily / weekly / monthly / yearly) with **Pass** / **Flag** buttons. Flagging an item opens the issue form automatically — a flagged item needs a reported issue behind it.

Inspection walks. Start a walk and a sticky bar tracks it: tap problem assets as you circle the rink — **untapped assets are attested OK at sign-off** — and answer any due checklist items. **Sign off** opens walk notes and **Complete walk**. The sign-off is refused while severity-A issues are unacknowledged or due items are unanswered. The done screen shows pass/fail/issue counts, a diagram colored with this walk's results, and **Download PDF / Print / Send Report** buttons.

Permission tiers: view = see the map; submit = report issues, run walks, resolve B/C issues; edit = acknowledge and resolve severity A; admin = configure the perimeter (admin console).

Offline: everything queues — starting walks, checks, issues, even the sign-off (validated when it syncs). Acknowledge/resolve and history need a connection.

2.9 Facility Paperwork

What it's for. A read-only library of documents your admins uploaded — policies, handbooks, safety documents, manuals.

Getting there. The Facility Paperwork tile.

Using it. Filter by category (only categories that actually contain documents are offered), then tap **Download** on any card. Download links are short-lived and generated per request, so always download from the page rather than bookmarking a file URL. Staff can't upload or edit here — that's the admin console's Facility Paperwork module.

2.10 Communications

What it's for. Your inbox: facility-wide operational **alerts** raised by other modules, **direct messages** to you, and — if you can send — a **Sent** view with read and acknowledgement receipts.

Getting there. The Communications tile or sidebar item (with an unread badge).

Alerts tab. Every alert for your facility, newest first — alerts are operational safety signals, so everyone with communications access sees all of them. Open one to read the full body; if it **requires acknowledgement**, an **Acknowledge this alert** card appears with an optional notes field. Once acknowledged, the alert shows when you did it.

Messages tab. Messages delivered to you. **Opening a message marks it read automatically** — acknowledgement, when required, stays a deliberate button press. Messages can carry a **PDF attachment** (Download PDF attachment button). **Reply** is available when the sender is a real person (not a system-generated message).

Sent tab (senders only). Your messages with per-recipient receipts: who read it and when, who acknowledged.

Composing. Tap **New message**: optionally start from a **template** (fills subject, body, and the ack flag), write your message, pick one or more **recipient groups** (staff only see groups flagged for staff messaging), optionally toggle **Requires acknowledgement**, and **Send message**. The confirmation screen shows the recipient count and receipts list. Replies lock the recipient to the original sender and prefill "Re: ...".

2.11 Working offline

Rink Reports is built for rinks with dead zones. The rule of thumb: **submitting works offline; browsing needs a connection.**

What happens when you submit offline. The report is saved on your device ("Saved on this device") and queued. When you're back online it submits automatically — and the server runs the exact same validation and threshold checks it would have run live. The amber offline banner at the top of every reports page tells you how many submissions are waiting.

The sync badge and queue. The header badge shows "N pending" (amber) or "N failed" (red); tap it to open the **Pending Sync Queue**. Each queued item shows its module, status ("pending", "retry in 45s", "failed", or "won't retry"), and the last error. Buttons: **Retry failed** and **Sync now** — worth knowing on iPhones and iPads, which don't sync in the background, so open the app and tap **Sync now** after a shift in a dead zone.

Offline pages. Three lightweight pages work with no connection at all, using data saved the last time you were online: **/offline-daily** (your daily-report areas), **/offline-forms** (custom report forms — you can create and fill these offline), and **/offline-schedule** (your published shifts). Each shows an amber "Offline — showing last-synced data" banner and refuses to show stale data as if it were today's.

What can go wrong (and what it means):

- "You're offline and the offline queue isn't ready yet. Keep this page open and try again." — the device has never synced for your account; the app refuses to fake a save. Your entries are still on screen.
- A queue item marked **"won't retry"** hit a permanent error (for example, the day locked before it synced). It's parked, never silently dropped — contact your administrator if you're unsure why.
- Queued items belong to the account that created them; someone else signing in on the same device can't flush your queue.

3. Scheduling — For Staff

3.1 The one concept to learn first: draft vs. published

Every shift starts life as a **draft** while managers build the schedule. Staff never see drafts. Only when a schedule window is **published** — a two-person approval among admins — do shifts appear in your app, your calendar feed, and your notifications. A published shift is locked: managers can't quietly edit it; any change re-publishes it and notifies you, and a cancellation notifies you too.

Everything you see in scheduling is shown in your **facility's local time**, and your "week" starts on whatever day your facility configured (usually Sunday).

3.2 The scheduling hub

Open **Reports** → **Scheduling** (the tile's badge counts your unread notifications plus open shifts you could pick up). Top to bottom:

1. **NEXT SHIFT hero** — your next shift with a countdown ("NEXT SHIFT · IN 6H"), the time range, and department · role. Buttons: **Request swap** and **Full schedule**.
2. **7-day week strip** — a dot marks each day you work.
3. **Upcoming** — your next shifts over the coming 28 days.
4. **Open · Pick up** — shifts available to claim (hidden when there are none).
5. **Your claims · Awaiting approval** — claims you've made that a manager hasn't decided yet.
6. **Quick links** — My schedule, Time off, Availability, Shift swaps, Notifications.

3.3 My schedule

The full personal schedule with a **LIST | WEEK** toggle. List view has From/To date filters and a status filter (Published, or All — which adds cancelled shifts). A header link, **Available offline** →, opens the offline copy of your schedule; bookmark it if you work in a dead zone.

Dropping a shift. Each shift you can still drop shows a **Drop** button — it only appears when the shift is published and far enough out to satisfy your facility's drop-notice window. Tapping it opens a confirm dialog whose wording tells you exactly what will happen:

- If your facility requires manager approval (the default): "Your manager will be asked to approve. You're still on the shift until they do." → **Request drop**. The row then shows **Drop pending** with a **Withdraw** button.
- If it doesn't: "This releases the shift immediately — it goes to the open list for a coworker to pick up." → **Drop shift**.

You can add an optional reason for your manager. Dropping is online-only — the server re-checks everything the moment it runs.

Calendar sync. At the bottom of My schedule, the **Calendar sync** card gives you a personal subscription link for Google or Apple Calendar. Tap **Turn on calendar sync**, then **Copy link** and subscribe in your calendar app; your published shifts appear automatically and update within a few hours. The feed covers the last week and the next 60 days. Treat the link like a password — anyone with it can see your shifts. If you ever share it by mistake, tap **Reset link** and old links stop working immediately.

3.4 Open shifts: claiming extra work

Shifts that need coverage appear under **Open • Pick up** on the hub and My schedule — only shifts starting within the next 14 days, soonest first, up to five per list. Tap **Claim shift**:

- If your facility runs first-come-first-served, the shift is yours immediately.
- Otherwise the card says "• Approval req." and your claim goes to a manager; it appears under **Your claims • Awaiting approval** until decided.
- If a coworker beat you to it: "That shift is no longer available."
- The system re-checks your certifications, overlaps, and hours — a claim can be refused on the rules even if the shift is open.

Open shifts appear when a schedule is published with unassigned shifts, when a drop is approved, or when approved time off frees up a shift.

3.5 Availability

Availability is a recurring weekly pattern, not a one-off: what you set for a Tuesday applies to every Tuesday. That's the single most common misunderstanding — the day page even says so.

From the hub, open **Availability**, pick a day, and add blocks:

- **Start / End time**, and a **Type**: **Available**, **Preferred**, or **Unavailable**.
- **Area / department you want to work** — only your assigned job areas are offered, plus "No preference".
- **Effective from / to** dates for temporary patterns (a semester, a season).
- Optional notes. Edit or delete blocks any time.

Know what each type actually does: only **Unavailable** blocks affect enforcement (a manager scheduling over one gets warned). Available and Preferred are planning information for whoever builds the schedule. Availability submission can be turned off facility-wide; you'd see a notice if so.

Adding and editing availability works offline and syncs later.

3.6 Time off

From the hub, open **Time off** → **New request**. Pick the start and end (interpreted in facility-local time), add an optional reason for your manager, and **Submit request**.

Requests move through statuses: **pending** → **approved**, **denied**, or **cancelled**. You can **Cancel** your own request while it's pending or even after approval — but not once it's denied or cancelled. When a manager decides, you get a notification (and usually an email) including any note they wrote.

Time-off requests can be submitted offline.

3.7 Shift swaps

From the hub, open **Shift swaps** → **New swap request**:

- **Your shift to give up** (required) — your published shifts over the next 60 days.
- **Coworker (optional)** — leave it as "Anyone" to post an open request.
- **Their shift to take (optional)** — appears after picking a coworker; leave blank to "just cover mine (no trade)".

- An optional note, then **Send swap request**.

Your **Outgoing** and their **Incoming** lists track every request with a status badge:

Status	Meaning
Pending	Filed, not yet accepted
Accepted	The coworker accepted (or a manager accepted on their behalf)
Approved & applied	A manager approved — the shifts have actually moved
Denied	A manager refused
Cancelled	Withdrawn by the requester or a manager
Expired	Nobody decided in time (default 72 hours) and it lapsed automatically

Accepting a swap does not move the shift. A manager must give final approval, and that approval re-checks both people against all the scheduling rules — so a swap can still fail at the last step if, say, it would double-book someone. Until you see **Approved & applied**, plan to work your original shift.

3.8 Notifications

The hub's **Notifications** link (red badge = unread) collects everything scheduling tells you: schedule published, shift changed, open shift available, swap request received/approved/denied, time off decided, overtime warnings, and automatic shift reminders 24 hours before each published shift.

Most rows have a **Mark read** button. One is special: a **Schedule published** notification has an **Acknowledge schedule** button instead — pressing it tells your manager you've seen the new schedule, and managers see the roll-up ("acknowledged 7/12") on their side. Make it a habit after every publish.

3.9 What works offline

Action	Offline?
Viewing your schedule (/offline-schedule)	Yes — last-synced copy
Requesting time off	Yes — queued
Adding/editing availability	Yes — queued
Claiming an open shift	No — online only
Dropping a shift / withdrawing a drop	No — online only
Swap requests and accepts	No — online only
Notifications / acknowledging	No — online only

The online-only actions all depend on live shift state that could change while you're offline — the app refuses to let you act on a world that may have moved on.

4. Scheduling — For Admins

Scheduling administration requires both admin-console access and the scheduling module's admin grant — console access alone lands on the Forbidden page. A sticky tab bar runs across every page: **Overview · Shifts · Templates · Publish history · Publish requests · Time-Off · Availability · Swaps · Compliance · Job areas · Settings · Notifications.**

4.1 The rules engine (read this first)

One shared validation engine checks every path that puts a person on a shift — grid edits, open-shift assignment, swap approvals, publish approvals, and staff self-claims. The checks:

Check	What it flags
Certification missing	The job area requires a cert the employee lacks (or it expires before the shift)
Minor overtime	Exceeds the weekly hour limit for minors
Overtime	Pushes past the overtime threshold
Break required	The shift is long enough to require a break that isn't scheduled
Min rest	Doesn't leave the required rest between shifts
Double-booked	Overlaps another shift for the same employee
Unavailable	Falls within a time the employee marked Unavailable
Time off	Overlaps approved time off
Not qualified	The employee isn't assigned to the shift's job area
Hour cap	Puts the employee over their personal weekly cap
Operating hours	Falls outside the facility's configured open hours

Two tiers of severity:

1. **Certification gaps always hard-block**, regardless of settings. A facility manager can override — the popover shows "Blocked — missing required certification", an override-reason field, and an **Override & assign** button — and every override is written to the audit log on the Compliance page.
2. **Everything else is advisory**: an amber "Heads up — this assignment:" box and a **Confirm & save** button. If the facility turns on **block on violations**, the same box becomes "Blocked by facility policy" and the save is disabled.

Unassigned (open) slots never gate — there's nobody to validate.

4.2 Overview

The scheduling command center: four stat cards (**Scheduled hours** with published/draft split, **Shifts this week**, **Open shifts** — with a red "N unassigned (no listing)" warning when shifts exist that nobody can see or claim — and **Pending requests** split by swap/time-off/drop), plus the three approval queues inline and a module card grid with live counts.

The approval panels (also available in the grid's right rail):

- **Pending drops** — Approve or Deny with an optional note. The approve copy is blunt on purpose: "Approving unassigns the shift and opens it for claims. Nobody is covering it until a coworker picks it up."
- **Pending swaps** — Approve or Deny with a note visible to the requester.
- **Pending time-off** — Approve or Deny. **The conflict branch matters:** if approval would overlap shifts the employee already holds, the app stops and lists them, then offers **Approve & unassign shifts** (drafts are cleared; published shifts go through the governed edit path, return to the open-claim pool, and the employee is notified), **Approve anyway**, or **Cancel**.
- **Open shifts** — **Assign** (pick an employee; the rules engine hard-blocks unqualified fills) or, for claimed listings, **Approve claim** / **Decline**.

4.3 The Shifts grid

The main canvas. Drag in a day column to create a shift; drag a block to move it or its edges to resize; click a shift to assign, duplicate, or delete.

Toolbar: week navigation, **DAY | WEEK | MONTH** views (month is read-only — click a day to edit that week), **Add shift**, and **Request publish**. A KPI strip shows scheduled hours, shift count, **labor cost** (hours × each employee's wage, falling back to the facility default rate), and open shifts. A sub-toolbar offers color-by (job area or person), a coverage heatmap, density options, **Apply template...**, **CSV Export**, **Print** (a clean weekly roster), and a job-area filter.

Creating and editing shifts. Drawing a block (or **Add shift**) opens the shift popover: start/end times, **Employee** (or "Open / unassigned"; minors are marked), **Job area**, and a **Repeat weekly** toggle with day-of-week chips and an end date — with a live count ("Will create 12 shifts") and caps of 84 days / 62 occurrences. A **Save as template** link saves the block's times and job area as a reusable one-slot template. While the engine checks the assignment, the save button adapts: **Save**, **Confirm & save** (advisory warnings), or **Override & assign** (cert gap, manager-only, audited).

For recurring batches, every occurrence is checked: a cert gap anywhere blocks the batch until overridden; advisory issues aggregate into one confirm; and per-date conflicts skip only that date, with a toast reporting exactly what was skipped and why.

Published shifts are different. Editing one carries the notice "This shift is published — saving republishes it and notifies affected staff." Drag-moving is disabled. Deleting a published shift is actually a governed **cancel** (the employee is notified); draft deletes are immediate with an **Undo** toast. Deleting a shift in a recurring series offers **Delete series (drafts)** — published occurrences are left for individual cancellation.

The right rail shows either the selected shift's detail (employee/job-area dropdowns that re-validate instantly, duration/break/pay chips, Edit/Duplicate/Delete) or, with nothing selected, the open-shift, swap, time-off, and drop queues plus a per-person hours crew list.

4.4 Templates

Reusable weekly patterns. Build one from scratch on the Templates tab (name, slug, description, then slots by day: job area, times, break minutes, role label, and a staff count — one draft shift is created per count), or capture one from the grid with **Save as template**.

Apply template (from the grid): pick the template and the week — any date snaps to that week's start. Applying always produces **unassigned draft shifts**; a template can never place people directly, so the rules engine is never bypassed. Archived templates must be reactivated before applying.

4.5 Publishing — always two people

Publishing is a deliberate two-admin handshake:

1. **File a request** from the grid (**Request publish · week of ...**), optionally with notes for the approver. Guards: there must be draft shifts in the window, and overlapping pending requests are refused.
2. **A different admin approves** on the **Publish requests** tab. If you filed it, your buttons are replaced with "You filed this request — a different admin must approve or reject it." — the separation is enforced server-side too.
3. **Approve & publish** does everything in one transaction: re-validates every assigned draft, publishes them, opens claim listings for unassigned shifts, notifies staff (in-app and email), and records the event. Rejections require a reason.

Publish history is the read-only log: when, range, shift count, who published — and an **Acknowledged** column ("7/12") counting how many staff pressed **Acknowledge schedule** on their notification. That's your read-receipt for the schedule.

4.6 Time-off, availability, and swaps queues

Time-off — filter by status (default Pending). Approve/Deny with a note the employee sees; the conflict branch from §4.2 applies. Admins can also cancel an approved request, which notifies the employee.

Availability — a read-only planning grid: employees down the side, the facility week across the top, blocks colored **Available** (blue) / **Preferred** (green) / **Unavailable** (red) with job-area preferences. A card below lists employees who submitted nothing. Remember the footnote: only Unavailable blocks affect enforcement — the rest is planning information.

Swaps — filter by status (default Open = pending + accepted). Actions:

- **Assign target** (pending, no target yet): picking an employee accepts on their behalf — they're notified that it still needs final approval.
- **Approve**: runs the atomic swap — both shifts locked, both directions re-validated, then the trade (or one-way coverage) is applied and both employees notified. Failures are specific about which direction broke which rule.
- **Deny / Cancel** with notes.

Undecided swaps auto-expire after the configured window (default 72 hours, capped at the shift's start).

4.7 Compliance and job areas

Compliance — the facility rules that drive the warnings: Minor max hours, Overtime, Break required, Certification required, Min rest between shifts, plus custom rules. Add, edit, reorder, enable/disable. At the bottom, the **certification override audit log**: every "Override & assign" ever pressed — when, who, which employee, which missing certs, and the reason typed. Show this in training; it's the accountability half of the override button.

Job areas — the assignable areas (Front Desk, Concessions...). Each employee can hold up to four. Add, rename, reorder, deactivate (delete is blocked if assigned — deactivate instead), plus CSV bulk upload. Per area, add **required certifications** — these are exactly what produces the hard cert block at assignment time.

4.8 Settings

Seed defaults on first run (also seeds three starter compliance rules). The fields:

Setting	Effect
Week start day	Changes everyone's week window, everywhere
Default shift minutes	Length of a freshly drawn shift (default 480)
Minor max hours / Overtime hours / Break rules	Feed the corresponding warnings
Swap request expiry (hours)	Undecided swaps lapse after this (default 72)
Default hourly rate	Labor-cost fallback when an employee has no wage
Opens at / Closes at (+ midnight checkbox)	The grid's visible hours; outside-hours shifts still save with an advisory
Shift-drop notice (hours)	How far ahead staff must drop; 0 = any time before start

Toggles: swaps require manager approval · open shifts first-come-first-served · notify on publish · notify on overtime · allow staff availability submission · require job-area qualification · **block grid saves that raise warnings** · **shift drops need manager approval**.

Four of these deserve their own minute in training because they visibly change what staff see: the week start day; first-come-first-served (off = every claim needs a manager); block-on-violations (turns "Confirm & save" into a wall); and the drop-approval toggle plus notice hours (changes the staff Drop dialog's wording and whether the button appears at all).

4.9 Notifications (admin view)

A read-only feed of every scheduling notification sent to staff, filterable by type, recipient, read state, and date. Also a **Send shift reminders** card: reminders go out automatically 24 hours before each published shift; this sends them earlier for a wider window (1–168 hours), de-duplicated so no shift gets two.

5. The Admin Console — Setup & System

The Admin Center is where a facility gets built: people, roles, permissions, module visibility, and the system-level tools (exports, retention, audit). This chapter covers everything except the per-module report consoles (chapter 6) and scheduling (chapter 4).

5.1 Getting in, and the console layout

Any `/admin` page requires admin access: you're an admin if you're a **super admin**, hold the **Admin Center permission** (the Admin module's Admin action in the permission matrix), or — as a fallback — have an active admin-tier role. Everyone else lands on the **Access denied** page, which shows exactly which account is signed in (useful when someone's in the wrong account) with Go-home and Sign-out buttons.

The sidebar groups the console into:

- **Setup** — Facility, Modules, People, Departments, Facility Spaces, Permissions
- **Module Admin** — one console per report module, plus Facility Paperwork
- **System** — Lists, PDF/Export Settings, Data Retention, Audit Log, Super Admin

Two navigation gotchas to teach on day one:

The sidebar is **not** permission-filtered — every admin sees every link, and clicking one you lack the grant for lands on Access denied. A visible link is not proof of access. And the **Roles** page isn't in the sidebar at all: reach it from the Dashboard setup checklist or directly at `/admin/roles`.

The header always offers a **Dashboard** button back to the staff app, the theme toggle, and your account menu.

5.2 Admin Dashboard

The console home: a **setup checklist** (facility info, roles seeded, role defaults set, first admin linked, staff added, invites sent — six steps with a "Ready" badge when complete), **stat cards**, a **recent report activity** grid showing 7-day and 30-day counts per module, and an **offline sync queue** health card (pending / synced / failed counts — failed turns red).

Super admins get a **facility switcher** here that carries a `?facility=` context through to the other facility-scoped pages.

5.3 People (Employees)

The roster. Search by name, email, role, or code; filter Active / Inactive / All; and use **Add employee**, **Bulk add**, or the per-row actions: **Edit**, **Invite**, **Preview**, **Deactivate/Reactivate**, and (super admins only) **Delete**.

How accounts actually get created. There is no signup page — this is the front door:

1. **Add employee** opens a form: name, **role** (required), a live preview of that role's default permissions, **job areas** (up to 4, with a primary), employee code, hire date, **max weekly hours** (the scheduling hour-cap), **hourly wage** (admin-only; powers labor-cost estimates), email/phone, a **minor** checkbox, and emergency contact (required unless the employee is a minor).

2. The **"Create a login & apply these permissions"** checkbox (on by default) sends an email invite and seeds the role's permission defaults. Uncheck it for schedule-only staff — you can send an **Invite** later from the list.
3. The invite email lands the person on a page where **they set their own password** — admins never see or set passwords. If the email already has an account, no email is sent; the existing login is simply linked ("Existing account linked. No email sent.").

Bulk add (up to 100 at once) is a spreadsheet-style grid with **Paste from spreadsheet** (column order: First name, Last name, Email, Hire date, Role, Job areas — pipe-separated), a downloadable CSV template, and a **"Send login invites & apply role permissions"** switch. Watch the per-row status: **Added** (green), **Partial** (amber — the record was created but the invite or permission seeding failed; send a manual Invite later), **Failed** (red, with the reason). Bulk add can't set departments, emergency contacts, or employee codes — follow up in the single-edit form.

Deactivating vs. deleting. Deactivate removes access immediately and pulls the person out of shift assignment and routing — reversible from the same list. Delete is super-admin-only and permanent.

Preview. The **Preview** button impersonates an employee's view for one hour, with an amber banner ("Previewing as {name}") and a **Stop preview** button. Important caveat, printed right in the banner: preview narrows what the app renders, not what the database returns — it's a navigation sanity check, not a security test. Starting and stopping preview are both audited.

Employee detail page (click a name): read-only profile, **Certifications** (add with expiry dates; badges show "Expired 3d ago" / "Expires in 12d" — always pick cert names from the suggestions, since a typo breaks scheduling cert enforcement), **Module Access** (a quick per-module view with a legacy 9-level override picker — prefer the Permissions matrix for precision), **Communication Groups** membership, and recent **Activity**.

5.4 Permissions (Module Access Control)

The authoritative answer to "what can this person do." Pick a user, and you get a matrix: 11 modules down the side (the nine report modules, Scheduling/Communications, and **Admin**), four actions across — **View / Submit / Edit / Admin**. Each checkbox saves instantly. A user with no rows has **zero access**.

- **View** — can see the module. **Submit** — can create entries. **Edit** — can modify submissions. **Admin** — can configure the module and use its admin console.
- **Presets** (Full Access, Submitter Only, Viewer Only, No Access) fill the whole matrix in one click.
- **The one checkbox to treat with respect:** the **Admin module's Admin action** is the Admin Center key — it makes someone a facility admin. Only super admins can grant it; presets applied by a facility admin silently leave it off.
- Facility admins manage only their own facility's users; the target must be an active member of the facility.
- A **Bulk CSV import** (user, facility, module, action, enabled) exists for migrations; invalid rows are skipped with per-line reasons.

Overrides are sticky. Every edit made in this grid is stamped as a manual override, and role re-seeding never overwrites it. If you hand-tune here and later expect a role change to fix things — it won't.

Also note: the matrix shows all modules whether or not the facility has them enabled. Permissions and module visibility are independent systems.

5.5 Roles

Roles bundle permission defaults so you don't hand-toggle 44 checkboxes per hire. The resolution rule, stated on the page: explicit override → role default → no access.

Canonical roles (seeded automatically, undeletable): Super Admin (level 0), Administrator (1), Manager (2), Staff (3). You can add **custom roles** (e.g. "rink_lead", "driver") with a display name, an immutable lowercase key, a hierarchy level (lower = higher rank), and a description.

The defaults matrix: roles × modules, four checkboxes per cell (V · S · E · A). Flipping a box **immediately re-applies to that role's current staff** — manual overrides are preserved, but this is a many-people-at-once screen; change it deliberately. A **Copy permission defaults** panel bulk-copies one role's defaults to another.

Safety rails: you can't create or edit a role that outranks your own; the Admin-Center cell is super-admin-only (and copying skips it); deactivating a role that people still hold warns you — those employees fall back to only their explicit overrides until reassigned.

5.6 Departments, Facility Spaces, Modules, and Lists

Departments — group shifts for the schedule's department filter and shift assignment (e.g. Ice Crew, Front Desk). Name, slug, color, sort order; reorder with arrows; deactivate rather than delete (existing shifts keep their department; it just leaves new-assignment pickers).

Facility Spaces — the shared list of physical areas feeding the location pickers in **Incident Reports, Accident Reports, and Air Quality**. Seed a starter set (Main Rink, Lobby, Locker Room...), add manually, or CSV bulk-import (duplicates are skipped, never overwritten). Deleting a space referenced by reports is blocked — "Cannot delete; in use by N report(s). Deactivate instead." Teach "deactivate, never delete" here.

Modules — per-facility on/off switches controlling which modules appear in **staff navigation**. The page says it plainly: this is a **visibility switch only** — it does not change per-employee permissions, and disabled modules stay protected by their own access rules. Missing settings default to enabled, and Dashboard/Admin Center are never toggleable. The #1 misconception to head off: turning a module off is not a security control; the permission matrix is.

Lists — per-facility option lists for dropdowns. Today that's one list: **Timezones** (the IANA zones offered in the Facility settings picker). Seed the canonical set (idempotent — safe to re-run, never overwrites your edits), add/edit/deactivate. Keys must be real IANA zone names; deleting anything in use is blocked.

5.7 Facility settings

What you see depends on who you are:

- **Facility admins get a read-only view.** The card says it outright: "Only super admins can edit facility settings." If the rink's phone number changes, route it to a super admin.
- **Super admins** manage every facility: create (which also seeds the canonical roles automatically), edit name/slug/timezone/address/contact, and set active status. A nice touch worth demoing: typing a **zip code auto-selects the timezone**, with a note to override only if the rink sits on a boundary.

5.8 Facility Paperwork (admin side)

Upload the documents staff browse in the Facility Paperwork module. **Bulk Upload:** pick a **Category** (Emergency Action Plan, Employee Handbook, Staff Manual, Policy Document, Safety Document, Other), select multiple files (PDF, Word, Excel, PowerPoint, text, or images — 25 MB each; invalid files are rejected in the browser before upload with the reason), add an optional shared description, and **Upload documents**.

Manage the list per document: **Hide/Show** (takes it off the staff page without deleting — prefer this over Delete), **Edit** (title, category, description), and **Delete** (permanent, confirmed).

One error message worth knowing by heart: "Your account has admin console access but not the facility admin permission required to manage documents." That's not a bug — the account got into the console via a role fallback but lacks the explicit Admin/Admin grant. Fix it in the Permissions matrix (super admin required).

5.9 PDF / Export Settings

Two tools on one page:

Run an export. Pick a module, format (**CSV** or **PDF**), and a date range, then download. Exportable modules: Daily Reports, Incident Reports, Accident Reports, Refrigeration, Air Quality, Ice Depth, Ice Operations, Communications. Limits: 2,000 rows and a 366-day window per export. The download re-checks permissions independently — you need the module's View action, so an admin who can open this page may still be refused a specific module.

Export settings (one set per facility): **Branding** (logo URL, header text like "Max Ice Center — Confidential", footer text), **Layout & format** (Letter/A4, date format, CSV delimiter, and include-on-every-export checkboxes for facility name, export date, and submitted-by), and **Column visibility per module** — unchecked columns are hidden from exports but remain in the database. A module never configured exports all its columns.

Most report consoles also carry a quick **Export** dropdown in their header (Download CSV / Download PDF for the last 30 days, plus a link here for custom ranges).

5.10 Data Retention

Per-module rules for how long submitted data is kept, measured from submission date. Each module row shows its window, whether **nightly auto-purge** is on, and when it last ran; expand to edit ("Keep for (days)" with preset chips from 30 days to 10 years, or **Forever**) and to run a **Manual purge** immediately (double-confirmed; cannot be undone; runs regardless of the auto-purge flag).

Regulatory floors are enforced in the database, not just the form: 30 days for daily/ice modules and communications, 90 for refrigeration and air quality, **365 for incident and accident reports**, and **7 years for the audit log** — the audit window can be raised, never shortened. Setting a module to Forever turns auto-purge off automatically and hides the purge button. Scheduling has no retention row on purpose — it can't be purged from here.

Audit-log destruction is the product's one two-person operation. Expired audit rows are staged into a batch, not deleted. The pending-destruction card shows "Awaiting first approval" / "1 of 2 approvals"; the second approval must come from a different admin, and **Cancel & restore** puts every row back untouched. Every retention change and purge is itself audited.

5.11 Audit Log

An immutable record of create, update, delete, and authentication events across the facility. Filters (all URL-driven, so a filtered view is shareable): action, entity type, actor, date range (From defaults to 30 days back), and an IP/entity-ID search. The table shows up to 300 rows — narrow with filters rather than scrolling.

Click a row for the detail: actor (or "System" for automated changes), IP address, user agent, and collapsible **Before / After** JSON showing exactly what changed. App-level events like preview start/stop and retention changes are recorded alongside row-level changes.

5.12 Super Admin

Cross-facility platform management — the link is visible to all admins, but the page itself is super-admins only. Panels:

- **Facilities** — every tenant, with Activate/Deactivate per facility.
- **Super admins & users** — every platform user. **Promote/Revoke** super-admin status (you cannot revoke your own), and **Reset password** — generates a one-time recovery link to hand to the user.
- **Invite service health** — a diagnostic that probes the email-invite service and reports the exact failure mode (key missing, invalid, or wrong project) so broken invites can be diagnosed without a developer.

Consolidated: what only super admins can do. Create/edit facilities; delete employees; grant Admin Center access (by any path — matrix, preset, CSV, role default, or copy); assign admin-tier roles; promote/revoke super admins; generate password resets; switch facility context; and delete individual ice-depth sessions, refrigeration reports, and air-quality reports.

6. The Admin Console — Module Administration

Each report module has its own admin console under **Module Admin** in the sidebar. Two rules apply to all of them:

Console access is not module access. Every module console requires the module's **Admin** action in the permission matrix on top of Admin Center access. Without it, the standard error is: "Your account has admin console access but not the {module}'s admin permission. Ask an administrator to grant it under Admin → Permissions."

Original submissions are immutable. Admins never edit what staff submitted. You append follow-up notes (which can't be edited or deleted), change statuses where a lifecycle exists, or — in Daily Reports only — file a supersession-style correction that leaves the original on record.

Most consoles carry an **Export** dropdown in the header (last-30-days CSV/PDF, plus a link to custom ranges in PDF/Export Settings).

6.1 Daily Reports Admin

Configure the daily checklist program and review submissions. Tabs: **Areas** · **Templates** · **Checklist Items** · **Form Builder** · **Area Access** · **Assignments** · **Submissions**.

- **Areas** — the tabs staff pick from: name, slug, color, order, active. Capped at **30 active areas**; bulk CSV upload available.
- **Templates** — per area, the shift checklists ("Opening checklist"), each with its **Checklist Items** (label, description, order — with bulk import and the ability to move items between templates).
- **Form Builder** — versioned custom forms letting staff file multiple titled reports per day ("one 'Event Set Up' report per event"). Field types: text, textarea, number, select, multiselect, checkbox, time, date (max 100 fields). Publishing an edit creates a **new version** — existing report instances keep their frozen snapshot, so history never mutates. This tab also hosts the **End-of-day lock**: locking a day makes every report instance for that day read-only, enforced at the database; you can unlock later.
- **Area Access** — a staff × area checkbox matrix controlling who may submit where (CSV import/export supported). Admins always have access.
- **Assignments** — turn on **area assignment routing** and set the pre-close warning window. When on, assignment resolves: manual override → published schedule → default owners → open. Per area, set default owners and map scheduling job areas.
- **Submissions** — filter by area/employee/date (default last 14 days). Open one to toggle checklist items, add notes, **Delete submission**, or **File correction**: adjust the items, give a required reason, and the correction supersedes the original — which stays on record with a "Corrected" banner linking to the correction.

Note the number staff see: the module advertises that daily reports auto-delete after **14 days** (the retention page's configurable floor is 30 — verify your deployed retention row before quoting a number in training).

6.2 Incident Reports Admin

Review incidents, drive them through a lifecycle, and maintain the dropdowns. Tabs: **History** · **Incident Types** · **Severity Levels** · **Activities**. Locations come from the shared **Facility Spaces** list (a "Manage locations" header button jumps there).

- **History** — filter by status, type, severity, employee, location, and dates. The detail view shows the full original report (read-only) — including the ambulance flag, people involved, witnesses — plus:
 - **Change status**: Submitted → In review → Resolved → Archived.
 - **Follow-up notes** (append-only).
 - **Change log** — the audit trail of every edit, including the reporter's own 24-hour-window edits.
- **Incident Types / Severity Levels / Activities** — the dropdown vocabularies, each with colors and sort order. Seed-defaults buttons appear when a list is empty; activities support bulk import.

6.3 Accident Reports Admin

Tabs: **History** · **Dropdowns** · **Workers' Comp**.

- **History** — filters include severity, body part, medical attention, and a Workers' Comp Yes/No. The detail is fully read-only (there is deliberately no status lifecycle for accidents) with append-only follow-up notes and a change log.
- **Dropdowns** — five vocabularies: **Injury Type**, **Body Part**, **Activity**, **Medical Attention**, **Severity** (location comes from Facility Spaces). The important one: medical-attention values have a **"Triggers communication alert"** checkbox — the seeded defaults fire alerts for Medical Office Visit, Emergency Room, and Hospitalization. This is how a serious injury automatically notifies managers.
- **Workers' Comp** — the instruction text staff must read and acknowledge when filing a comp claim. One textarea, newlines preserved, **Save**.

6.4 Air Quality Admin

Tabs: **Setup** · **Compliance** · **History** · **Settings**, plus the printable **Monitoring Log**.

- **Setup** — locations come from Facility Spaces; per location (or facility-wide) manage **Equipment** (monitors: name, model, serial) and the **Reading types** (label, key, unit, decimals, required). Reading types carry no thresholds themselves — thresholds live in the compliance profile.
- **Compliance** — pick a **jurisdiction profile** (badged Binding or Guidance, 1-hour TWA or Single-sample method), choose the metrics tracked, and set **threshold overrides** per tier — **Corrective**, **Notification**, **Evacuation** — which you may tighten but never loosen below the regulatory floor. Write the **escalation steps** text operators see at each tier. A separate panel keeps a free-text library of compliance rules by jurisdiction.
- **History** — filter by employee, location, equipment, reading type, exceedance, dates, and notes text. Exceedance flags are computed at submit time and frozen. Append-only follow-up notes; deleting a report is super-admin-only.
- **Settings** — testing frequency text shown to staff, default jurisdiction, the alerts on/off switch, and the default alert severity.
- **Monitoring Log** (Printable monitoring log → from History) — an inspector-ready table (date, equipment, time, resurfacer count, CO, NO2, alert level, maintenance, recorded by) over any date range, with **Download PDF**, **Print**, and **Send...** — which emails the PDF straight to recipients you type (e.g. an inspector). Bold red values exceeded a threshold at submit time.

6.5 Ice Depth Admin

Tabs: **Rinks** · **Diagrams** · **Overlays** · **History** · **Analytics** · **Settings**.

- **Rinks** — each sheet of ice, with a default flag.
- **Diagrams** — per rink, the measurement layouts. The point editor places each numbered point by label and X/Y position, with renumbering support.
- **Overlays** — per rink: place **door markers** (typed and colored — door types are seedable) on the diagram, and upload a **center-ice logo** (transparent PNG/SVG/WebP, max 2 MB) rendered as a watermark on every report.
- **History** — sessions filtered by layout, employee, low/high flags, dates. Sessions are immutable; notes are append-only; deleting a session is super-admin-only.
- **Analytics** — average depth, below-min and above-target counts, a **problem-spots heat map** (each point colored by its most common condition, showing its average depth), a per-point breakdown sorted by how often each point reads below minimum, and a daily activity trend.
- **Settings** — measurement unit (inches/mm), low and high thresholds, **alert on** (low / high / any), the three severity colors, alerts on/off, and default severity. Existing sessions snapshot the unit and thresholds at submit time — changing settings never reclassifies history.

6.6 Ice Operations Admin

Tabs: **Setup** · **History** · **Settings**.

- **Setup** — **Rinks**; **Equipment** (name, type — Ice Resurfacer, Edger, Blade Set, Hand Edger, Other — model, serial, hours count, **tank capacity** which enables the %-of-tank water unit, and fuel type); **Circle check items** (scoped to all equipment or one unit); **Fuel types**; and **Circle-check templates** (fuel-type-matched checklists, capped at 4 per facility, with bulk item import). A seed button creates a starter configuration.
- **History** — filter by operation type, employee, rink, equipment, failed-check, dates, and notes search. Detail views are read-only with append-only notes.
- **Settings** — alerts on/off, default severity, and **Visible operations**: which of the five built-in operations staff can log (the types themselves are fixed; at least one must stay on).

6.7 Refrigeration Admin

Tabs: **Setup** · **History** · **Settings**.

- **Setup** — the reading structure: **Sections** (a seed button creates the six standard ones: Compressors, Pumps, Condensers, Supply/Return, Machine Hours, Alarms) → per-section **Equipment** → **Fields** (numeric, text, boolean, select, computed; with units and select options) → **Thresholds** per field: scope (all equipment or one unit), min, max, and severity (warn / high / critical). An equipment-specific threshold overrides the section-level one — the same precedence staff see in their range hints.
- **History** — filter by employee, dates, out-of-range (Any/Yes/No), notes. Detail shows every value with **Out of range** badges; notes are append-only; deletion is super-admin-only.
- **Settings** — out-of-range alerts on/off (off by default), default alert severity, and **readings per shift** (caps the staff Round # field and shows "of N" in their form).

6.8 Dasher Boards Admin

The spatial module. **Labels are permanent identity — the editor never renumbers existing assets.** Tabs: **Perimeter** · **Checklist** · **Lists** · **Walks**, with a first-run wizard (rink name, template, dimensions, perimeter anchor, direction, inspection weekday) generating the whole perimeter from what you enter.

- **Perimeter** — tap a position to edit it: relabel, convert board↔door, set the door subtype, toggle glass, and enter **replacement specs** (dimensions, thickness, material — a "No spec on file" badge flags gaps, and a bulk glass-spec tool fills ranges at once). You can move the start point and change the glass numbering scheme — the numbers printed on the diagram change, but nothing is relabeled.
- **Checklist** — the inspection weekday plus cadenced items (weekly / monthly / yearly; daily ships empty by design — the tap-the-problem model carries daily coverage). Due items must be answered before a walk can sign off, and flagged items create issues in the same pipeline as spatial reports.
- **Lists** — door subtypes (Bench, Scoreboard, Zamboni...) and per-asset-type issue categories (boards, glass, doors).
- **Walks** — read-only inspection history with fail counts, rendered in the rink's glass numbering so it matches what the crew used on the floor.

One permission nuance: entering glass **specs** is available to the edit tier (managers), while structural perimeter changes stay admin-only.

6.9 Communications Admin

The hub that ties every module's alerts together. Tabs: **Inbox** · **Broadcast** · **Templates** · **Groups** · **Routing** · **Reminders** · **Deliveries** · **Audit Log**.

- **Inbox** — two views. **Alerts**: every alert from every module, filterable by source module, severity (info/warn/high/critical), open/resolved, and dates; drill in to **Resolve**, **Re-open**, or Delete (resolving is usually the better close-out), and see the per-person **acknowledgements** list. **Messages**: all staff messages with read/ack rollups.
- **Broadcast** — send to **groups**, **everyone with a role**, or the **whole facility**; start from a template; optionally **Send later** (scheduled broadcasts queue and can be cancelled until they go out); optionally require acknowledgement.
- **Templates** — reusable subject/body/ack presets, used by broadcasts, staff compose, and reminders.
- **Groups** — named recipient lists with members; the **"Staff can message this group"** toggle controls which groups staff see in their compose screen.
- **Routing** — the heart of the alerting story. Each rule: name, **source module**, severity filter, optional area (daily-report area or air-quality location), a single **target** (group, role, employee, or department), a **timing** (Immediate, End-of-day digest, Weekly, or Manual), and two checkboxes — **Attach PDF of the submission** (renders a PDF of the source record onto the email and message) and **Require recipient acknowledgement**. A **Preview Recipients** action shows who a rule will actually hit before it fires. With no rules, alerts still land in this inbox — they just don't fan out to people.
- **Reminders** — recurring scheduled sends: a 5-field cron schedule (e.g. `0 9 * * 1` = 9:00 AM every Monday, evaluated in the facility's timezone), a template, and a group/role target.
- **Deliveries** — failed email and notification sends that exhausted their automatic retries, each with a **Retry** button that re-queues it.
- **Audit Log** — the communications-scoped audit trail.

6.10 How a submission becomes an alert (the pipeline)

Worth a whiteboard minute in any admin training:

1. A staff member submits a report. If it trips a configured trigger — an out-of-range refrigeration value, an air-quality exceedance, a failed circle check, an alert-flagged medical-attention level, a severity-A dasher issue — the module raises an **alert**.
2. **Routing rules** match it (module, severity, area) and fan it out to their targets on their timing — immediately, in an end-of-day digest, or weekly.
3. Recipients get an **in-app message** and an **email**; if the rule attaches a PDF, a rendered PDF of the submission is linked in-app and attached to the email.
4. If the rule requires acknowledgement, recipients must acknowledge in their inbox — and admins can watch the ack list on the alert.
5. Background jobs run every few minutes to render PDFs, drain the queue, and send emails, with automatic retries; anything that fails permanently surfaces in **Deliveries** for a manual retry.

Individual accident/incident PDFs reach people exactly two ways: as a routing-rule attachment, or inside a date-range export. There's no per-report download button on those admin detail pages.

7. Quick Reference

7.1 Status lifecycles at a glance

Object	Statuses
Schedule shift	draft → published → cancelled
Time-off request	pending → approved / denied / cancelled
Swap request	pending → accepted → manager approved (applied); or denied / cancelled / expired
Shift drop request	pending → approved / denied / withdrawn
Publish request	pending → published / rejected
Incident report	Submitted → In review → Resolved → Archived
Offline queue item	pending → synced / failed / won't retry

7.2 Who approves what (scheduling)

Request	Approver
Time off	Any scheduling admin
Swap	Scheduling admin, after the target accepts
Shift drop	Scheduling admin (unless approval is turned off — then instant)
Open-shift claim	Automatic if first-come-first-served; otherwise a scheduling admin
Publishing a schedule	A different admin than the requester — always two people
Certification override	A facility manager only — always audit-logged

7.3 Edit windows and immutability

Report type	After submitting
Incident report	Reporter can edit for 24 hours, then read-only
Accident report	Reporter can edit for 24 hours, then read-only
Daily report	Corrections only (original stays on record)
Ice depth / Ice operations / Refrigeration / Air quality	Immutable immediately
Admin follow-up notes (all modules)	Append-only — can never be edited or deleted

7.4 What works offline

Works offline (queued)	Online only
All report submissions (daily, incidents, accidents, ice depth, ice ops, refrigeration, air quality, dasher issues & walks)	Editing an existing report
Time-off requests, availability changes	Claiming, dropping, and swapping shifts
Composing messages	Acknowledging alerts, marking read
Viewing /offline-daily, /offline-forms, /offline-schedule	All admin console work

On iPhone/iPad, open the app and tap **Sync now** in the Pending Sync Queue after working offline — iOS doesn't sync in the background.

7.5 Permission actions

Action	Means
View	Can see the module
Submit	Can create and submit entries
Edit	Can modify own or facility submissions (e.g. acknowledge severity-A dasher issues, supervisor tiers)
Admin	Can configure the module and use its admin console

Special cells: **Admin module × Admin action** = the Admin Center key (super-admin-only to grant). Module consoles need the module's Admin action on top of console access.

7.6 Super-admin-only operations

Create/edit facilities · delete employees · grant Admin Center access · assign admin-tier roles · promote/revoke super admins · generate password-reset links · switch facility context · delete ice-depth sessions, refrigeration reports, and air-quality reports.

7.7 Training checklists

New staff member — first session (30 minutes):

1. Sign in from the invite email and set a password.
2. Tour the dashboard tiles and status bubbles; hide unused tiles.
3. Fill in Account settings (phone, address, emergency contact).
4. Submit a practice report in your primary module.
5. Scheduling: set availability for each weekday, set up calendar sync, find the Drop and Swap buttons.
6. Turn on airplane mode and submit a report — watch it queue and sync.
7. Find the Pending Sync Queue and the offline schedule page.

New admin — first session (60 minutes):

1. Tour the console: Setup vs. Module Admin vs. System groups; know that the sidebar isn't permission-filtered and Roles isn't in it.
2. Walk the setup checklist on the admin dashboard.

3. Create a test employee with an invite; watch the role defaults apply.
4. Open the Permissions matrix; find the Admin Center cell and understand why it's special.
5. Configure one module end-to-end (e.g. refrigeration sections → fields → thresholds → submit a test report → see the out-of-range flag → watch the routing rule fire).
6. Scheduling: draw shifts, hit a cert block, file a publish request, approve it as a second admin.
7. Review retention floors and the audit log.

Part 4 — Install, Setup & Deployment

Getting the app onto a phone, standing up a facility, and running your own copy.

Install Rink Reports on Your Phone

Rink Reports works like a regular app once you add it to your home screen.

You only need to do this once. After that, tap the Rink Reports icon like any other app — no browser, no typing the address, and you stay logged in.

Install it on the phone you carry during your shift. It keeps working even when the wifi is bad rink-side: anything you submit is saved on your phone and sends itself when the signal comes back.

Tip: the first time you open the site on your phone, a navy banner may appear at the bottom offering to install. Following that banner does the same thing as the steps below.

iPhone (Safari)

<!-- SCREENSHOT: Safari open on the login page, Share button highlighted -->

1. Open **Safari** and go to the Rink Reports website (use the link your manager sent you).
2. Log in with your email and password.
3. Tap the **Share** button — the square with an arrow pointing up, at the bottom center of the screen.
4. Scroll down the list and tap **Add to Home Screen**.
5. Tap **Add** in the top-right corner.
6. Close Safari. You'll see the **RinkReports** icon (white snowflake on navy) on your home screen.
7. From now on, open the app from that icon — not from Safari.

Important: this only works in Safari. If you use Chrome on your iPhone, open the site in Safari for this one step.

Android (Chrome)

<!-- SCREENSHOT: Chrome open on the login page, install banner visible -->

1. Open **Chrome** and go to the Rink Reports website (use the link your manager sent you).
2. Log in with your email and password.
3. A banner should appear offering to **Install app** — tap it and confirm.
4. If you don't see the banner: tap the **⋮** menu (three dots, top-right), then tap **Add to Home screen** (on some phones it says **Install app**), then confirm.

5. You'll see the **RinkReports** icon on your home screen.
 6. From now on, open the app from that icon.
-

After Installing

- **Stay logged in.** Log in once and the app remembers you. If it ever asks you to log in again, just sign back in — your saved reports are not lost.
 - **Weak signal is okay.** Submit your reports normally. If you're offline, the app saves them on your phone and shows a pending count; they send automatically when you're back in signal.
 - **Open the app at least once per shift.** This keeps everything synced and up to date.
 - **Don't delete the icon** — removing the app from your home screen can delete reports that haven't synced yet. If you need to reinstall, make sure the pending count is zero first.
- Questions? Ask your facility admin.

RinkReports 5-6 — Admin Setup Guide

A practical, step-by-step guide for getting your ice facility operational with RinkReports. Written for facility managers setting up the system for the first time at **Tennity Ice Skating Pavilion** (the same steps apply to any facility).

Before you start: You need Super Admin credentials. If you don't have them yet, contact whoever installed RinkReports — they create the initial Super Admin account directly in Supabase.

1. Log in as Super Admin

1. Open your RinkReports URL in a browser (Chrome or Edge recommended).
2. You will land on the **Login** page. If you go to a different page, you'll be bounced to `/login` automatically.
3. Enter your **Super Admin email and password**.
4. Click **Sign In**. You'll be redirected to `/dashboard`.

If sign-in fails:

- Check that you're using the email tied to your Super Admin record.
- Use **Forgot password?** to reset via email.
- If you see a "Forbidden" page after login, your account exists but isn't flagged as an admin yet — see Section 3.

2. Access the Admin Control Center

1. From the dashboard, click your name (top right) → **Admin**, or go directly to `/admin`.
2. The Admin Control Center is your home for everything in this guide. Modules include:
 - **Facilities** — facility info, areas, equipment
 - **Employees** — staff, roles, permissions
 - **Scheduling** — shifts and assignments
 - **Reports** — Daily, Incidents, Accidents, Ice Depth, Ice Operations, Refrigeration, Air Quality, Communications
 - **Retention** — data retention policies
 - **Exports** — CSV/PDF exports for compliance

Only Super Admins, Admins, and GMs see the Admin Control Center. Regular staff are redirected to `/forbidden` if they try to open it.

3. Set Facility Information

Set this up **first** — almost everything else (employees, reports, scheduling) is scoped to a facility.

1. In the Admin Control Center, open **Facilities**.
2. Click **+ New Facility** (or open the existing record if RinkReports created one during install).
3. Fill in:

- **Name:** Tennity Ice Skating Pavilion
 - **Address, City, State, ZIP**
 - **Time Zone:** select your local zone (this drives report timestamps)
 - **Contact phone / email**
 - **Operating hours** (per day of week)
4. Add **Areas** — the physical zones inside the facility (each becomes a slug used in report URLs):
 - main-rink
 - studio-rink (if applicable)
 - locker-rooms
 - mechanical-room
 - lobby
 - pro-shop
 5. Click **Save**. The facility now appears in the dropdown at the top of the admin shell.

Tip: Set the facility's time zone before staff start submitting reports. Changing it later will shift the displayed timestamps on historical reports.

4. Create Roles and Permissions

RinkReports ships with a role-based permission model. Roles control **what someone can see and do**; permissions are attached to roles, not individuals.

Built-in roles

Role key	Typical title	Sees Admin Center?
super_admin	Owner / IT lead	Yes
admin	Facility Director	Yes
gm	General Manager	Yes
supervisor	Shift Supervisor	No (sees Reports + Scheduling)
staff	Rink Attendant / Skate Guard	No (submits reports only)

Adjust permissions

1. Open **Admin → Employees → Roles**.
2. Click a role to open its permission matrix.
3. Toggle module permissions per role. Common starting points:
 - **Staff:** can submit Daily, Incidents, Ice Depth. Cannot view others' reports.
 - **Supervisor:** can view all reports for their facility, can edit same-day submissions.
 - **Admin / GM:** full read/write on all modules for their facility.
4. Click **Save**.

Create a custom role (optional)

1. **Roles → + New Role**.

2. Give it a key (lowercase, no spaces — e.g. `ice_tech`) and a display name.
3. Set permissions, save.

Permissions are enforced both in the UI and at the database (Row-Level Security). A staff member cannot bypass the UI to view another facility's data.

5. Configure the Ten Daily Report Tabs

The Daily Report is what most staff will touch every shift. Configure the ten tabs once for Tennity, and they apply to every daily submission going forward.

1. Open **Admin → Reports → Daily → Templates**.
2. Select the **Tennity Ice Skating Pavilion** facility from the top filter.
3. You'll see ten tabs. Configure each one in order:
 1. **Opening Checklist** — doors unlocked, lights on, restrooms cleaned, lobby walk-through.
 2. **Ice Conditions** — ice surface check, edge condition, debris, photo upload.
 3. **Ice Depth** — depth readings at predefined points (add the points under the **Ice Depth** module first).
 4. **Refrigeration** — brine temp, header pressure, compressor run hours, ammonia alarm test.
 5. **Air Quality** — CO and NO₂ readings (set thresholds that trigger an alert).
 6. **Resurfacer Log** — flood times, water temp, blade condition, fuel/charge level.
 7. **Locker Rooms & Restrooms** — cleaned, restocked, damage noted.
 8. **Pro Shop / Skate Rental** — inventory snapshot, sharpening log.
 9. **Public Skate & Programs** — attendance counts per session, incidents flag.
 10. **Closing Checklist** — ice covered, doors locked, alarm armed, lights off, equipment shut down.
4. For each tab:
 - Click **Edit Fields**.
 - Add the questions/fields you want staff to answer (text, number, yes/no, photo, signature).
 - Mark required fields with the red asterisk toggle.
 - Set **threshold alerts** where applicable (e.g. ice depth < 1.0" pages the GM).
5. Click **Publish Template**. The new layout is live for the next submission.

Tip: Keep required fields to the minimum that satisfies your insurance/health-dept requirements. Staff abandon long forms — every extra field is a missed entry on a busy Saturday.

6. Add Staff Members

1. Open **Admin → Employees → + Add Employee**.
2. Fill in:
 - **Full name**
 - **Email** (this is their login)
 - **Phone**
 - **Facility:** Tennity Ice Skating Pavilion

- **Role:** pick from Section 4
 - **Start date**
 - **Active:** yes
3. Click **Send Invite**. The staff member receives an email with a link to set their password.
 4. After they accept, their row shows **Active** with a green dot.

Bulk add

For a full roster, use **Employees → Import CSV**. Required columns: `email`, `full_name`, `role_key`, `phone`. Each row triggers an invite email.

Day-one checklist for new staff

- ☐ Account created and invite accepted
- ☐ Role assigned (and verified — try logging in as them in an incognito window if unsure)
- ☐ Added to at least one scheduled shift in **Scheduling**
- ☐ Walked through submitting one Daily Report on their phone (it's a PWA — they can "Add to Home Screen")

You're Operational

At this point Tennity Ice Skating Pavilion has:

- A configured facility with areas and hours
- A role/permission model that matches your org chart
- A ten-tab Daily Report template tailored to your operation
- Staff accounts that can log in and submit reports — including offline, since RinkReports queues submissions in the service worker and syncs when the connection returns

Where to go next

- **Scheduling** — build recurring shift patterns so the Daily Report knows who's on the floor.
- **Retention** — set how long submissions, photos, and exports are kept (defaults are conservative; check with your insurer).
- **Exports** — schedule a monthly CSV/PDF export for compliance and insurance.
- **Incidents / Accidents** — review the templates; these are the reports your insurer will actually ask for.

Getting help

- Forgot your Super Admin password? Use the **Forgot password?** link on `/login`.
- Locked out entirely? Whoever has Supabase access can re-flag your user as Super Admin in the `users` table (`is_super_admin = true`).
- Bug or feature request? Open an issue on the project's GitHub repo.

Duplicating RinkReports — Step-by-Step Setup Guide

This guide explains how to stand up a **complete, independent copy** of RinkReports — your own running instance with its own database, its own login, and (optionally) its own public web address. Follow it top to bottom for a fresh production deployment, or stop after Part 4 if you only need a local copy for development or evaluation.

What you are duplicating. RinkReports is a Next.js (App Router) progressive web app backed by a Supabase project (PostgreSQL + Auth). The web app is stateless and hosts on Vercel; all data, accounts, and security rules live in Supabase. So "duplicating the software" really means three things: (1) get the application code, (2) create a fresh Supabase backend and load its database structure, and (3) connect the two with a handful of settings, then deploy. Brand reference for any copy you customize: **#4DFF00** (primary green), **#002244** (navy).

A copy is fully isolated: each instance has its own facilities, employees, and reports — you only ever see data for your own instance and facility; this is automatic.

Part 0 — Prerequisites (install these once)

Tool	Version	Why	Get it
Node.js	20 LTS or newer	Runs the build and tooling	nodejs.org
pnpm	latest	Package manager (this repo's lockfile is <code>pnpm-lock.yaml</code>)	corepack enable pnpm
Git	any recent	Clone the code	git-scm.com
Supabase CLI	latest	Create/link the database, run migrations locally	supabase.com/docs/guides/cli
Docker Desktop	latest	Required by supabase start for the local stack	docker.com
psql (PostgreSQL client)	15+	Run the database security test	bundled with PostgreSQL

Accounts you'll need for a **production** copy (skip for local-only):

- A **Supabase** account (Pro plan recommended for production — log retention and point-in-time recovery).
- A **Vercel** account (to host the web app).
- A **Resend** account (optional) — only if your copy should send invitation/notification emails.
- A **domain name** (optional) — only if you want a custom web address.

Part 1 — Get the application code

```
git clone <your-fork-or-repo-url> rinkreports
cd rinkreports
pnpm install
```

`pnpm install` reads `pnpm-lock.yaml` and installs the exact dependency versions this app was built and tested against (Next.js 16, React 19, the Supabase client libraries, and the test/build tooling).

Tip — make it truly yours. If this is a brand-new copy rather than a contribution back to the original, create your own empty Git repository and push the code there (`git remote set-url origin <your-new-repo>; git push -u origin main`). Your new Supabase project and Vercel project will connect to that repo.

Part 2 — Create the Supabase backend

You have two tracks. Do **Track A** to develop/evaluate on your laptop. Do **Track B** to run a real, always-on copy. For a production copy you'll typically do A first (to verify everything works), then B.

Track A — Local backend (laptop copy)

1. Start Docker Desktop.
2. Boot the local Supabase stack. This **also applies every migration** in `supabase/migrations/` automatically:

```
supabase start
```

It prints local connection details. The committed `supabase/config.toml` pins the local ports: **API 54321, DB 54322, Studio 54323, Inbucket (test email) 54324**. Studio (the database UI) is at `http://127.0.0.1:54323`.

3. The local database now contains the full RinkReports schema — every table, security rule, and helper function — because the migrations ran during `supabase start`.

Track B — Remote backend (production copy)

1. In the Supabase dashboard, **create a new project**. Choose a strong database password and a region near your users.
2. From **Settings → API**, copy these three values (you'll need them in Part 3):
 - **Project URL**
 - **anon public key**
 - **service_role secret key** (server-only — never exposed to the browser)
3. Link your local checkout to the new remote project and push the database structure:

```
supabase link --project-ref <your-project-ref>
supabase db push      # applies all migrations in numeric order
```

This creates the entire RinkReports schema on your new cloud database. The migration set is a flat, numerically ordered series of files under `supabase/migrations/` (there are 160+); `supabase db push` runs any that haven't been applied yet.

Important: never hand-edit the cloud database structure. All schema changes go through the migration files so any copy can be rebuilt identically.

Part 3 — Connect the app to the backend (environment variables)

Copy the template and fill it in:

```
cp .env.example .env.local
```

For **local** development, point at the local stack values that supabase start printed. For a **remote/production** copy, use the values from your Supabase project's **Settings → API**.

Variable	Required?	What it is
NEXT_PUBLIC_SUPABASE_URL	Yes	Your Supabase Project URL
NEXT_PUBLIC_SUPABASE_ANON_KEY	Yes	The Supabase anon public key
SUPABASE_SERVICE_ROLE_KEY	Server features	Service-role secret. Powers the email-invite flow and the notification cron. Leave blank in local dev — the Invite button then shows a friendly error and the cron returns a harmless 503.
NEXT_PUBLIC_SITE_URL	Recommended	Base URL for links inside invitation / password-reset emails. Local: <code>http://localhost:3000</code> . Production: your canonical site URL, no trailing slash.
CRON_SECRET	Production	A long random string (openssl rand -hex 32) that authenticates the scheduled background jobs.
RESEND_API_KEY / RESEND_FROM	Optional	Email delivery (see Part 7). Leave blank to disable — outgoing messages simply queue until configured.
RESEND_ENABLED	Optional	Override the "production only" email gate (true to force-enable, false as a kill switch).
NEXT_PUBLIC_POSTHOG_KEY / _HOST / _ENABLED	Optional	Error/analytics reporting (see Part 8). Leave blank to disable entirely.

Never prefix a secret with NEXT_PUBLIC_. Anything with that prefix is shipped to the browser. The service-role key and cron secret must stay server-only.

Part 4 — Generate database types, then run it

1. Regenerate the TypeScript database types so the app's types match your schema. Point DATABASE_URL at a fully-migrated database (the local stack is easiest):

```
DATABASE_URL="postgresql://postgres:postgres@127.0.0.1:54322/postgres" pnpm types:write
```

(pnpm types:check is the read-only version that CI uses to confirm the types are in sync.)

2. Start the app:

```
pnpm dev          # development server at http://localhost:3000
```

For a production-style local run: `pnpm build` then `pnpm start`.

At this point you have a working copy — but no one can log in yet, because RinkReports has **no public signup**. Create the first account in Part 6.

Part 5 — Verify the copy is healthy (tests & checks)

Run these against your new instance before trusting it:

```
pnpm lint          # code/style checks
pnpm test          # unit tests (vitest) – pure submission/compute logic
pnpm check:cron    # validates the scheduled-job configuration
```

Database security regression test — confirms each facility's data stays isolated (the core guarantee of the app). Run it against a migrated database; it executes in a single transaction and rolls back at the end, so it changes nothing:

```
psql "postgresql://postgres:postgres@127.0.0.1:5432/postgres" \
-v ON_ERROR_STOP=1 -f supabase/tests/rls_isolation.sql
```

Against a **remote** database, point `psql` at that project's connection string instead. Any failure raises and exits non-zero.

Part 6 — Bootstrap the first administrator

There is no public sign-up page, so the very first account is created directly in Supabase. This person becomes the platform owner (super-admin) who can then create facilities and invite everyone else through the app.

1. Create the user. In **Supabase Studio** → **Authentication** → **Users**, click **Add user** and set an email + password. (Locally, Studio is at `http://127.0.0.1:54323`; test emails land in Inbucket at `http://127.0.0.1:54324`.)
2. Promote that user to super-admin. In the **SQL editor**, run:

```
update public.users
set is_super_admin = true
where id = '<the-new-user-uuid>';
```

3. Sign in to the running app with that email/password. You now have the **Super Admin** console and the **Admin Center**.

From here, everything else is done **inside the app** — no more SQL. Follow

ONBOARDING-ADMIN.md to create your first facility, add employees, assign roles, and turn on the modules you need. The detailed configuration for each module lives in its chapter under `modules/` (see the Master Manual).

You can stop here for a local or internal copy. Parts 7–10 are only needed to put your copy on the public internet.

Part 7 — Deploy the web app (Vercel)

1. In Vercel, **Add New → Project** and import your Git repository. Framework preset: **Next.js** (auto-detected).
2. Add the environment variables from Part 3 under **Settings → Environment Variables**:

Variable	Scope
NEXT_PUBLIC_SUPABASE_URL	All environments
NEXT_PUBLIC_SUPABASE_ANON_KEY	All environments
SUPABASE_SERVICE_ROLE_KEY	Production + Preview
CRON_SECRET	Production + Preview
NEXT_PUBLIC_SITE_URL	Per environment (production = your canonical URL)
RESEND_API_KEY / RESEND_FROM	Production + Preview (if using email)

3. **Scheduled background jobs** are committed in `vercel.json` and picked up automatically — no dashboard setup. Your copy runs **five** crons, all authenticated with `CRON_SECRET`:

Schedule (UTC)	Job	Purpose
* / 5 * * * *	/api/cron/drain-notifications	Fan out in-app inbox notifications
2-59 / 5 * * * *	/api/cron/send-communications	Send queued emails via Resend (staggered after drain)
4-59 / 10 * * * *	/api/cron/expire-scheduling	Expire stale scheduling requests
3-59 / 5 * * * *	/api/cron/run-reminders	Send scheduled reminders
17 3 * * *	/api/cron/run-retention-purge	Daily data-retention cleanup, off-peak

4. Deploy (push to main, or trigger from the Vercel UI) and confirm a green build.

Part 8 — Email delivery (Resend) — optional

Only needed if your copy should email staff (login invitations, notifications). Without it, the app works fully; outgoing messages just stay queued and flush automatically once email is configured.

1. In Resend, **verify your sending domain**. The exact DNS records (SPF, DKIM, DMARC) are listed in `.env.example` next to `RESEND_FROM`; verification fails unless all three are present.
2. Generate a Resend API key → `RESEND_API_KEY`.
3. Set `RESEND_FROM` to a sender on the verified domain, e.g. RinkReports `<noreply@send.example.com>`.

Email only sends in the **production** environment by default, so a preview or dev copy that inherits production secrets can never email real staff. Override with `RESEND_ENABLED=true/false` if you need to.

Part 9 — Custom domain & analytics — optional

- **Domain:** add it under Vercel **Settings → Domains**, point your DNS as instructed, and set `NEXT_PUBLIC_SITE_URL` to that canonical address (no trailing slash) so email links resolve correctly. Vercel issues TLS automatically.

- **Analytics/error reporting (PostHog):** set NEXT_PUBLIC_POSTHOG_KEY (and _HOST only for EU/self-hosted). Leave blank to disable — PostHog is then never loaded, so there's no bundle or network cost. Capture is production-gated and PII-scrubbed.

Part 10 — Confirm the live copy

After the first production deploy, run the smoke tests. The first checks are also automated by the post-deploy-smoke GitHub workflow; the full list is in ../DEPLOY.md §5.

```
curl -s https://<your-domain>/api/health # 200, {"ok":true,...}
curl -sI https://<your-domain>/login | head -1 # 200
curl -sI https://<your-domain>/sw.js | head -1 # 200 (service worker — enables offline)
curl -sI https://<your-domain>/robots.txt | head -1 # 200, body "Disallow: /"
```

Then sign in as your super-admin and walk the first-facility setup in ONBOARDING-ADMIN.md.

Quick duplication checklist

```
[ ] Install Node 20+, pnpm, Git, Supabase CLI, Docker, psql
[ ] git clone ... && pnpm install
[ ] Supabase: supabase start (local) OR create project + supabase link + supabase db push (
remote)
[ ] cp .env.example .env.local → fill SUPABASE URL + anon key (+ service role / cron secret
for prod)
[ ] DATABASE_URL=... pnpm types:write
[ ] pnpm dev (verify it loads)
[ ] pnpm lint && pnpm test && pnpm check:cron
[ ] psql ... -f supabase/tests/rls_isolation.sql (security test passes)
[ ] Create first user in Studio → set is_super_admin = true
[ ] (prod) Import repo to Vercel + set env vars; crons load from vercel.json
[ ] (prod) Configure Resend + DNS; add custom domain; set NEXT_PUBLIC_SITE_URL
[ ] (prod) Run /api/health + smoke tests
[ ] Sign in as super-admin → follow ONBOARDING-ADMIN.md to stand up a facility
```

Troubleshooting

Symptom	Likely cause / fix
supabase start fails	Docker isn't running, or ports 54321-54324 are taken. Start Docker; stop the conflicting service.
App loads but every page bounces to /login	NEXT_PUBLIC_SUPABASE_URL / ANON_KEY missing or wrong in .env.local.
Can't sign in at all	No account exists yet — RinkReports has no public signup. Do Part 6.
Signed in but /admin shows "Forbidden"	The account isn't an admin. Set is_super_admin = true, or have an admin grant Admin-Center access.
Invite employee button errors	SUPABASE_SERVICE_ROLE_KEY not set (expected in local dev).
pnpm types:check fails in CI	Schema and generated types drifted — rerun pnpm types:write against a fully-migrated DB and commit the result.

Symptom	Likely cause / fix
Emails never arrive	Resend not configured, domain unverified, or you're not in the production environment. See Part 8.
Background jobs not running	CRON_SECRET missing on Vercel, or you're testing on a plan without Cron. Crons are defined in <code>vercel.json</code> .

Companion documents: Master Manual · Admin onboarding · production operations runbook
../DEPLOY.md · local-dev quickstart in the repository README.md.